# Unintended Environmental Policies:

## The Impact of European Agricultural Subsidies on Pollution*

Emma Du Puy†

October 19, 2026

**Abstract:** I study the environmental impact of production subsidies in polluting markets. I develop a dynamic model where the elasticity of substitution between clean and polluting inputs signs the correlation between firm profitability and pollution intensity. Applied to the Common Agricultural Policy, the EU's largest industrial policy, estimation shows more profitable farms pollute more, making marginal incumbents greener. Subsidies generate unintended environmental benefits by retaining marginal farms and shifting within-farm input use. Relative to no intervention, the optimal chemical tax lowers chemical-use intensity by 64%, optimal subsidies by 31%, but at similar abatement costs and without consumer and producer surplus losses.

JEL Codes: L11, Q18, Q52, H23, D24
Keywords: Environmental Pollution, Agricultural Policy, Industry Dynamics, Production Function

*I thank my advisors Suresh Naidu, Pietro Tebaldi, Wolfram Schlenker, Gautam Gowrisankaran and Jeffrey Shrader for their guidance during this project. I am also indebted to Matthew Backus, Mathias Reynaert, Bernard Salanié, Sylvain Chabé-Ferret, Eric Verhoogen, Jack Willis, Allan Hsiao, Robert Metcalfe, Rodrigo Soares, Karam Kang, Sebastián Otero, Diego Restuccia, Marguerite Obolensky, Nicolas Longuet-Marx and Thierry Mayer for their generous feedback, as well as Yulia Kim for her support. I am grateful for comments from participants in seminars and conferences at Columbia University, the Toulouse School of Economics, the CUNY Stone Center on Socio-Economic Inequality, Sciences Po, the University of Saskatchewan, the Paris School of Economics, CIRED, NC State, CREST, the Northeastern Workshop on Energy Policy and Environmental Economics, the Bozeman Applied Economics Summer Conference, Camp Resources, NAREA, WEAI, the 8th Social Cost of Water Pollution Workshop, IIOC, EARIE. All remaining errors are my own. I would like to acknowledge funding from the Columbia Center for Political Economy and the Columbia Center for Environmental Economics and Policy, and thank INSEE, the CASD and the French Ministry of Agriculture for data access. A previous version of this paper was circulated under the name "An Empirical Model of Agricultural Subsidies with Environmental Externalities".

†Du Puy: Department of Economics, Sciences Po (emma.dupuy@sciencespo.fr)

# 1 Introduction

The design of policies that aim to reduce pollution is influenced by the characteristics of their targeted market, from the nature of production technologies to the heterogeneity of producers (Goulder and Parry (2008)). Just as these features shape the optimal design of environmental regulation, they can also give rise to environmental side effects for policies without explicit environmental goals. Such side effects can be expected to be large for industrial policies that substantially distort production within polluting markets. These distortions are typically economically inefficient, but are they also harmful to the environment? This paper studies the unintended environmental consequences of such policies, focusing on producer subsidies in settings where market participation is endogenous to policy, and producers differ both in profitability and pollution intensity.

To understand how subsidies shape environmental outcomes, one needs to account not only for their effect on within-firm production decisions (an intensive margin), but also for whether they shift market selection (an extensive margin). On the intensive margin, subsidies can alter both how firms produce and what they produce, with environmental consequences governed by technology. First, subsidies can shift input mixes by changing the relative prices of clean versus polluting inputs—either directly, or indirectly by subsidizing complements of or substitutes for polluting inputs. Second, output-biased subsidies alter product mix choices, whose environmental effect depends on how goods differ in pollution intensity. On the extensive margin, subsidies distort market participation by shifting the profit threshold for exit, retaining marginal incumbents that would otherwise leave the market. The environmental consequence of this selection effect depends on the correlation between firm profitability and pollution intensity. If marginal firms are the most pollution-intensive, for example because they are more resource-intensive, softening selection is both economically and environmentally costly. Conversely, if profitability is driven by the intensive use of polluting inputs, subsidies that retain less profitable, marginal firms may inadvertently reduce aggregate pollution. Which of these cases prevails is an empirical question.

I account for these different margins by developing a dynamic model of heterogeneous multiproduct producers, which I apply to the European Union's Common Agricultural Policy (CAP) and the French grain market. The CAP is the EU's largest industrial policy, having accounted for 45% of the EU budget since 1980, and while most of its subsidies have not historically targeted the environment, their potential unintended environmental effects have been a great source of concern for policymakers (European Court of Auditors (2024)). Agriculture offers an ideal setting to study the unintended environmental effects of production subsidies. It is an industry which combines substantial producer heterogeneity with extensive government support, nearly $750 billion annually (FAO (2024)), which heavily shapes mar-

ket participation and distorts production. It is also an industry responsible for widespread and loosely regulated air and water pollution that stems from fertilizer and pesticide use, a pollution on par with the energy sector in terms of magnitude, but one that has proven more difficult to abate (Tschofen et al. (2019)). Finally, my focus on the French grain market is justified by its rich administrative data, in the form of a detailed representative panel of producers and its farm-level Census of Agriculture.

I start by documenting the effect of CAP subsidies on farm behavior and aggregate chemical use. I rely for this on the 1992 MacSharry reform, which replaced crop-heterogeneous price support with more uniform land subsidies, simultaneously lowering per-unit-of-output subsidies and altering incentives for crop and input choices. Relying on a shift-share analysis, I show that within the most exposed municipalities, the reform led both to farm exit and to a homogenization of land allocations across crops. It also lowered within-farm chemical expenditure as well as visible fertilizer-driven pollution on surface water. This exercise highlights how agricultural subsidies can interact with the intensive and extensive margins of aggregate chemical-use intensity, but falls short of adjudicating the respective role of these margins. To quantify their respective importance, as well as to compare the performance of subsidies relative to direct chemical regulations, I then turn to a model of the French grain market.

The empirical model builds on the dynamic single-agent industry framework of Hopenhayn (1992) and captures the main drivers of farm chemical use. Farms differ both in total factor productivity (TFP) and chemical use efficiency, and crops vary in their reliance on chemicals. Finally, farms are multiproduct producers that accumulate capital over time, a feature which helps me to better capture the determinants of market exit and its relation with farm profitability. The estimation of the model proceeds in three steps. First, I recover iso-elastic demand curves using weather shocks as supply shifters. Second, I estimate the production parameters. This step faces two identification challenges: producers have unobserved heterogeneity in both TFP and chemical-use efficiency, and while farms are also modeled as multiproduct, all inputs in the data except for land are observed at the farm rather than the farm-crop level. I address the endogeneity issue stemming from the unobserved farm type by inverting first-order conditions for profit maximization and specifying a process for this type. In so doing, I recover the moments needed to both identify the production parameters and to recover the distribution of unobserved productivity shocks, building on Doraszelski and Jaumandreu (2018). To account for multiproduct producers, I follow De Loecker et al. (2016) and restrict estimation to single-product farms, for which the input-output allocation is known. I control for the induced estimation bias with an extension of the correction developed by De Loecker et al. (2016). Third, I use a simulated method-of-moments estimator to recover the dynamic costs of capital adjustment, entry, and incumbency.

The estimation yields an elasticity of substitution between land and chemicals of 1.79. The implied substitution pattern is intuitive if one thinks of organic farming where the high yields of intensive chemical use are compensated for with more intensive land use. Using the estimated model in counterfactuals, I start by showing that past and existing CAP subsidies have unintended environmental benefits, and rely on an extension of the Melitz and Polanec (2015) decomposition to show how these benefits are achieved. Due to the substitutability of land and chemicals, a land subsidy lowers the market's average chemical-per-output ratio through the intensive margin of input choice. The estimated elasticity also implies that lower profit farms have a lower chemical-per-output ratio. As a consequence, a lump-sum payment to low-profit farms reallocates production towards greener marginal incumbents. The exercise further illustrates how selection effects tend to be smaller than input choice ones, in line with the MacSharry shift-share results where combined farm exit and the introduction of a land subsidy lowered water pollution. These subsidies are compared to direct environmental regulations which have been considered by the EU, first a chemical tax, and second a cap on farms' crop-level chemical-to-land ratios. While the tax treats all farms by shifting input use, the threshold acts by reallocating production within infra-marginal incumbents away from the highest pollution farms.

I then study the efficiency of optimal policies at curbing environmental damages. I propose different parameterizations of an environmental damage function which account for environmental damages stemming both from the intensity of local chemical use, and for the ecological opportunity costs of allocating more land to agriculture. I search for the welfare-maximizing levels of each policy design. Optimal land subsidies and lump-sum payments generate sufficient environmental damage reductions to achieve overall welfare gains, the lump-sum payment doing best when the ecological opportunity costs of agricultural land use are large, while the land subsidy does best when the convexity of damages in local chemical-use intensity is high. A comprehensive subsidy combining a land subsidy heterogeneous across crops as well as a small farm lump-sum payment does best, subsidizing relatively more the crop grown by the most chemical-intensive farms, and further allowing for some production reallocation towards marginal incumbents. Yet, while the tax reduces overall chemical-use intensity by 64%, the optimal subsidy only reaches 31%. It does so, however, with comparable abatement costs, and while avoiding the large consumer and producer surplus losses generated by the tax. This comparison highlights a tension between the environmental efficiency of direct chemical regulation, and its acceptability to consumers and producers. I finally propose a simple extension of the model where France produces alongside the rest of the world to meet world demand. This exercise generates two additional lessons: first, environmental regulations increase foreign production and generate pollution leakages while subsidies pool additional production in a now greener French agriculture, second, as integration in world markets raises the elasticity of the residual demand faced by France,

the incidence of policy shifts and producers bear increasingly more of the costs of regulation and receive more of the benefits of subsidies.

Within environmental economics, I contribute to a large literature studying the optimal design of environmental regulation (Carlton and Loury (1980); Bovenberg and Goulder (1996)). Recent work has highlighted the dual role of policies in generating within-firm changes in pollution, as well as across-firm reallocations (Fowlie et al. (2016); Shapiro and Walker (2018)). Here, I show how policies can reallocate production towards marginal incumbents. I further show how these marginal incumbents can systematically differ from infra-marginal firms in pollution intensity, quantifying the environmental effect of these reallocations. I also complement a literature discussing the incidence of abatement costs (Ganapati et al. (2020); Dechezleprêtre et al. (2025)), revealing a tension between the efficiency and acceptability of environmental regulations relative to second-best production subsidies.

I draw on a literature which has documented the unintended environmental effects of generic industrial policies within polluting markets. Previous work has studied cases where subsidies have negative effects, such as fuel subsidies (Davis (2017); Englander et al. (2023)) or trade policy (Shapiro (2021)). In contrast, I show that the EU's largest industrial policy carries unintended environmental benefits. These benefits rely first on the substitutability between clean (land) and polluting (chemical) inputs, creating a role for land subsidies in reducing chemical-use intensity. They rely second on the presence of heterogeneity in chemical-use efficiency across producers, which, combined with substitutability, implies that larger farms pollute more. This pattern matches observational studies (Larsen and Noack (2017)) and a well-documented yield gap between organic and non-organic agriculture (Seufert et al. (2012)). It stands in opposition to the case of manufacturing where larger firms are usually found to be cleaner due to heterogeneity in regulation (Becker and Henderson (2000)), and because of their higher resource efficiency and capacity to cover the fixed costs of abatement technologies (Qi et al. (2021)). My paper focuses on the well-documented case of agricultural pollution (Chabé-Ferret et al. (2021); Dias et al. (2023)), tracking the effect of fertilizer pollution directly through satellite data (Taylor and Heal (2023)), and proposing a dynamic heterogeneous-agent industry model to explain the overall pollution intensity of the industry.

Finally, this paper contributes to the substantial literature documenting the welfare effects of agricultural policies, often highlighting their role in generating misallocations and production inefficiencies (Gollin et al. (2014)). A growing subset of this literature focuses on the interactions between the environment and agricultural policies. Hsiao et al. (2026) and Obolensky (2025) for example study how such policies can shape adaptation to climate change. Because agriculture is also a primary source of land use, accounting for 38% of all EU land, it is also important to understand how agricultural policies shift both the overall land footprint of agriculture, and how they change the chemical-use intensity with which

this land is cultivated.[1] Both Aspelund and Russo (2024) and Grupp et al. (2024) study the efficiency of conservation policies in pooling land outside of cultivation, while Coeurdacier et al. (2026) study the role of the CAP in sustaining the land footprint of European agriculture against reforestation and urbanization dynamics. Their analysis complements mine, as I focus on how the CAP shifts production across heterogeneous producers within agriculture. Like them, I take into account the ecological opportunity costs of drawing more land into agriculture, which I weigh alongside both the benefits of reducing the intensity of chemical use, and the economic efficiency costs of government intervention in agriculture.

This paper proceeds as follows: Section 2 describes the data and provides background on the Common Agricultural Policy. In Section 3, I present motivating facts that will guide the design of the empirical model. Section 4 outlines the model and the estimation strategy, and Section 5 presents the equilibrium analysis of different agricultural policies. Finally, Section 6 concludes the analysis.

## 2 Data and Context

### 2.1 Data

**Farm Accountancy Data Network (FADN):** I use the French subset of the European Farm Accountancy Data Network (FADN) for 1980-2020, which I restrict to 1980-2015 for the model estimation, to match other data sources. It is a rich farm-level panel survey designed to be representative of commercial agriculture in the EU.[2] The FADN is an ideal dataset for the analysis due to its detail and panel structure. While the FADN is a sample (approx. 8,000 farms per year in France), it represents the vast majority of French production. In 2000, the commercial farms it represented accounted for 95% of France's total agricultural production.[3] The dataset's richness stems from its foundation in farm accounting records. For each farm, it provides annual data on crop-specific sales, produced quantities, and prices. It also contains detailed information on input expenditures (e.g., fertilizer, pesticides), capital stocks, and crop-specific land allocations. Finally, farms are geocoded at the department level for the period of analysis, enabling me to match them with local input prices and weather data.

**Census of Agriculture:** The French Census of Agriculture provides farm-level data for the universe of French farms. The census is fielded approximately every ten years (for

[1] See for EU land use: https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Farms_and_farmland_in_the_European_Union_-_statistics.

[2] My analysis focuses on France, as many of the datasets I match the FADN to only exist for France.

[3] The FADN's definition of a "commercial farm" has evolved but consistently targets farms that are professionally managed and market-oriented. Prior to 2010, it included requirements on the manager's work hours and a minimum land area of 5ha for non-specialized farms. Since 2010, the threshold is based on a minimum standard production capacity of €25,000.

my purposes 1970, 1979, 1988, 2000, and 2010 waves) and covers virtually every agricultural establishment in France.[4] While the Census lacks the detailed financial data of the FADN, its comprehensive scope makes it ideal for measuring market structure. Its primary variables for my purposes are farm municipality-level location and crop-specific acreage. A key limitation is the absence of a stable farm identifier, which prevents me from tracking individual farms across census waves. I therefore use the repeated cross-sections of the Census to construct precise, municipality-level measures of farm count and crop-mix composition. Other sources of data used in this paper are described in Appendix B.

### 2.2 The Common Agricultural Policy and Environmental Pollution

The Common Agricultural Policy (CAP) was established after World War II to ensure food security, boost production, and support farm income. It quickly became the largest policy administered by the EU, accounting for 45% of its total budget since 1980, and reaching over 50 billion euros in annual spending since 2005. At the same time, the environmental footprint of European agriculture has become a growing concern for policymakers: three reports published by the European Environment Agency in 2025 document for example that agriculture accounts for 12% of all EU greenhouse emissions, a significant fraction of which comes from nitrous oxide emissions from fertilizer use, that agriculture represents 94% of all EU ammonia emissions, and that between 2013 and 2023 at least 19% of monitored water bodies had pesticide levels exceeding safety thresholds (EEA (2025a),EEA (2025b),EEA (2025c)). Levels of pollution in European agriculture are high, and have also been declining at a much slower rate than emissions from other industries. While harmful air pollutants from energy and manufacturing such as fine particulate matter (PM2.5) and sulphur oxides have steadily decreased in the last decades, neither has ammonia in air pollution, nor nitrates in water pollution. At the same time, the first CAP subsidy directly targeting the environment was only introduced in 2007, and most CAP spending is not conditioned on the adoption of sustainable environmental practices.[5] More aggressive efforts to condition CAP payments on environmental outcomes have recently been scrapped. In 2024, the EU Commission for example withdrew the Sustainable Use Regulation that targeted agricultural pesticide use, and would have aimed for a 50% reduction in pesticide use. Similar efforts to introduce a tax on agricultural chemicals have not been fruitful.[6] In a context where CAP spending remains high, but a direct targeting of agricultural environmental pollution is unlikely, it is important to understand how the design of existing subsidies can influence this pollution

[4] Farms are defined as production units with independent management that meet a minimum size of 0.2ha, with exceptions for specific agricultural sub-markets. The original list of establishments filtered with these requirements is built out of a combination of land registries and data from the farmers' family allowances mutual savings societies.

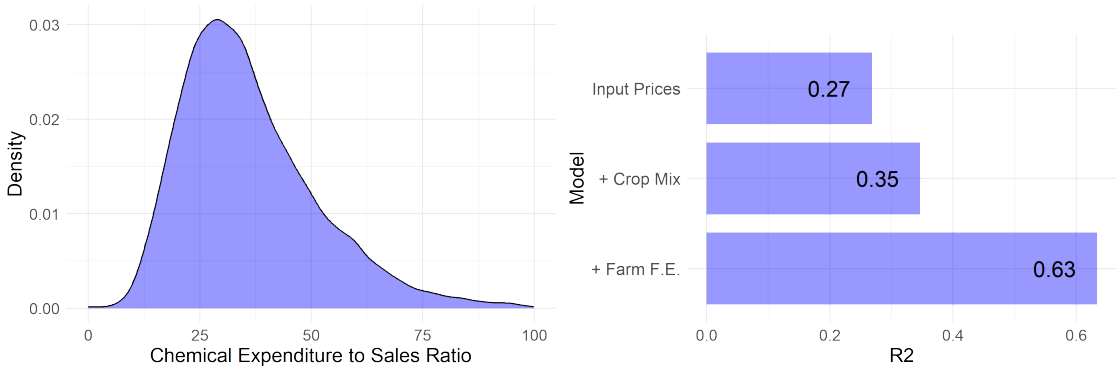
[5] See Figure A1 for a breakdown of spending, and https://agriculture.ec.europa.eu/data-and-analysis/financing/cap-expenditure_en for the underlying data.

[6] See https://www.politico.eu/article/commission-revives-talks-over-eu-wide-tax-on-pesticides/

and whether these can serve as second-best environmental policies.

# 3 Motivating Facts

## 3.1 A Framework to Understand Farm-Level and Aggregate Chemical Use




a) Heterogeneity in Chemical Use b) Decomposing the Heterogeneity

Figure 1: Heterogeneity in Chemical Use Intensity (1980-2020)

*Notes:* These figures rely on 1980-2020 FADN data for grain farms, for which at least 90% of sales can be attributed to grains. I compute their chemical expenditure to revenue ratio using total farm-level chemical expenditure (fertilizers plus pesticides), and revenues are computed using farm-crop sold quantities and French average crop prices. This is done in order to focus the analysis on heterogeneity in chemical use, rather than also heterogeneity in output prices. The distribution of this ratio is plotted in panel a). In panel b), I plot the $R^2$ of three linear regressions where the outcome is the ratio, and explanatory variables are iteratively added on top of each other: 1) an intercept and input prices, 2) adding the share of land allocated to the different grain crops, 3) adding farm fixed effects. These regressions suggest that input prices, crop mix decisions and persistent farm heterogeneity in production practices drive much of the overall heterogeneity in chemical use intensity.

In this paper, I model three distinct mechanisms through which agricultural subsidies impact chemical use: 1) an intensive margin of input substitution, 2) an intensive margin of crop choice, and 3) an extensive margin of market selection. Figure 1 illustrates the overall heterogeneity in chemical use across French farms and decomposes this heterogeneity in a way that relates to these three mechanisms. Relying on grain farms in the FADN, I measure chemical use intensity at the farm level using the ratio of chemical expenditure to sales.[7] To focus the analysis on cross-farm heterogeneity in chemical use rather than output price variation, revenues are computed using quantities sold at the farm level multiplied by average French prices. Over 1980-2020, the standard deviation of this ratio is 15.3%, with an average of 35.7%, as shown in Panel a). Panel b) decomposes this heterogeneity by plotting the $R^2$ of linear regressions where the outcome is the chemical expenditure to revenue ratio.

[7]For the purpose of this exercise, grain farms are farms for which at least 90% of their annual revenue comes from the selling of grains.

Combined, spatial and temporal variation in input prices (land prices and price indices for fertilizers and pesticides) yields an $R^2$ of 0.27, suggesting that input prices are an important driver of chemical-use intensity. Adding the shares of land allocated to different grain crops as additional regressors raises the $R^2$ to 0.35, indicating that crops vary in their chemical intensity and that crop mix decisions influence farm-level chemical use. Finally, the inclusion of farm fixed effects raises the $R^2$ to 0.63. This substantial residual heterogeneity represents persistent differences across farms in chemical-use intensity, beyond differences in input prices and crop mix. This persistent heterogeneity likely captures differences in soil, weather, or management practices that dictate a farm's efficiency and propensity to use chemicals. These three sources of heterogeneity can interact with agricultural subsidies in different ways.

The main EU subsidy for grain agriculture is a land use subsidy, which decreases the effective price of land. If land and chemicals are substitutes in production, this incentivizes farms to use relatively more land and fewer chemicals, decreasing a farm's chemical use intensity. Subsidies further interact with a farm's crop mix. For example, if crops have different land requirements, a homogeneous land subsidy will disproportionately incentivize farms to produce land-intensive crops. Subsidies might also be differentiated across crops, further shifting the profitability of growing different crops. Finally, as subsidies impact market profitability, they shift the threshold at which farms select into being a market incumbent. If persistent farm heterogeneity in chemical use is correlated with farm profitability, this selection effect will alter aggregate chemical-use intensity. If low-profitability farms are those that rely the least on chemicals, subsidies that soften market selection will inadvertently decrease aggregate chemical-use intensity by retaining these more sustainable producers. In the structural model I develop, the correlation between farm profitability and persistent chemical use intensity is not imposed ex ante but rather determined by the estimated parameters.

### 3.2 Learning from the 1992 MacSharry Reform

Next, I rely on the variation in CAP subsidies generated by the 1992 MacSharry reform to illustrate the relevance of the channels discussed in the previous section. Prior to 1992, the CAP supported farmers through purchasing guarantees at price floors called intervention prices (implemented at the level of agricultural intermediary buyers), and export subsidies, which kept French farm-gate prices significantly above world market levels. Driven internally by an EU budgetary crisis and externally by the GATT Uruguay Round negotiations, the 1992 MacSharry Reform fundamentally altered this structure by replacing price support with direct land-based payments (land subsidies), which are still the dominant form of agricultural subsidy in the EU today.

The reform had three main components relevant to farm incentives. First, it drastically reduced intervention prices for cereals and removed them for oilseeds. Following the re-

form, French farm-gate prices converged to lower world market levels, but the magnitude of this decrease varied substantially by crop, creating cross-farm differences in exposure based on a farm's pre-reform crop mix (see Figure A2 for the evolution of prices). Second, the EU introduced compensatory land subsidies.[8] Third, it implemented a mandatory set-aside requirement, forcing larger farms to leave a portion of land fallow to receive subsidies. Effectively, this set-aside requirement reduced the payments per hectare received by farmers through land subsidies. This reform provides an ideal quasi-experiment to highlight how subsidies interact with the determinants of chemical use. The variation in subsidies it generates is of two forms: a decrease in the generosity of subsidies measured in euros per unit of output, and a change in the incidence of subsidies from a differentiated output price subsidy towards an input-biased land subsidy.[9]

I summarize the total government support for each crop $c$ in year $t$ as a 'revenue cushion' ($cushion_{ct}$), measured in euros per unit of output. This cushion is the sum of the pre-reform price support and the post-reform direct subsidies, and writes:

$$\begin{aligned} cushion_{ct} &= \underbrace{\mathbb{1}\{Intervention_{ct}\}\left(P_{ct}^{inter} - \overline{P}_{ct}^{I}\right)}_{\text{Price Support}} + \underbrace{\left(\frac{\overline{Sub}_{ct}}{\overline{Yield}_{ct}} - setaside_{ct}\right)}_{\text{Effective Land Subsidy}} \\ Exp_j &= \sum_{c \in \mathbb{C}_{j,t_0}} \frac{s^l_{j,c,t_0}}{\sum_{c'} s^l_{j,c',t_0}} \left(cushion_{c,t=1991} - cushion_{c,t=1995}\right) \end{aligned} \tag{1}$$

In the expression for $cushion_{ct}$ given in Equation (1), the price support wedge captures the pre-reform system of demand-side government intervention. It is defined as the difference between the EU intervention price ($P_{ct}^{inter}$) and the world market price (proxied by North American farm-gate prices: $\overline{P}_{ct}^{I}$), active only when an intervention price exists.[10] The land subsidy captures the post-reform system. It consists of the new average French land subsidy ($\overline{Sub}_{ct}$), converted to a per-output measure through division by an average French yield ($\overline{Yield}_{ct}$), minus the opportunity cost of the compulsory set-aside requirement

[8] Figure A4 shows their evolution over time in the FADN data.

[9] Figure A3 details the evolution of French grain imports in the years before and after the MacSharry reform. Post-reform, wheat has a small increase in imports, which remains quite insignificant compared to national production levels. Imports reach 400k tons post-reform, for an average of 33 million tons of national production. Colza seed had a very precise spike in 1994, also reaching 400k tons, but immediately falling to almost zero in ensuing years. Sunflower is the only grain crop with a more marked increase between 1996 and 1999 at 294k tons, or 15% of national production. Sunflower, however, represents a small fraction of the overall grain market. These small trade effects justify the focus on a closed economy for the shift-share analysis, and the model section will consider an extension with France producing for and exporting to the world grain market.

[10] The U.S. and Canada accounted for most of the export market over the 1980s to 2000s, making these prices a relevant reference point for the analysis. I use USA prices except for colza, for which I use Canadian prices—given that the USA was not a significant producer of colza at that time, but Canada was. All these prices are average farm-gate prices from FAOSTAT.

($setaside_{ct}$).[11][12] Figure A5 plots the evolution of these revenue cushions. The cushions for all major crops decline sharply between 1991 (the final pre-reform year) and 1995 (the final reform implementation year), tracking a significant loss in subsidies for French farms. Critically, the magnitude of this decline varies substantially across crops, with oil crops and durum wheat experiencing a much larger negative shock than the remaining grains. I then measure each farm's exposure to the reform with variable $Exp_j$ also shown in Equation (1), using a weighted average of these crop-specific shocks, where the weights are the farm's pre-reform land allocations.[13] In this expression, $s^l_{j,c,t_0}$ stands for the share of farm j's land allocated to crop c in the pre-reform period, $t_0 = 1991$. These cropping decisions were made in the fall of 1990, before the reform's design was finalized. The farm-specific reform exposure variable $Exp_j$ is then the total change in this revenue cushion over the reform period. This variable captures how hard each farm was hit by the reform, based entirely on its pre-reform state.[14]

**Empirical Design & Identifying Assumptions:** The facts documented in this section rely on three sources of data: 1) the farm level FADN panel dataset where I can observe farm-gate prices, profit and chemical use, 2) the Census of Agriculture which I use to track municipality level farm counts and land use, 3) remote sensing data paired with the Census aggregated at the county level where I can track algal blooms, a form of agricultural chemical pollution, on surface water. Together, these datasets allow me to show how subsidies can impact farm profit and market participation, local crop mix compositions and finally chemical use and environmental pollution. To understand these effects, I make use of the following regression equation:

$$Y_{jt} = \alpha_j + \eta_{d(j)t} + \sum_{\tau \neq t_0} \theta_\tau Exp_j \mathbb{1}\{t = \tau\} + \sum_{\tau \neq t_0} \delta_\tau X_j \mathbb{1}\{t = \tau\} + \varepsilon_{jt}. \tag{2}$$

When running regressions at the farm level, $j$ stands for a farm, it otherwise stands for a municipality or a county, when regressions rely on Census and remote sensing data. At these two aggregated levels, $Exp_j$ denotes the average farm exposure within resp. the municipality or the county. In all regressions, $d(j)$ is the department in which $j$ is located, an adminis-

[11] Subsidy data in the FADN at times feature large positive outliers, yielding a decreasing average subsidy per hectare that does not match actual EU policy. To address this issue, the median land subsidy in the FADN tracks EU policy better, and I use it instead of the average in the calculation of the $cushion_{ct}$ variable. In the restricted sample I later use to estimate my production function, I winsorize land subsidies at the top 2.5% centile in order to address this mis-measurement.

[12] The per unit cost of the set-aside is recovered as 10% of the amount of land required to produce one ton of the given crop, times the price of land to which is removed the crop-specific land subsidy. This subsidy is removed from the price of land, as farms would still receive land subsidies for set-aside areas.

[13] I include the following crops to compute the farm-level cushion: wheat, barley (winter), corn, rye, oats, sunflower, colza, barley (spring), durum, sorghum. In the appendix, I test another construction of $Exp_j$ where land shares are used to aggregate over the land subsidy, and output weights over the price subsidy, and show that results are similar.

[14] The distribution of $Exp_j$ in the FADN sample is shown in Figure A8.

trative unit roughly the size of a US county.[15] $\alpha_j$ is a fixed effect for the unit of observation, either a farm, municipality or county, $\eta_{d(j)t}$ a department-by-year fixed effect, and $X_j$ a set of controls set to pre-reform levels, which are allowed to have heterogeneous effects over time. The coefficients of interest $\{\theta_t\}_t$ capture the year-specific effect of the reform relative to $t_0$, which is set to 1991 for the FADN and the remote sensing-based regressions, and to 1988 for others relying on the Census. In both cases $t_0$ is the last observed year pre-reform. As the whole sample is treated, the $\{\theta_t\}_t$ will capture the effect of additional reform exposure, and miss the effect of average exposure to the reform. Standard errors are clustered at the department level. FADN regressions use sampling weights, and Census regressions weight municipalities by their 1988 share of overall French agricultural land.

The causal interpretation of $\{\theta_t\}_{t>t_0}$ requires that pre-reform crop shares are uncorrelated with unobserved factors that drive differential growth in the outcome variables post-reform, conditional on the controls. The pre-$t_0$ coefficients $\{\theta_t\}_{t<t_0}$ provide a direct test of this assumption. The inclusion of fixed effects and controls builds towards this conditional exogeneity. Unit-level fixed effects $\alpha_j$ account for the time-constant influence of any time-constant unobserved unit-level confounder, for example unobserved soil quality driving certain crop choices and profit rates. A second threat comes from spatially correlated shocks, and shocks correlated over time: local climate conditions can for example create strong spatial cropping patterns. Local areas are also likely subject to unobserved, time-varying shocks (e.g., evolving local demand or input prices), that could induce a spurious correlation between exposure and the outcome. The inclusion of department-by-year fixed effects $\eta_{d(j)t}$ controls for any such shocks that are common to all farms within a department in a given year. Finally, I include a rich set of observables set to pre-reform levels in $X_j$, relying on the respective set of variables available in the FADN and the Census. These are meant to capture the state of farms and municipalities pre-reform, to the extent that for dynamic farms, initial pre-reform conditions can influence growth, chemical use and exit in the following years.[16] Results are shown in Figure 2.

The first set of results relates to how the MacSharry-driven decrease in subsidies reduced profits and led to farm exit. The first two panels in Figure 2 rely on FADN data, and show the impact of a one standard deviation increase in reform exposure (a loss of €103 in subsidies

[15] Metropolitan France on which I focus has over 36,000 municipalities, 2,000 counties and 96 departments.

[16] For farm-level regressions, $X_j$ contains the following elements: farm 1991 capital stock, total labor used, total land use, total profit, their chemical use, the number of crops they grew and the evenness of their land allocation across these crops, their fertilizer-to-land and pesticides-to-land ratios, the share of their production which corresponds to oil crops (colza and sunflower), and the farms' 1983-1984 (the start of my period of analysis) adoption trends in chemicals measured as the evolution in their chemical use. For regressions relying on Census and remote sensing data, $X_j$ includes moments of the farm size distribution (mean, minimum, maximum and standard deviation), the total municipality-level agricultural area and cultivated area, the minimum cultivated area across farms within the municipality, the average crop count across farms, and the crop count at the municipality level, the average crop evenness across farms, and crop evenness at the municipality level, the fraction of farms in the municipality growing grains in 1988, and the fraction of municipality land allocated to cereals (versus oil crops) in 1988.

per ton of grain) on farm-gate prices and farm profit.[17] The pre-reform coefficients $\{\theta_t\}_{t<1992}$ in these regressions are statistically indistinguishable from zero. After 1992, the coefficients become negative and statistically significant. In panel a), a one-standard-deviation increase in exposure leads to an average post-reform 24% larger decline in a farm's average output price relative to the 1991 baseline—farms that are more exposed to this reform which reduced output prices hence experience a larger decrease in output prices. Panel a) can hence be understood as a first stage for the analysis.[18] Panel b) then shows these translated into long-term decreases in profit, with a one-standard-deviation increase in exposure leading to a 31% decrease in profits post-reform. In the last step, I turn to the Census of Agriculture, and find that these reduced profits at the farm level were accompanied by a reduction of the farm population. In panel c) of Figure 2, I find that by 2000, a one-standard-deviation increase in a municipality's 1988 reform exposure led to a net decrease of 0.7 farms per municipality, growing to a 1.1 decrease by 2010.

In a second step, I build a measure of crop evenness at the municipality level, and use it to study how changes in subsidies impacted the allocation of land across crops.[19] The event study in panel d) of Figure 2 shows that a one standard deviation increase in municipality reform exposure expanded land use evenness by 0.016 in 2010 (3.6% of the average). Here the effect combines intensive within-farm changes in crop-mix composition, reallocations of land across farms and potentially the exit of smaller more specialized farms.[20]

I finally study the effect of subsidies on the use of chemicals and visible environmental pollution. At the farm level, panel e) of Figure 2 shows that a one standard deviation increase in reform exposure was accompanied by a 24% decrease in chemical expenditure. This result could be the joint effect of: 1) introducing a land subsidy increasing the relative price of chemicals, 2) changes in crop mix composition, 3) changes in production scale. Finally, I connect the reform directly to water pollution. Fertilizer pollution is a primary cause of algal blooms in waterways. I measure algal blooms with a county-year index, relying on Landsat 5 satellite data, and following the methodology of Taylor and Heal (2023). The method is further detailed in Appendix B.5. I aggregate the analysis at the county level to match the coarser geographic precision of Landsat 5. Results in panel f) of Figure 2 show that by 2001, a one-standard-deviation increase in reform exposure leads to a 10% decrease

[17] I measure the farms' output price as the weighted average of its crop-specific prices, using relative land allocations as shares. I use the following crops to compute the average (those for which I observe a balanced time series): wheat, winter barley, corn, rye, oats, sunflower, colza, spring barley, durum.

[18] This estimated effect is consistent with the aggregate average 1991-1995 33% price decrease observed in the FADN data.

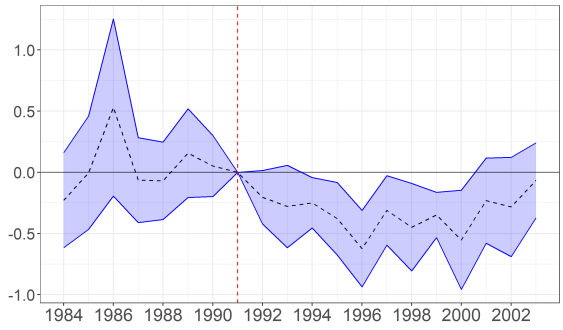
[19] Crop evenness is measured using the Pielou diversity index frequently used in ecology to measure species diversity. It corresponds to the observed Shannon diversity index normalized by the maximum Shannon index for that same community. Here a value of 1 indicates a maximum evenness in the allocation of land across crops.

[20] I show results for additional outcomes in Figure A11. I cannot identify any significant movement in the municipality farm-size distribution. A potential interpretation is that farm exit was relatively more driven by pre-reform crop mix composition, than by farm size, and that small farms were sufficiently heterogeneous in their crop mixes.

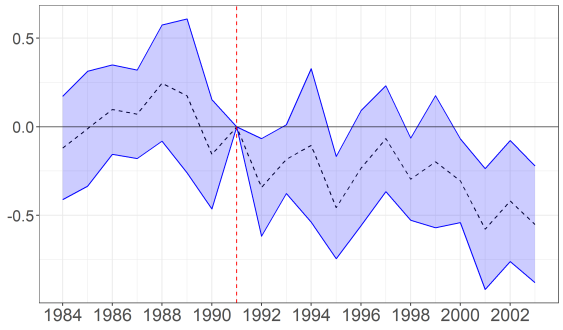
in the algal bloom index relative to its 1991 level. This result provides evidence that the reform had a positive impact on environmental quality.

This analysis illustrates the effects of subsidies on farm behavior and chemical use, and highlights how subsidies interact with the different drivers of chemical use from Section 3.1: following the reform I observe farm exit potentially associated with a change in market selection, I further observe changes in municipality-level crop mixes and within-farm chemical use. This reform finally lowered water pollution as observed in satellite data. In order to obtain this observed decrease in pollution, in the case where smaller farms pollute less than large farms, the intensive margin shifts must dominate extensive margin effects. Specifically the land subsidy raising the relative price of chemicals, the changes in crop mix and the overall decrease in production scale must dominate the extensive-margin selection effect of farm exit. When running counterfactuals, I later find that selection effects on aggregate chemical-use tend to be lower in magnitude than intensive effects of input use. In Appendix G, I use this model to simulate the effects of a MacSharry-like change in subsidy design, holding government expenditure constant. In this case, the reform generates a reduction in overall chemical-use intensity and intensive-margin effects dominate.

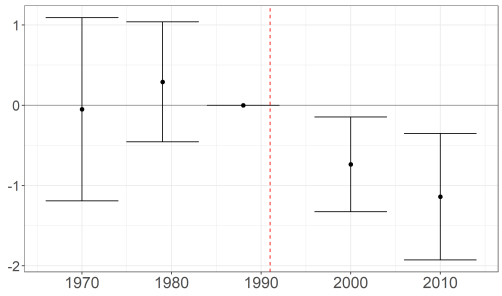
**Robustness & Remaining Questions:** My analysis relies on a unique policy shock which treats the entire grain market at the same time. And when decomposing the variation of the exposure variable into crop-by-crop sources following the Rotemberg weights of Goldsmith-Pinkham et al. (2020), I find that variation in $Exp_j$ within the FADN is dominated by shocks to a few crops. This justifies my identification approach which relies on the conditional exogeneity of exposure shares, rather than a law of large numbers in shocks and endogenous shares (Borusyak et al. (2022)). Further robustness analysis relying on these Rotemberg weights, as well as results using a difference-in-difference design, and relying on an alternative exposure instrument which combines 1991 land and output shares, respectively to aggregate over pre-reform output subsidies and post-reform land use subsidies are shown in Appendix C. The MacSharry reform was jointly a decrease in subsidy levels, and a change in subsidy design. This composite nature of the reform is useful to the extent that it produces enough variation to highlight the different ways through which subsidies impact chemical use. However, because these channels interact, the analysis falls short of being able to isolate the respective contributions of these channels. Further, because the MacSharry reform impacted all French farms, the design relies on variation across farms in exposure intensity, missing the average effect of the reform, and as such much of its potential general equilibrium effect. The empirical framework developed in the following section will capture general equilibrium effects, isolate and quantify the individual contributions of the competing channels, and allow me to study optimal subsidy designs.



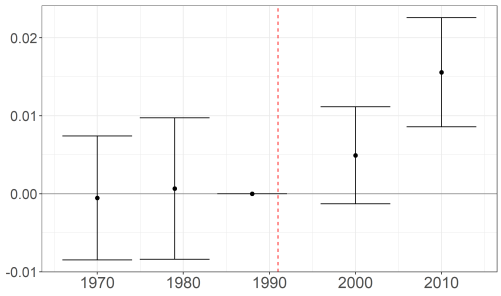

a) Within Farm Average Prices



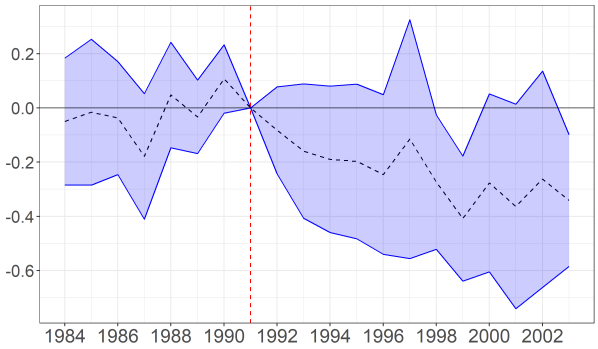

b) Within-Farm Profit



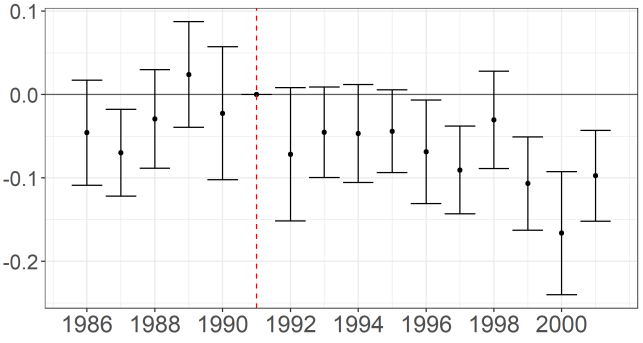

c) Within Municipality Farm Count




d) Within Municipality Overall Crop Evenness




e) Within-Farm Chemical Use




f) Within County Algal Blooms

Figure 2: Effect of MacSharry Reform on Farm Count, Crop Mixes, Chemical Use and Pollution

*Notes:* Regressions in panel a) and b) and e) are at the farm-level using the FADN, regressions in panel c) and d) are at the municipality level using the Census of Agriculture, regression in panel f) is at the county level using the Census combined with Landsat 5 remote sensing data. Outcomes are: log average farm-gate price (a), log farm profit (b), municipality farm count (c), municipality evenness of land use across crops (d), farm-level log deflated chemical expenditure (e), county-level log algal bloom index (f). All regressions have department-year and unit of observation (farm, municipality, county) fixed effects. Standard errors are clustered at the department level. Controls include unit of observation characteristics set to pre-reform levels. Result tables are Table A3, Table A4 and Table A5.

# 4 Empirical Model

## 4.1 Static Demand

I model the respective demands for grain crops $c \in \mathbb{C}$. Farms are price takers, and face a downward-sloping iso-elastic aggregate demand curve for each crop, where $Q_{ct}^D$ is the total quantity demanded and $P_{ct}$ is the market price:

$$Q_{ct}^D = \exp(\gamma_c + e_{ct}) P_{ct}^{\beta_c} \ , \ \forall c \in \mathbb{C}. \tag{3}$$

The parameter $\beta_c < 0$ is the constant price elasticity of demand for c. The term $exp(\gamma_c + e_{ct})$ represents a demand shifter, composed of a crop-specific mean intercept $\gamma_c$ and an idiosyncratic, mean-zero demand shock $e_{ct}$. While the supply-side focus of the paper justifies a simple demand structure, Appendix D.8 discusses an alternative model that incorporates monopolistic competition through horizontal differentiation across grain crops, following the framework of Mayer et al. (2014).[21]

## 4.2 Supply

**Farm State:** Farm $j$ is characterized by an individual state vector $\Upsilon_{jt} = \{\omega_{jt}^h, \omega_{jt}^{ch}, K_{jt}, \mathbb{C}_{jt}, \sigma_j\}$. This state captures farm heterogeneity along two dimensions of productivity, both of which are exogenous to the farm's decisions and follow a persistent stochastic process. The first is $\omega_{jt}^h$, a farm-level TFP shock, and the second is $\omega_{jt}^{ch}$, a shock to the farm's efficiency at using chemicals. Farms are further characterized by their capital stock $K_{jt}$. Farms are also multi-product producers. Farm $j$'s crop mix at $t$ is denoted $\mathbb{C}_{jt} \subseteq \mathbb{C}$. The last element of $\Upsilon_{jt}$ is a time-invariant 'competence ladder' $\sigma_j$. This ladder is a permutation of $\mathbb{C}_{jt}$ and represents a farm-specific ranking of crop-specific production competence, where the first crop on the ladder is produced with greatest efficiency, and the last one with least efficiency.[22] This ladder, drawn upon entry, interacts with the farm-level TFP shock $\omega_{jt}^h$ to determine a crop-specific TFP. The ladder, whose interaction with farm TFP is parametrized below, is a parsimonious way to introduce within-farm heterogeneity in TFP across crops, where the gap in production efficiency between two crops is kept constant over time.

**Market State & Timing:** Farms are single-agent producers in the sense that they take the aggregate market state as given. $\Omega_{jt} = \{\mu_t, N_t^e, \{P_{jct}\}_{\mathbb{C}}, \{P_{jt}^x\}_x, Policy_t\}$ is the market state

[21] This addition allows for price dispersion across producers growing the same crop: high efficiency producers selling a large quantity at a low price, and low efficiency ones selling a small quantity at a high price—in this case, the gains from a low effective cost of chemicals are larger than in my perfectly competitive model, which hence offers a lower bound for the potential consequences of farm selection.

[22] For example, if wheat is the crop produced with the highest TFP within farm $j$, then the ladder encodes a rank 1 to wheat, and writes $\sigma(1) =$ wheat.

faced by $j$ at $t$, where $\mu_t$ is the measure describing the mass of incumbents at each point of the farm state space, $N_t^e$ the mass of entrants, and then vectors of output ($\{P_{jct}\}_{\mathbb{C}}$) and input prices ($\{P_{jt}^x\}_x$), as well as the state of agricultural subsidies $Policy_t$. For an incumbent farm, the timing within period t is as follows. First, farms observe their shocks $\{\omega_{jt}^h, \omega_{jt}^{ch}\}$, and the market state $\Omega_{jt}$. Second, they make their static production decisions, i.e. choosing variable inputs to produce quantities $Q_{jct}$ for all crops in their mix, and profits are realized. Third, they make dynamic decisions: they decide whether to stay on the market ($\xi_{jt}^x = 0$), or exit ($\xi_{jt}^x = 1$). If a farm remains on the market, it pays a fixed cost of incumbency $f_{inc}$, representing annual fixed costs of operation, and chooses its capital stock and crop mix for the next period. Short-lived potential entrants face an initial entry period before they can start producing. Observing the market state $\Omega_{jt}$, potential entrants decide whether to pay a fixed entry cost $f_e$ to enter the market ($\xi_{jt}^e = 1$). If the entry cost is paid, a farm realizes its initial shock vector ($\{\omega_{jt}^h, \omega_{jt}^{ch}\}$) and a competence ladder ($\sigma_j$). After observing its type, a farm can confirm its entry decision, and make initial investment and crop mix decisions, or definitively leave the market. The Bellman equations for the incumbent and entrant problems, as well as the law of motion for the distribution of firms are detailed in Appendix D.1. Diagrams outlining timing assumptions are also shown.

**Production & Profits:** The production function for quantity $Q_{jct}$ of crop $c$ for farm $j$ at time $t$ is Cobb-Douglas in its main variables:

$$\begin{aligned} Q_{jct} &= e^{\omega_{jct}^h} K_{jt}^{\alpha_{k,c}} L_{jct}^{\alpha_{l,c}} X_{jct}^{\alpha_{x,c}} e^{\varepsilon_{jct}}, \\ X_{jct} &= \left\{\delta_{s,c} S_{jct}^{\rho} + (1-\delta_{s,c})\left(e^{\omega_{jt}^{ch}} Chemicals_{jct}\right)^{\rho}\right\}^{\frac{1}{\rho}}, \end{aligned} \tag{4}$$

with $L_{jct}$ as labor, $X_{jct}$ a composite input detailed below. $\alpha_{k,c} + \alpha_{l,c} + \alpha_{x,c} = 1$, such that there are decreasing returns to scale for the flexible inputs. $\varepsilon_{jct}$ is a mean-zero, i.i.d. shock realized after input decisions are made.[23] The TFP for crop c is given by $\omega_{jct}^h = \omega_{jt}^h \lambda^{\sum_{n=0}^{dim(\mathbb{C})-1} n\mathbb{1}\{\sigma_j(n)=c\}}$, with $n$ the rank of crop c on farm $j$'s ladder, and $\lambda \in (0,1)$ governs the strength of specialization.[24] The key mechanisms of substitution and heterogeneous chemical use efficiency are embedded in the composite input $X_{jct}$, which combines land and chemicals. This composite input is modeled as a constant elasticity of substitution (CES) function. The CES has constant returns to scale, and combines land ($S_{jct}$) with chemicals ($Chemicals_{jct}$), with an elasticity of substitution $\sigma = \frac{1}{1-\rho}$. Farms can be expected to face differences in soils, in local climates, and in managing styles, that are auto-correlated over time. These together imply that chemicals might be more efficiently used by some farms

[23] Because $\varepsilon_{jct}$ is realized ex-post, it only enters in expectation in the profit maximization problem. Because it is assumed to be mean zero, it will disappear from first order conditions and optimal expressions below.

[24] This structure allows for differentiated TFP shocks to be recovered from production data where input-to-output allocations are imperfectly observed, as I detail in the next section.

than others, for example because soils will absorb fertilizers more easily, or because farmers are better at timing pesticide application. The CES combined with $\omega_{jt}^{ch}$ allows me to model this phenomenon. An increase in the value of $\omega_{jt}^{ch}$ means a farm will be more efficient in its use of chemicals, and achieve a higher output volume with the same input levels. I finally assume that profit maximization occurs independently for each crop in a farm's portfolio, such that flexible input levels are decided independently for each crop, in order to maximize crop-specific profit. Here, all flexible inputs are chosen for, and fully allocated to a specific crop's production line. Capital ($K_{jt}$) is treated as a shared input, fully available to all crops grown by the farm without rivalry or cost.[25] This framework deliberately simplifies away certain economies of scope to focus on the primary mechanisms of input substitution and productivity.[26]

**Dynamics:** The final components of the model govern the farm's dynamic decisions. To capture observed frictions in agricultural investment, I introduce convex costs of capital adjustment ($C_k^Q$) alongside partial irreversibilities ($\tau_k$). Convex costs reflect the reality that large investments disrupt production more heavily than small ones. Partial irreversibilities account for the specialized nature of agricultural capital, which can force farms to accept discounts on the secondary market. The cost of adjusting capital from $K_{jt}$ to $K_{jt+1}$ via investment $i_{jt}$, is given by:

$$C(i_{jt}, K_{jt}) = P_{jt}^K i_{jt} + C_k^Q K_{jt} \left( \frac{i_{jt}}{K_{jt}} \right)^2 - \tau_k P_{jt}^K i_{jt} \mathbb{1}\{i_{jt} < 0\}. \tag{5}$$

Finally, farms face a fixed cost of entry $f_e$, and a fixed cost of incumbency $f_{inc}$ paid every period and capturing general administrative and maintenance costs. Here, a farm decides to include in its crop mix $\mathbb{C}_{jt+1}$ every crop for which it expects to make a positive profit in period $t+1$, given its expectations at the end of period $t$. I model a unique fixed cost of incumbency, which, because of the decreasing returns to scale in flexible inputs of $Q_{jct}$, implies that farms will always include all crops in their crop mix. However, farms still differ in their crop mixes, in the sense that their total land use $\sum_{c \in \mathbb{C}_{jt}} S_{jct}$ is heterogeneously distributed across crops. This approach approximates the discrete crop-mix decisions that would be obtained by including differentiated fixed costs across crops, but reduces the set of parameters to estimate.

[25] The capital law of motion is $K_{jt+1} = (1 - \delta_k)K_{jt} + i_{jt}$, where $\delta_k$ is the rate of capital depreciation, and $i_{jt}$ is investment. A rationalization for shared capital is that harvest seasons are not perfectly aligned for all crops, and that machinery and buildings can be used to harvest and store these different grains without rivalry.

[26] The assumption that flexible inputs are fully attributable to production lines is common in the literature (De Loecker et al. (2016); Orr (2022)). The assumption that crop-specific profits can be maximized independently captures the assumption of flexible input allocations, of exogenous output prices and of limited economies of scope. It is also necessary for input prices to not depend on the quantity purchased, and that productivity shocks are exogenous to the size and composition of the crop mix.

**Equilibrium:** To make the full dynamic equilibrium model computationally tractable, I introduce a few simplifying assumptions at this stage. These assumptions are used to solve for the equilibrium, both in the estimation of dynamic parameters and in counterfactuals, but are not imposed during the estimation of the production function itself, which does not require a simulation of the market equilibrium. First, I abstract from farm-level heterogeneity in crop specialization ability by assuming a homogeneous competence ladder ($\sigma_j = \sigma, \forall j$). Second, input and output prices are assumed to be common across all farms in a given period. As such $\Omega_t$ is common to all farms. I finally focus on stationary equilibria where the aggregate market state is constant ($\Omega^\star$). The equilibrium is such that, given input prices and the state of agricultural policy, the zero ex ante profit condition holds, $\mu^\star$ is constant across periods, and crop markets clear. Appendix D.1 provides the formal definition.[27]

**How Subsidies Impact Input Use & Crop Mix:** To understand how subsidies impact within-farm input use, I express the optimal farm-crop chemical-to-output ratio as:

$$\frac{Chemicals^\star_{jct}}{Q^\star_{jct}} = \frac{P_{jct}}{P^{ch}_{jt}} \frac{\alpha_{x,c}\big(1-\delta_{s,c}\big)}{\delta_{s,c}\left[\frac{P^s_{jt}(1-\delta_{s,c})e^{\omega^{ch}_{jt}}}{P^{ch}_{jt}\delta_{s,c}}\right]^{\frac{\rho}{\rho-1}} + \big(1-\delta_{s,c}\big)}.$$

A land subsidy directly impacts $P^s_{jt}$ in the denominator. A larger subsidy will impact the chemical-to-output ratio in a way that depends on the parameter of substitution $\rho$. If land and chemicals are substitutes, or $\rho > 0$, a larger land subsidy will reduce this ratio. On the contrary, a larger subsidy will increase this ratio if the inputs are complements. For $\rho \approx 0$, the input nest would reduce to a Cobb-Douglas function where the ratio would not depend on land prices. Next, I can use the log ratio of crop-level land use within a farm to gain some intuition about how subsidies can impact crop mix decisions. The log ratio of the land allocated to crop $c$ relative to the one allocated to crop $c'$, within farm $j$, writes as:

$$\begin{aligned} log\left(\frac{S^\star_{jct}}{S^\star_{jc't}}\right) = &+ \left[\frac{1}{1-\alpha_c}\log P_{jct} - \frac{1}{1-\alpha'_c}\log P_{jc't}\right] + \left[\frac{\alpha_{l,c}}{1-\alpha_c}\log(\frac{\alpha_{l,c}}{P^l_{jt}}) - \frac{\alpha_{l,c'}}{1-\alpha_{c'}}\log(\frac{\alpha_{l,c'}}{P^l_{jt}})\right] \\ &+ \left[\big(\sigma - 1 - \frac{\alpha_{x,c}}{1-\alpha_c}\big)\log(P^X_{jct}) - \big(\sigma - 1 - \frac{\alpha_{x,c'}}{1-\alpha_{c'}}\big)\log(P^X_{jc't})\right] + \left[\frac{1}{1-\alpha_c}\omega^h_{jct} - \frac{1}{1-\alpha_{c'}}\omega^h_{jc't}\right] \\ &+ \left[\frac{\alpha_{k,c}}{1-\alpha_c} - \frac{\alpha_{k,c'}}{1-\alpha_{c'}}\right]\log(K_{jt}) + A_{cc'}, \end{aligned}$$

with $\alpha_c = \alpha_{x,c} + \alpha_{l,c}$ for the returns to scale to flexible inputs, $P^X_{jct}$ the effective price index of $X_{jct}$ which is a function of $\omega^{ch}_{jt}$, $A_{cc'}$ a constant term equal across farms that accounts

[27] Holding input prices fixed makes sense for the French context where: 1) land prices (both rental prices and sales prices) are heavily regulated by local boards (SAFER), 2) grain agriculture represents a small fraction of labor demand, and hence should have minimum impact on wages. On the chemicals side, fertilizer prices mostly respond to oil prices.

for the respective input elasticities of the crops.[28] Here the first three terms reflect that variations in input and output prices across farms will shift their relative crop mix. The third and fourth terms express how $\omega_{jt}^{ch}$ and $\{\omega_{jct}^{h}\}_c$ also impact this mix: $\omega_{jt}^{ch}$ because different crops rely on chemicals and land to a different extent, and as such are differentially impacted by an increase in chemical-use efficiency, the second because different crops can have different returns to scale in their flexible inputs. Finally the first term in the last line highlights the role of capital: crops can differ in their capital elasticities, implying that more heavily capitalized farms will grow more of the capital intensive crops. Overall, this expression implies that heterogeneous prices, TFP, chemical-use efficiency and capital stocks all will make crop mixes vary across farms. Here, both input and output subsidies, even if homogeneous across crops, influence a farm's crop mix composition, simply because crops differ in their input elasticities.

**Profitability and the Selection Channel:** Following the literature on industry dynamics (Foster et al. (2008)), farm entry and exit are determined by profitability. It is then useful to consider the expression for the farm-crop level optimal profit:

$$\pi_{jct}^{\star} = (1-\alpha_c) P_{jct}^{\frac{1}{1-\alpha_c}} \left[ e^{\omega_{jct}^{h}} K^{\alpha_{k,c}} \left( \frac{\alpha_{l,c}}{P_{jt}^{l}} \right)^{\alpha_{l,c}} \left( \frac{\alpha_{x,c}}{P_{jct}^{X}} \right)^{\alpha_{x,c}} \right]^{\frac{1}{1-\alpha_c}}.$$

Crop-level profit is always increasing in $\omega_{jt}^{ch}$, regardless of the value of $\rho$ (see Appendix D.2). This is because the price index of the composite input $P_{jct}^{X}$ is decreasing in $\omega_{jt}^{ch}$, which in turn raises profits. As the farm-level profit is simply the sum of crop-level profits, total farm profit is also increasing in $\omega_{jt}^{ch}$. The expression given for the chemical-to-output ratio above, however, shows that this ratio only increases with $\omega_{jt}^{ch}$ if land and chemicals are substitutes, while it decreases with $\omega_{jt}^{ch}$ if they are complements. Hence, at the crop-level, farms that are more profitable will use more chemicals per unit of output if and only if chemicals and land are substitutes. The aggregation of this correlation at the farm level will be complicated by the fact that variations in crop mixes will shift the overall chemical-to-output ratio of the farm, depending on the crops' relative chemical intensities. In the case where all crops share the same chemical intensity, $\rho$ again determines the correlation between a farm's total profit and its chemical use intensity. This is how the third channel of selection will relate subsidies to chemical use.

**Extensions & Robustness:** Expressions relevant to the farm's static profit maximization are in Appendix D.2. I also extend the production function in several ways. First, in Appendix D.3, I model pesticides and fertilizers as separate inputs bound in a second CES nest. This second nest makes the estimation more non-linear and yields noisier coefficients,

[28] The full expression for the log-ratio, as well as the derivation and expression for $P_{jct}^{X}$ are given in Appendix D.2.

yet these remain close to the ones estimated here, and importantly characterize the two chemicals as complements. This implies that their quantities will change in lock-step, and that the simplified modeling used here is without loss of generality. Second, in Appendix D.4, I replace the Cobb-Douglas relation between capital, labor and the composite input with a CES. This section shows that the conditions for an increase in $\omega_{jct}^{ch}$ to raise the chemical-to-output ratio are more restrictive in the Cobb-Douglas case. Under the CES case, a slight complementarity between chemicals and land, paired with substitutability between labor, capital and the land nest can still generate a positive relation between $\omega_{jct}^{ch}$ and the chemical-to-output ratio. I also discuss the conditions under which the overall relation between farm profit and chemical-use-intensity are maintained in a CES case. In the CES case, chemical-use intensity is also impacted by a farm's TFP. Estimating this model, point estimates imply that a higher farm TFP further increases both farm profit and its chemical-use intensity. In that sense, the correlation between profit and chemical-use intensity obtained in the Cobb Douglas case is a lower bound for the more general CES case. I keep the Cobb-Douglas case as my main parametrization, first because the CES estimation cannot reject a Cobb-Douglas elasticity between labor, capital and the land-chemical nest, and second because it yields a profit-chemical-intensity correlation that is more readily interpretable, as the result of the elasticity between clean and polluting inputs in production. Finally, I estimate an alternative joint production function system where inputs are shared publicly across products with a penalty in Appendix D.5. This model allows for more general economies of scope, but also leads both to lumpier input decisions and to non-linear estimation steps.

### 4.3 Estimation & Results

For both estimation and counterfactuals, I aggregate crops into two categories: $\mathbb{C} = \{wheat, other\}$.[29] This aggregation is necessary to recover the production function using only data from single product farms. The estimation is done using the 1980-2015 FADN.[30]

#### 4.3.1 Demand

The challenge in estimating demand is that market prices are endogenous, as they are affected by unobserved demand shocks. To address this endogeneity, I employ an instrumental variable strategy, using weather shocks as instruments that shift crop supply but are plausibly uncorrelated with demand. This approach has a long precedent in the estimation of agricultural commodity demand (Roberts and Schlenker (2013)). In my preferred speci-

[29] List of crops in *other* category: sunflower, colza, oats, barley, rye, triticale, corn, sorghum and durum. Finer or different aggregations dramatically reduce the number of single-crop farms available for identification. This aggregation is also a reasonable representation of the French grain market. As shown in (Table A2), wheat is the dominant crop, and it is the one most frequently grown in monoculture. Even among the two-thirds of farms that are multi-crop, approximately 90% include wheat in their portfolio.

[30] Prior to 1980, I do not observe output price data in the FADN, and my hourly wage data stops in 2015.

fication, the instrument set includes national yearly growing and heat degree day (GDD and HDD) realizations, the construction of which is detailed in Appendix B.4. I also include the national average of department-level deviations of yearly GDD and HDD from their thirty-year historical averages. These deviations isolate local weather shocks, which are unlikely to be correlated with EU-wide demand shocks. The identifying assumption is that these weather variables are orthogonal to aggregate demand shocks. All specifications use national output and price data aggregated from the FADN. They include a linear time trend to capture smooth changes in demand, and I compute Newey-West standard errors to account for autocorrelation in the error terms. Resulting demand elasticity estimates are in Table A6. For the structural model, the crop-specific mean intercepts ($\gamma_c$) are calibrated to their 2015 values by adding the estimated time trend to the regression intercept. This ensures the simulated market aligns with the contemporary scale of the French agricultural sector.

As a robustness check, I estimate an alternative specification that excludes GDD and HDD in levels, relying exclusively on weather deviations and their interactions with levels, alongside the yearly EU land subsidy as an additional supply shifter. This instrument set is more robust to potential exclusion restriction violations arising from spatial weather correlations across the EU. While the demand elasticity estimates remain comparable to the baseline, the first-stage F-statistics indicate potential weak instrument concerns. Consequently, I rely on the estimates from the baseline specification for the remainder of the paper.

#### 4.3.2 Production Function

The estimation strategy for the production function is designed to address the two main challenges caused by the model specification and the structure of the data: 1) the unobserved productivity shocks that are correlated with input choices, 2) the allocation of inputs across product lines for multiproduct farms. To address the first point, the parameters of the production function are estimated in a multi-step process, and the estimating equations are derived from the first-order conditions of the farm's static profit-maximization problem as in Doraszelski and Jaumandreu (2018).[31] The aggregation of crops into two categories along with a restriction of the sample addresses the second (De Loecker et al. (2016)).

**Parametric Expressions of Productivity Shocks:** I use first order conditions (FOCs) from profit maximization to express $\omega_{jt}^{ch}$ and $\omega_{jct}^{h}$ as functions of observed input levels, parameters and shocks. First, I combine the farm's FOCs for land and chemical use to derive an

[31] Because variable input choices have no dynamic implications in the model, the static first-order conditions are equivalent to those from the full dynamic problem. Independence of crop profits, and the fact that all flexible inputs are fully allocated to product lines finally means that I will be looking at the first order conditions for crop-specific static profit maximization. As frequently done in the production function literature, I assume interior solutions and therefore restrict the estimation sample to observations with non-zero flexible input use (De Loecker et al. (2016)).

expression that inverts for the unobserved productivity shock:

$$\rho\omega_{jt}^{ch} = (1-\rho)\log\left(\frac{Chemicals_{jct}}{S_{jct}}\right) - \log\left(\frac{P_{jt}^{s}}{P_{jt}^{ch}}\right) - \log\left(\frac{1-\delta_{s,c}}{\delta_{s,c}}\right) \tag{6}$$

This equation expresses the unobserved shock $\omega_{jt}^{ch}$ as a function of observed data and the structural parameters. It also provides some intuition into what drives a farm's chemical-to-land ratio: relative input prices, relative input shares for the relevant crop, and finally the farm's chemical-use efficiency shock. In turn, this equation implies that $\omega_{jt}^{ch}$ will be recovered as the residual driver of the farm-crop-level chemical-to-land ratio. In that sense, this exercise gives structure to the simple decomposition performed in Figure 1.[32]

In a similar fashion, relying on the FOC for labor, the TFP can be expressed as a function of observables, shocks and parameters. I write, with lower case terms denoting logs:

$$\omega_{jct}^{h} = p_{jt}^{l} - p_{jct} - \alpha_{k,c}k_{jt} - \log(\alpha_{l,c}) + (1-\alpha_{l,c})l_{jct} - \alpha_{x,c}x_{jct}. \tag{7}$$

The expression for the TFP relies on $\omega_{jt}^{ch}$, which is included in $x_{jct}$. This dependency yields a natural estimation path used by Doraszelski and Jaumandreu (2018): in a first stage the parameters of the composite input $X_{jct}$ along with $\omega_{jt}^{ch}$ will be recovered, the parameters of the Cobb-Douglas and TFP residual will be recovered in a second.[33]

**Input Observability:** While crop-specific land use is observed in the FADN, all other inputs are measured at the farm level. This makes it impossible to directly estimate a crop-level production function for multiproduct firms. To address this, I follow De Loecker et al. (2016) and estimate the production function using data from single-product farms for which the input-to-output allocation is known. I then use the estimated parameters and the model's FOCs to recover the unobserved productivity shocks for the full sample of multiproduct farms. Using only single-product farms for estimation introduces a potential selection bias, as these farms are likely systematically different from multiproduct farms. To correct for this, I extend the selection-correction methodology of Olley and Pakes (1996), adapted by De Loecker et al. (2016), for my specific context. The procedure explicitly models the farm's choice to be a single-product producer and includes a correction term in the production function estimation. As argued by De Loecker et al. (2016) and Backus (2020), this bias affects the coefficients of variables correlated with the threshold rule used for the introduction of a second product. Because the first stage of my estimation relies entirely on a static,

[32] Relative to the chemical-to-output expression which is highly non-linear in parameters, the chemical-to-land ratio yields an expression which is close to linear, providing a more appropriate estimating equation.

[33] The estimation could be done in a single step, potentially yielding efficiency gains. My approach has the advantage of increased transparency, where Equation (6) provides a linear equation tightly linking chemical-use efficiency to input prices, input ratios and input shares.

intra-period optimality condition that holds regardless of the broader crop portfolio, it is immune to this dynamic selection bias. The variables correlated with the threshold rule—specifically, the farms' capital stocks and chemical-use efficiency—enter the estimation solely in the second step. I therefore include the selection correction only in the second stage of my estimation routine.[34]

**First Stage:** The first stage aims to estimate the parameters of $X_{jct}$ ($\{\rho, \{\delta_{s,c}\}_{\mathbb{C}}\}$) and to recover the unobserved chemical heterogeneity shock ($\omega_{jt}^{ch}$) . Here I rely on Equation (6). This equation expresses $\omega_{jt}^{ch}$ as a function of observed data and the structural parameters. To separate the shock from the parameters, I introduce the assumption that $\omega_{jt}^{ch}$ follows a time-inhomogeneous first-order Markov process. This allows me to decompose the shock into its conditional expectation and a random innovation $\zeta_{jt+1}^{ch}$, such that:

$$\omega_{jt+1}^{ch} = \mathbb{E}_t[\omega_{jt+1}^{ch}|\omega_{jt}^{ch}] + \zeta_{jt+1}^{ch} = g_{ch,t}(\omega_{jt}^{ch}) + \zeta_{jt+1}^{ch}.$$

The innovation $\zeta_{jt+1}^{ch}$ is mean independent of any information available at time $t$. The goal is to obtain a robust approximation for $g_{ch,t}(.)$, to isolate the innovation term and use it to form moments for the estimation. This is achieved first by using Equation (6) to provide an expression for $\omega_{jt}^{ch}$, and second by employing a third-order polynomial to numerically approximate $g_{ch,t}(.)$. Relying again on Equation (6) to express $\omega_{jt+1}^{ch}$, I can finally isolate an expression of $\zeta_{jt+1}^{ch}$ by relying on the expression:

$$\begin{aligned}\zeta_{jt+1}^{ch} = \frac{1-\rho}{\rho}\log\left(\frac{Chemicals_{jct+1}}{S_{jct+1}}\right) - \frac{1}{\rho}\log\left(\frac{P_{jt+1}^{s}}{P_{jt+1}^{ch}}\right) - \frac{1}{\rho}\log\left(\frac{1-\delta_{s,c}}{\delta_{s,c}}\right) \\ - g_{ch,t}\left(\frac{1-\rho}{\rho}\log\left(\frac{Chemicals_{jct}}{S_{jct}}\right) - \frac{1}{\rho}\log\left(\frac{P_{jt}^{s}}{P_{jt}^{ch}}\right) - \frac{1}{\rho}\log\left(\frac{1-\delta_{s,c}}{\delta_{s,c}}\right)\right)\end{aligned}$$

Because technological trends are likely to shift overall chemical-use intensity over time, the polynomial used for $g_{ch,t}(.)$ is time inhomogeneous and includes year fixed effects. This relaxes the identifying assumptions, as the chemical-use efficiency innovation is only assumed to be a mean-zero deviation from a year-specific common shock affecting all farms. Finally, with $A_{jt+1}^{z,ch}$ denoting a matrix of instruments orthogonal to the innovation term (detailed below), the parameters of the first step are estimated via a generalized method of moments (GMM) using moments of the form: $\mathbb{E}\left[\zeta_{jt+1}^{ch} A_{jt+1}^{z,ch}\right] = 0$.

**Second Stage:** The second stage relies on the results from the first stage to estimate the output elasticities ($\{\alpha_{k,c}, \alpha_{l,c}, \alpha_{x,c}\}_{\mathbb{C}}$), and to recover the distribution of TFP shocks. It

[34] De Loecker et al. (2016) argue one should keep observations related to both always single-product firms, and sometimes single product firms when they happen to be single product. I follow their dataset construction guidelines for all my estimation.

does so by using the parametric expression for the TFP shock obtained in Equation (7). The standard approach would then be to assume the TFP also follows a first-order Markov process, and form moments in a similar way as I did for the first step. However, there is a potential bias caused by the restriction of the data to single-product farms. The decision to grow a second crop likely depends on the farm's state variables, particularly its capital stock and its chemical-use efficiency. Because these variables also appear in the second-stage estimating equation, the selection process induces a correlation between the regressors and the error term, likely biasing the estimates of $\{\alpha_{k,c}, \alpha_{x,c}\}_{\mathbb{C}}$. To address this bias, I explicitly model the crop selection process and incorporate a correction term. The true conditional expectation for TFP must account for the information that the farm possessed when deciding to remain a single-crop operation. Let $\Xi_{jct}$ be a dummy variable indicating whether farm $j$ only produces $c$ in $t$. Under the assumption that the TFP shock follows a first order Markov process, I can then account for the truncated TFP process: $\omega^h_{jct+1} = \mathbb{E}[\omega^h_{jct+1}|\omega^h_{jct}, \Xi_{jct+1} = 1] + \zeta^h_{jct+1}$.[35] Appendix D.6 details the correction procedure. With the inclusion of the correction, I can build a matrix of instruments $A^{z,h}_{jct+1}$ orthogonal to the full residual of the production function composed of the TFP innovation and the idiosyncratic shock to production $(\zeta^h_{jct+1} + \varepsilon_{jct+1})$. I use them to form the moment conditions for the GMM: $\mathbb{E}\left[\left(\zeta^h_{jct+1} + \varepsilon_{jct+1}\right) A^{z,h}_{jct+1}\right] = 0$.

**Additional Considerations:** Three final data considerations are worth noting. First, a key advantage of FADN data is that it provides crop-level output in volumes, allowing me to use physical quantities in production and avoid the biases associated with using revenue data to measure production (Foster et al. (2008)). Second, expenditures on chemical inputs must be deflated to recover quantities, while I directly observe labor in hours, and land in hectares. I use national-level price indices for this purpose. Store-level data suggests minimal regional price dispersion in France, and no dataset is readily available to build representative regional agricultural input prices.[36] Third, I observe aggregate input bills for both fertilizers and pesticides, and cannot address issues of varying consumption basket composition—across farms or across time. Homogeneous input quality updates at the country level might be reflected in price changes. Any upgrade in quality that comes with an increase in price not reflected in the national price index—for example a farm starting to buy better and more expensive products—will be manifested as a higher volume of chemicals used rather than an increase in quality. As shown above, a higher chemical-to-land ratio unexplained by varying

[35] Note that this time, the Markov process is assumed to be time-homogeneous, which is standard for Hicksian processes. This assumption implies that technological progress in French agriculture over my period of analysis is focused on improving agricultural chemical efficiency, which tracks with the literature (Carpentier and Rainelli (1997)).

[36] These indices are built using an agricultural price survey. Labor and land prices are observed separately, and specified at the department-by-year level.

prices or a change in crop mix will be classified as an increase in chemical productivity.[37]

**Estimation:** I estimate the parameters for both stages using GMM.[38] The validity of the estimation relies on using a set of instruments that are orthogonal to the productivity innovations and random shocks, while being relevant to flexible input choices. Such instruments can include exogenous policy shocks or lagged input decisions (Ackerberg et al. (2015)), lagged output and input prices (Doraszelski and Jaumandreu (2013)), current input prices if they are believed to be exogenous (Griliches and Mairesse (1998)), or the lagged price of output (De Loecker et al. (2016)).

It is useful here to go back to Equation (6), which specifies the evolution of the unobserved chemical-use efficiency term. Importantly, the instruments need to be uncorrelated with the innovation of that process, but do not need to be uncorrelated with overall farm chemical-use efficiency. With this in mind, I use the following instruments for the first stage: lagged year dummies to match the time inhomogeneous structure of $g_{ch}(.)$, lagged capital stock, lagged input choices (the lagged log ratio of chemicals used to land used), and exogenous factor prices and policy variables (the lagged log ratio of chemical and land prices, the lagged price of land, the current land subsidy, and a measure of current-farm exposure to EU agricultural policy using lagged crop land shares as weights following my shift-share instrument). Lagged input decisions such as capital or the chemical-to-land ratio relate to the $t+1$ chemical-to-land ratio through persistence: in the case of capital simply because the remaining part of lagged capital post-depreciation accrues to $t+1$ production, for the lagged chemical-to-land ratio because static characteristics to the farm such as soil quality will influence its average ratio in any period. Persistence in local input prices also correlates lagged prices to current input choices, while being plausibly uncorrelated to the innovation. Finally, agricultural policy is set at the EU-level, in a way likely orthogonal to local farm shocks. This is even more likely to be the case after including year dummies in $g_{ch}(.)$.

Instruments for the second stage must be correlated with the farm's input decisions at $t+1$ but uncorrelated with the full residual ($\zeta^h_{jct+1}+\varepsilon_{jct+1}$). Production scale and additional inputs play a stronger role for second step instruments, as these should shift output and all input decisions. This justifies a more general set of instruments including the investment decision made at $t$, through including lagged and current capital stock, as well as additional lagged input levels and input prices. I use: a constant, $t+1$ capital, lagged input decisions (lagged capital, the lagged ratio of chemicals-to-land, the lagged composite input), input prices (the lagged hourly wage, the lagged price of land, the lagged log price ratio of chemicals-to-land,

[37] In that sense, the chemical-use efficiency shock will not discriminate between input upgrading, learning or changes in farm management that lead to increased chemical use.

[38] Following the approach of Wooldridge (2010), the parameters of the third order polynomial functions used to approximate the conditional expectation of productivity ($\{g_{ch,t}(.), g_h(.)\}$) are absorbed and not estimated directly. I use Amemiya (1974) GMM weights and a Nelder-Mead minimization algorithm for the estimation.

the lagged output price), and finally subsidy shifters ($t+1$ land subsidy and farm exposure as built in the first step). Estimates are shown in Table 2 and discussed below.

**Sensitivity of Results to Moments & Robustness:** To understand how my choice of instruments shapes the production function estimates, I implement the sensitivity analysis of Andrews et al. (2017). This procedure quantifies how a violation of each moment condition would bias each parameter estimate, making explicit which exclusion restrictions the parameters lean on most heavily. This is specifically relevant for the estimate of $\rho$, which is central to the policy analysis. The estimate of $\rho$ is found to be sensitive to violations of moments involving: the lagged chemical-to-land price and quantity ratios, the lagged land price, and the two subsidy instruments. The estimated sensitivity matrix shows how, in the presence of an omitted variable positively correlated with the chemical-to-land quantity ratio, my estimate of $\rho$ would be upward biased. Additionally, any bias shifting the moment associated to the chemical-to-land input price ratio upward would generate a downward bias in the estimated $\rho$. These results also relate to the identifying intuition one can draw from Equation (6), where $\rho$ is estimated by studying how the chemicals-to-land ratio shifts with the chemical-to-land input price ratio, controlling for crop type and the unobserved chemical-use efficiency. Complete derivations and results are shown in Appendix F.[39]

Because the estimates are sensitive to the subsidy moments, and we might be worried about mis-measurement in subsidy values, I re-estimate the production function without them. In both estimation stages, I replace them with two instruments composed of a lag dummy for whether the farm was growing wheat or the second crop interacted with the crop's farm-level lag price value. Lagged output prices are often used to estimate production function (e.g. Doraszelski and Jaumandreu (2013)), and the lagged crop decision is assumed to be orthogonal to the innovation of the chemical-use efficiency process. Estimates are shown in Table A17 and comparable to the main specification, with a slightly larger $\rho$ value of 0.55, compared to 0.44 in the baseline, but their difference is not statistically significant.

#### 4.3.3 Dynamic Parameters

Four key parameters still need to be estimated: $\{f_{inc}, f_e, C_k^Q, \tau_k\}$. Together, these parameters determine firm survival thresholds and investment behavior. I estimate these parameters using a Simulated Method of Moments (SMM) procedure (Pakes and Pollard (1989); Gouriéroux et al. (1993)). The algorithm proceeds as follows: for a vector of potential parameters $\Theta = \{f_{inc}, f_e, C_k^Q, \tau_k\}$, I solve for the stationary equilibrium of the market, and

[39] For space, the sensitivity table for the first stage reports the price, input, and subsidy instruments and omits the 35 year dummies from the first step, which serve as time controls and are of lesser independent interest for the exclusion-restriction discussion.

obtain the farms' policy functions for investment, crop choice and exit.[40] I then simulate a panel of farms, drawing their initial type from the equilibrium distribution. I use this panel to compute a vector of simulated moments. This model is run relying on the estimation of the demand curves from Section 4.1, and on the production function and associated parameters discussed previously.[41] The parameter estimates are those that minimize the distance between these simulated moments and their empirical counterparts. Following Gouriéroux et al. (1993), identification requires that the function mapping parameters to moments is injective (a relevance condition), and that the simulation procedure provides a consistent estimate of this function. The first condition is about the relevance of the chosen moments, shown in Table 1. I use bootstrapping and a large simulated panel to ensure that the second condition holds.[42]

Table 1: Targeted and Untargeted Moments

| **Moments for Estimation** | Moment | Observed Value | Simulated Value |
|---|---|---|---|
| Auto-Correlation Investment (FADN) | | 0.088 | 0.093 |
| $10^{th}$ to $50^{th}$ Centiles Revenue Ratio (FADN) | | 0.219 | 0.249 |
| Fraction Observations with Investment Absolute Value $< 5\%$ (FADN) | | 19.5% | 20.7% |
| Exit Rate in 7 Years (ESEA) | | 28.0% | 25.7% |
| **Untargeted Moments** | Moment | Observed Value | Simulated Value |
| Price (wheat) | | 158 €/t | 143 €/t |
| Price (other) | | 181 €/t | 196 €/t |
| Supply (wheat) | | 29.4 million t | 32.6 million t |
| Supply (other) | | 27.2 million t | 29.2 million t |
| Chemical Expenditure to Revenue Ratio | | 27.5 % | 27.4% |

*Notes: The exit rate is computed using the ESEA 2016 and 2023 datasets. All other targeted moments are computed using the FADN survey over the 1995-2015 period. The auto-correlation in investment tracks the correlation in investment rates within farms over time, the $10^{th}$ to $50^{th}$ centiles revenue ratio is computed for farms with positive revenue and tracks the size of the left tail of the farm revenue distribution, the fraction of observations with annual investment rates in absolute value below 5% captures the degree of inaction in investment. Untargeted moments are averages in the FADN for 1995-2015. Scatter-plots showing the obtained relationship between parameters and simulated moments for 2,048 Sobol draws of the parameter vector are shown in Figure A12.*

[40] I solve for an equilibrium with a 51% land subsidy, the currently implemented level in France.

[41] Table A7 details the additional parameters used when running the numerical model. After the production function estimation, I recover the distributional parameters for the TFP and chemical-use efficiency shocks, as well as the parameter $\lambda$ for the competence ladder, measured as the average ratio between minimum and maximum TFP within observed multi-product farms.

[42] Following Asker et al. (2014), I use an identity matrix in the minimization of the distance between observed and simulated moments, as moments are similarly scaled. To account for the influence of initial draws on estimation in this model with endogenous exit, I bootstrap this entire simulation B=100 times, drawing new entrants from the equilibrium entrant distribution to maintain a constant sample size. Within each simulation, I draw a panel of 1,000 farms, for T=200 periods. I first evaluate the model over 2,048 Sobol draws, and then use a local minimization routine to refine the estimation.

**Moment Relevance:** While all moments are used to estimate all parameters, the relevance condition is best discussed by going through the respective relation of each moment with each relevant parameter. Figure A12 illustrates the discussion that follows by showing the obtained relation between parameter draws and the relevant simulated moments over 2,048 runs in the model where parameters are drawn using a Sobol sampling method. Convex costs of capital adjustment relate most to the auto-correlation in investment, as high convex costs of capital adjustment lead to smaller more frequent investment bursts, smoothing investment over time. High capital irreversibility makes it less advantageous to sell capital, disincentivizing the resale of capital when TFP drops, and generating inaction bands. As such, a high irreversibility will increase the fraction of investment whose absolute value is below 5 percent. The fixed cost of incumbency relates most to the ratio of the $10^{th}$ centile to $50^{th}$ centile of the distribution of farm revenue. Indeed, larger fixed costs of incumbency require farms to have larger revenues and larger profits to survive, increasing this revenue ratio. Finally, the fixed cost of entry relates best to the exit rate. From the zero profit condition at entry, a high fixed cost of entry raises output prices. Conditional on entry, however, the profit threshold for exit is not affected by the fixed cost of entry. As such, because more diverse farms can survive, it lowers the exit rate of the market.

The first three moments are measured in the FADN using the production function estimating sample, the exit rate is computed by pairing two waves of the inter-Census ESEA survey for 2016 and 2023.[43] I compare the values of simulated and observed targeted and untargeted moments in Table 1 as a measure of model fit. Overall, the model evaluated at the estimated parameters provides a good fit of the four targeted moments. Of the four targeted moments, the revenue ratio and the exit rate show the largest deviations from the observed moments, implying that the left tail of farm revenues will be slightly smaller in my simulated data, and that the exit rate will be slightly smaller than observed. Finally, the model replicates observed output prices, production levels and the chemical-expenditure-to-revenue ratio well.[44]

#### 4.3.4 Estimation Results

I present the main parameter estimates in Table 2. Table A7 provides the values for the additional model parameters that I calibrate directly from the FADN or external sources. From $\rho = 0.44$ (0.19), I recover an elasticity of $\sigma = 1.79$. As derived in Section 4.2, this elasticity signs the crop-level correlation between farm profits and chemical use intensity: smaller farms will be both less chemical intensive and have lower yields, using less chemicals

[43] To proxy for a market exit rate, I rely on the ESEA survey and its 2016 and 2023 waves. Appendix B details its construction.

[44] The average chemical-expenditure-to-revenue ratio used here is measured in my FADN estimation sample, and hence differs slightly from the average shown in Figure 1 on a larger sample.

and more land per unit of output. [45] I also obtain a yearly fixed cost paid by incumbents of 7,230€, Industry estimates of administrative costs for the average French wheat farm place these around 9,400€ (Guillermet (2015)). The fixed cost of entry is estimated to be 223,000€. Industry estimates set the initial investment value to enter the agricultural market to be around 250,000€, relatively close to my estimate.[46]

Table 2: Estimated Model Parameters

| Coefficient | Parameter | Estimate | Std. Error |
|---|---|---|---|
| **Demand Parameters** | | | |
| Demand constant (wheat) | $\gamma_{wheat}$ | 18.94 | (0.41) |
| Demand constant (other) | $\gamma_{other}$ | 19.09 | (0.47) |
| Demand elasticity (wheat) | $\beta_{wheat}$ | -0.33 | (0.086) |
| Demand elasticity (other) | $\beta_{other}$ | -0.36 | (0.095) |
| **Production Function** | | | |
| Substitution Land-Chemicals | $\rho$ | 0.44 | (0.19) |
| Land Share (Wheat) | $\delta_{s,wheat}$ | 0.45 | (0.22) |
| Land Share (Others) | $\delta_{s,other}$ | 0.40 | (0.20) |
| Labor Share (Wheat) | $\alpha_{l,wheat}$ | 0.38 | (0.022) |
| Labor Share (Others) | $\alpha_{l,other}$ | 0.44 | (0.019) |
| Land-Nest Share (Wheat) | $\alpha_{x,wheat}$ | 0.40 | (0.024) |
| Land-Nest Share (Others) | $\alpha_{x,other}$ | 0.39 | (0.022) |
| **Capital & Fixed Costs** | | | |
| Convex Adjustment Cost | $C_k^Q$ | 2.72 | (0.97) |
| Capital Irreversibility | $\tau_k$ | 0.073 | (0.0083) |
| Fixed Cost of Incumbency (€1K) | $f_{inc}$ | 7.23 | (0.26) |
| Fixed Cost of Entry (€1K) | $f_e$ | 223.5 | (5.15) |

*Notes: 1) Demand parameters come from two-stage least squares regressions and their standard errors are corrected for auto-correlation using the Newey-West procedure. Demand constants are computed for $t = 2015$ as a combination of the regression intercept and time trend, and their standard errors are computed using the delta method. 2) Production function parameters are obtained from a two-step GMM estimation. Standard errors for the production function come from a block bootstrap procedure at the farm level, with $B = 1000$. The two steps of the production function estimation are included in the bootstrap so that second step standard errors account for first step uncertainty. 3) Costs of capital adjustment and fixed cost parameters are obtained with a simulated method of moments estimation. Their standard errors reflect only uncertainty from this last estimation step, and do not account for estimation error in the demand and production function parameters: propagating it would require re-estimating the dynamic parameters for bootstrap draws of the demand and production function estimates, each estimation requiring thousands of equilibrium solutions of the model.*

Finally, the production function parameters also imply that high-profit farms will allocate a larger share of their land to the aggregated crop, relative to lower profit farms, as illustrated

[45] The magnitude of my estimate is comparable to other key elasticities found in agriculture, such as the elasticity of 1.71 between labor and capital estimated for U.S. agriculture by Kislev and Peterson (1982).

[46] See https://www.agri-france.com/reprendre-une-ferme/, accessible on the Web Archive at June 13th 2025.

in Figure A13. This systematic heterogeneity in crop mix is driven by two forces. First the second crop has higher returns to flexible inputs than wheat (0.83 vs 0.78), and the log ratio for crop choice shown in Section 4.2 implies that higher TFP farms will then grow more of the high returns to flexible inputs crop. Second, the aggregated crop is also more chemical-use intensive, and so high chemical-use efficiency farms will also grow relatively more of this second crop. This systematic heterogeneity in crop mix along farm profit, which also tracks with farm chemical-use intensity, implies that differentiating policies across crops will be a good way to target high vs. low chemical-use intensity farms.

# 5 Equilibrium Evaluation of Agricultural Policies

I first illustrate how CAP subsidy designs have the capacity to generate unintended environmental benefits. I then turn to studying the optimal redesign of these subsidies when their environmental benefits are evaluated alongside their effects on producers and consumers.[47]

## 5.1 Unintended Environmental Effects of CAP Subsidies

I illustrate how different agricultural policies impact chemical-use intensity across farms using Figure 3, relying on illustrative policy intensity levels. The next section will tackle the question of optimal policy levels. This figure plots a farm's total profit in a baseline no-policy equilibrium on the X-axis. This variable serves as a one-dimensional summary of the multi-dimensional farm type (TFP, chemical-use efficiency and capital), which is fixed across equilibria. On the Y-axis, I plot farm-level chemical-use per unit of output, a summary measure of chemical-use intensity in each policy regime. The figure hence tracks how different policies impact a farm's use of chemicals. For the baseline no-policy equilibrium, this figure also plots the correlation between farm profit and average chemical-use intensity. Finally, the figure illustrates changes in selection patterns induced by policy. For any given farm type proxy, the data is plotted as a shaded square if no farm of that type survives in the relevant equilibrium. The data is plotted as a full triangle if all farms survive, and circles indicate farm type proxies with mixed survival levels.[48] The vertical red line shows the lowest profit level at which any farm survives in the absence of policy. Movements in the support for incumbency (the triangle and circle sections of the curve) relative to baseline help track the impact of policies on market selection. I also perform a total decomposition of the effect of each policy, relative to baseline, on the average market farm-level chemical-use intensity. Results are shown in Table A8. The decomposition follows the approach of Melitz

[47] Additionally, in Appendix G I simulate the effects of the policy design change at the heart of the MacSharry reform, moving from government purchasing to a land subsidy.

[48] The inclusion of capital stock in a firm dynamics model with entry and exit implies that there is no single profit cutoff for farm exit, but rather an exit surface.

and Polanec (2015), adapted to my context to include changes in crop mix within the farm, and the formula is described in Appendix E.

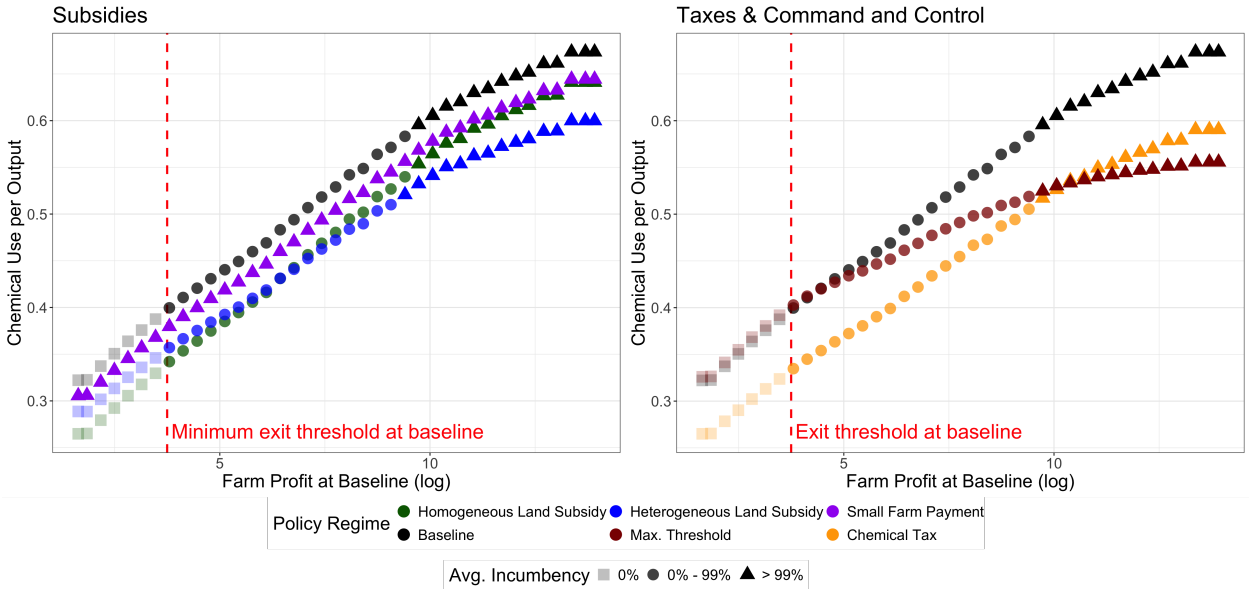



Figure 3: Intensive & Extensive Effects of Agricultural Policies

*Notes:* This figure plots on the x-axis a farm's total profit in the baseline no-policy equilibrium. This variable serves as a one-dimensional proxy for a farm's type (a combination of TFP, chemical-use efficiency and capital stock), and allows one to track farm behavior across equilibria. On the y-axis, the figure plots a farm's farm-level average chemical-use per unit of output under each relevant equilibrium. Finally, changes in incumbency behavior are highlighted by changes in the shading and shapes of the curves: full triangles indicate a farm type proxy for which all farms are incumbents, circles indicate types for which at least some farms are incumbents, and squares types for which no farms remain incumbents.

Following the black curve, the first thing to note in Figure 3 is that high-profit farms use more chemicals per unit of output. The estimated $\rho = 0.44$ implies a positive crop-level correlation between chemical-use per unit of output and profit (as illustrated in Figure A14 and Figure A15). These crop-level relations follow through at the farm level: while farms with large and small baseline profit levels do differ systematically in crop mix composition, the two crops are not found to differ enough in chemical-use intensity and so the crop-level relations scale to the farm-level.[49] On the left panel of Figure 3, I illustrate the effects existing or past CAP subsidy designs can have on chemical-use intensity. The first subsidy I model is a 33% land subsidy, homogeneous across crops. This homogeneous land subsidy is the main way through which the EU subsidizes agriculture today. This policy heavily decreases chemical-use per unit of output across all farms, and does not impact the selection margin—as a result the green curve is simply a downward shift of the no-policy black curve. Table A8 indicates that this policy reduces the average chemical-use intensity on the market by 7.7%, with all the reduction coming from within-farm within-crop changes in input mix. I also model a heterogeneous land subsidy, a 66% land subsidy only applied to the aggregated

[49] Figure A16 tracks farm-level chemicals per unit of land, a ratio I use when computing environmental damages.

second crop. Heterogeneous land subsidies existed in the EU after the MacSharry reform, and here the policy interacts with the heterogeneity in crop mix composition along the farm type distribution, high type farms growing more of the aggregated crop (see Figure A13). As a consequence, the blue curve associated to this subsidy is a clockwise rotation of the green curve associated to the homogeneous land subsidy. The overall effect of the heterogeneous land subsidy is a higher reduction in chemical-use intensity of 11.3%.

The last subsidy I model on Figure 3 is a lump-sum payment to low-profit farms, matching the design of the EU's 2014-2020 Small Farmers Scheme. Because lower type farms have lower profits, are less likely to participate in the market, and on average use less chemicals per unit of output, this policy has the capacity to lower the average chemical-use intensity of the market by reallocating production from infra-marginal incumbents to marginal incumbents. On Figure 3, I plot results for a 10,000 euro lump-sum payment for farms with profits below 3,000 euros, a lump sum slightly above the fixed cost of incumbency I estimated at 7,230 euros. This policy has a substantial effect on market selection: in the equilibrium with the lump-sum payment, all farm types take part in the market. This payment also lowers equilibrium output prices through the zero expected profit of entry condition, and hence lowers both production scale and chemical-use intensity for all incumbents. On Figure 3, the purple line features positive incumbency across the entire farm type distribution, and also stands below the no-policy black curve (this second feature being the result of reduced farm scale). Table A8 shows the policy reduces average chemical-use intensity by 6.2%, with 4.1% coming from reductions in scale and 2.5% from selection.[50]

On the right panel of Figure 3 I also model two chemical-use reduction policies which have been considered by European policy makers in the past but failed to be implemented into European law.[51] The tax modeled in Figure 3 is a 20% tax. The maximum threshold acts as an upper bound on the farm-crop-level chemical-to-land ratio, set to the median of the wheat chemical-to-land ratio observed in the no policy equilibrium. Figure 3 highlights how the tax treats all incumbent farms equally, while the maximum threshold only treats high type farms by flattening the curve associated to the policy. From Table A8, almost all of the 13.4% tax-induced decrease in chemical-use intensity can be attributed to within-farm within-crop changes, while half of the effect of the threshold is to reallocate production across infra-marginal incumbents away from the most chemical intensive ones (-6%), on top of reducing farm-crop chemical-use intensity for high chemical use intensity farms (a 6.0% decrease). In the next section, I study whether agricultural subsidies can substitute for these

[50] The lump-sum payment has a small positive effect on chemical use intensity through the reallocation of production within infra-marginal incumbents, explaining that these two effects sum to an effect larger than the overall policy's 6.2% effect.

[51] A chemical-use tax for agriculture was proposed in 2022 by Members of the European Parliament. The maximum threshold relates to the proposed 50% reduction in pesticide-per-hectare which was to be part of the now defunct 2024 Sustainable Use of Pesticides Regulation.

direct regulations of chemical-use.

### 5.2 Agricultural Subsidies as Second-Best Environmental Policies

To compare agricultural policies, I rely on the following damage and welfare objectives:

$$\text{Damages} = \int \theta_1 S_j \left(1 + \frac{\text{Intensity}_j - X}{X}\right)^{\theta_2} + \theta_3 S_j d\mu,$$
$$W = \text{CS} + \text{PS} - (1+\Lambda)\text{Gov. Expenditure} - \text{Damages}.$$

Here $S_j$ and $\text{Intensity}_j$ resp. track a farm's total land use, and average chemical-use per unit of land, and $\mu$ is the allocation of the mass of farms in a given equilibrium. $\Lambda$ is the social cost of EU budget funds, set to $\Lambda = 0.15$.[52] Environmental damages come from two sources: first from chemical use, and here with $\theta_2 > 1$, damages are convex in chemical-use intensity, and second from land use, because the ecosystem services performed by agricultural land are lower than for land under conservation. I parametrize $\theta_1 = 828$ euros, corresponding to the minimum cost of water purification for the chemical pollution generated by the cultivation of one hectare of land, at the observed chemical-use intensity per hectare in France (CGDD (2011)). Because this marginal cost is measured for the observed level of intensity in France, $X$ anchors this level at the predicted average chemical-use intensity per hectare of land in my model under the current 51% CAP land use subsidy. I set $\theta_3 = 500$ euros, using the assessed value of ecosystem services from one hectare of forest (CAS (2009)).[53] The damage function hence focuses on scenarios where local agricultural chemical-use generates convex environmental damages, but also where there are ecological opportunity costs to agricultural land use. The convexity can arise from a convex emission function, i.e. as chemicals are used more intensively a larger fraction of them leaches into soils or is emitted as gases, or from a damage function convex in the volume of pollutants.[54]

I then compare the performance of policies in maximizing the welfare criterion, considering first a scenario with $\theta_2 = 1.25$, and then a scenario with $\theta_2 = 1.75$. $\theta_2 = 1$ would imply that only aggregate chemical use matters for damages, a pattern which does not match the local pollutant nature of agricultural chemicals. I also note that I do not account for the other effects of subsidies which might be valued by governments such as guaranteeing domestic food sovereignty. The purpose of this exercise is to focus on the scale of the environmental gains relative to the abatement costs. I also track the impact of policies on

[52] For the chemical tax, I accordingly pre-multiply additional government revenue by $\frac{1}{1+\Lambda}$.

[53] The underlying assumption is that all land not used in agriculture would be transformed into forest. This is a conservative assumption, Coeurdacier et al. (2026) for example show that a majority of the land freed from agriculture in the last decades in France was built upon, rather than transformed into forests.

[54] Shcherbak et al. (2014) provides evidence for convex emission functions for nitrogen fertilizer, and Larsen et al. (2017) discuss the convexity of pesticides-derived damages on human health.

producer surplus, a measure which accounts for both flexible profit and fixed costs paid by producers. Policies that increase producer surplus are not necessarily Pareto improving, reallocating production across producers rather than increasing every incumbent's profit. The two regulations considered are the ones discussed in the previous section: a homogeneous chemical tax and a maximum farm-crop-level chemical-to-land ratio for production. The first subsidy is a land subsidy which is allowed to be heterogeneous across crops. The second is the lump-sum payment to low-profit farms. The evaluation of reallocation-oriented policies reflects recent policy debates around the potential environmental benefits of small agriculture (e.g. European Court of Auditors (2020)), as well as a legacy of economic analysis on the larger economic efficiency of large farms (Foster and Rosenzweig (2022)). The last subsidy considered is a combination of the heterogeneous land subsidy and the lump-sum payment.

Table 3: Optimal Policies (relative to no-policy in M€, Mt or %, and abatement cost in €)

**Panel A: $\theta_2 = 1.25$ Scenario**

| Policy Type | Policy Level | Δ Welfare | (Δ CS, Δ PS) | Gov. Expenditure | Δ Damage | Abatement Cost | Δ Intensity (%) |
|---|---|---|---|---|---|---|---|
| Land Subsidy | (10%, 21%) | 85M | (103M, 3M) | 142M | -121M | 0.30 | -27% |
| Lump-Sum | (3.4k€, up to 14.0k€) | 94M | (136M, 249M) | 431M | -140M | 0.33 | -9% |
| Combined Subsidy | ((7%, 24%), 2.3k€, up to 13.4k€) | 140M | (163M, 123M) | 315M | -168M | 0.17 | -31% |
| Max. Threshold | ($50^{th}$*centile*) | 384M | (-88M, 24M) | 0M | -448M | 0.14 | -45% |
| Chemical Tax | 74% | 1,395M | (-2,070M, 164M) | -1,406M | -1,895M | 0.26 | -64% |

**Panel B: $\theta_2 = 1.75$ Scenario**

| Policy Type | Policy Level | Δ Welfare | (Δ CS, Δ PS) | Gov. Expenditure | Δ Damage | Abatement Cost | Δ Intensity (%) |
|---|---|---|---|---|---|---|---|
| Land Subsidy | (31%, 51%) | 2,417M | (384M, -3M) | 743M | -2,779M | 0.13 | -64% |
| Lump-Sum | (3.4k€, up to 14.0k€) | 582M | (136M, 249M) | 431M | -628M | 0.07 | -9% |
| Combined Subsidy | ((32%, 48%), 2.4k€, up to 14.6k€) | 2,501M | (443M, 175M) | 972M | -2,856M | 0.12 | -65% |
| Max. Threshold | ($37^{th}$*centile*) | 3,452M | (-266M, 39M) | 0M | -3,679M | 0.06 | -66% |
| Chemical Tax | 94% | 4,430M | (-2,495M, 197M) | -1,612M | -5,116M | 0.13 | -70% |

**Panel C: Land Sparing Break-Even Thresholds ($\theta_3$) for $\theta_2 = 1.25$ Policies**

| Land Subsidy | Lump-Sum | Combined Subsidy | Chemical Tax | Max. Threshold |
|---|---|---|---|---|
| 609 | 1,353 | 653 | 1,416 | 757 |

**Panel D: International Market Integration, Pollution Leakages & Incidence**

| | $\theta_2 = 1.25$ Policies | | $\theta_2 = 1.75$ Policies | |
|---|---|---|---|---|
| Policy Type | Δ Q for both crops (ROW) | Δ PS (France) | Δ Q for both crops (ROW) | Δ PS (France) |
| Land Subsidy | -0.64M | 20M | -2.5M | 58M |
| Lump-Sum | -0.29M | 125M | -0.62M | 250M |
| Combined Subsidy | -0.95M | 140M | -2.6M | 230M |
| Max. Threshold | 0.60M | 6M | 1.7M | -9M |
| Chemical Tax | 10.8M | -155M | 12.7M | -185M |

*Notes: Given a damage function and welfare criterion, I search the policy space of each regulation type, and show its welfare consequences in this table. Producer surplus accounts here for both flexible farm profit and fixed costs. The government expenditure measure already accounts for the marginal cost of public expenditure Λ, and a negative value indicates the policy generates government revenue. Abatement costs correspond to the government expenditure and negative producer and consumer surplus effect of a policy, divided by the damage reductions it generates. The intensity column indicates the change in the average chemical-to-land ratio of the market. All changes are relative to a baseline no-policy equilibrium. Panel C considers the minimum value of $\theta_3$, the marginal environmental cost of land, such that the welfare consequence of a policy is over-turned. Finally, Panel D evaluates optimal policies within a model extension where France along with a rest-of-the-world representative producer produce for a calibrated world demand.*

Results in Table 3 show that, given the parametrization of the damage function, the environmental benefits of optimal subsidies are large enough to outweigh their combined economic and fiscal effects measured as the sum of consumer surplus, producer surplus and government expenditure. The optimal land subsidy rises in value with the convexity of chemical-use damages, and the subsidy always targets the second crop more than the first: this second

crop is the one found to be more chemical intensive, it is also grown relatively more by high chemical-intensity farms. As such, this asymmetry in the optimal land subsidy is an efficient way to reduce chemical use intensity specifically among the highest intensity producers. This asymmetry becomes more pronounced as the convexity of damages rises. In contrast, the optimal lump-sum payment is stable across damage specifications, revealing that once the more sustainable producers have been brought into the market, there is little more that this subsidy can do to curb the chemical-use intensity of the market. This optimal lump-sum payment is found to correspond to about half of the fixed cost of incumbency, for farms earning less than 14k€ in profits.[55] Because the land subsidy can scale with the convexity of damages, it performs relatively better in a high convexity scenario, while both subsidies are on par for the low convexity case. Finally, a paired subsidy combining land subsidies and a lump-sum payment always performs best.

Both the tax and the maximum threshold always outperform subsidies, both in terms of the reductions in average chemicals per unit of land, amount of environmental damages curbed, or in terms of welfare gains. The best subsidy achieves 8.9% of the damage reduction of the best regulation in the low convexity scenario, and 56% in the high convexity one, or respectively 10% and 56% of the welfare gains. However, the abatement costs of subsidies are on par with those of regulations, with a combined subsidy costing 0.17 € for each euro of damage abated in the low convexity scenario, for 0.26€ for the tax, and the lump-sum payment costing 0.07€ in the high convexity case when the chemical threshold policy costs 0.06€. The incidence of subsidies and regulations is also different: regulations decrease the joint consumer and producer surplus with insufficient government revenues to compensate them, while subsidies benefit them. In my context with a relatively inelastic demand, the costs and benefits of policies are mostly borne by consumers. Overall, subsidies and regulations have different political economies, subsidies appearing more beneficial to consumers.

In Panel C of Table 3, I search for the lowest $\theta_3$ such that each low convexity policy stops generating welfare gains. This addresses the tension between land sharing and land sparing: the benefits of a less chemically intensive agriculture relying on more land, versus the ecological opportunity cost of increasing the land footprint of agriculture. As freeing land from agriculture becomes more beneficial, the land subsidy and the threshold regulation perform relatively worse than other policies, because their chemical-use reductions come with larger increases in the agricultural land footprint.

In my closed economy with a relatively inelastic demand, subsidy benefits and regulatory costs are mostly borne by consumers. I consider a model extension, outlined in Appendix D.9, where France along with a rest-of-the-world (ROW) producer both cultivate grain for world demand. Similar crops from different origins are aggregated on the demand side with a

[55] 14k of farm profits sits at the $31^{st}$ centile of the no-policy equilibrium distribution of infra-marginal incumbents.

CES function, where the elasticity of substitution between origins is taken from Costinot et al. (2016), and the overall elasticity of demand from Roberts and Schlenker (2013). In this setting, France faces a residual demand curve which is more elastic than in my baseline model. As a result, agricultural policies have smaller effects on grain prices, and producers capture both more of the benefits of subsidies and bear more of the costs of regulation. This shift in incidence is highlighted in Panel D of Table 3, where subsidies now always generate producer surplus gains, while regulations either leave producer surplus relatively unchanged or reduce it. In this setting, French producers will welcome subsidies but oppose direct environmental regulations. This extension also allows me to address the question of leakages, i.e. the fact that part of the gains of environmental regulation can be offset if the polluting activity can be displaced outside of the zone of regulation. Panel D shows that regulations can generate a substantial increase in foreign production, displacing pollution, which is mostly local, outside of the boundaries of France. I note that the threshold regulation generates lower leakages than the tax, as its environmental gains are achieved by a reallocation among French producers rather than a homogeneous increase in production costs. Subsidies offer an environmental advantage against leakage: they draw global production into a domestic market that now operates more sustainably.

## 6 Conclusion

In this paper, I document how production subsidies in polluting markets can generate unintended environmental effects, interacting both with the intensive margin of within-firm production decisions, and the extensive margin of market selection. I develop for this a dynamic model of heterogeneous producers who differ in their propensity to pollute because they grow different product mixes, but also are heterogeneous in their efficiency at using polluting inputs. I apply the model to the EU's Common Agricultural Policy and find that CAP subsidies generate unintended benefits, first by subsidizing a clean input (land) which substitutes with a polluting input (chemicals) in production, second by favoring the incumbency decision of marginal farms which are greener than infra-marginal market incumbents. I finally compare optimal subsidies to optimal environmental regulations, first an optimal chemical tax, and second a cap on the chemical-to-land ratio which can be used by farms in production. The environmental damage abatement achieved by subsidies is large enough for them to generate welfare gains, however subsidies fall short of reaching the damage reductions of the optimal regulation. Yet, the abatement costs of subsidies and regulations are comparable, and subsidies achieve environmental gains without the large consumer and producer losses generated by regulations, highlighting a tension between the optimality of environmental regulations and their acceptability.

# References


Ackerberg, D. A., K. Caves, and G. Frazer (2015): "Identification Properties of Recent Production Function Estimators," *Econometrica*.

Amemiya, T. (1974): "The Nonlinear Two-Stage Least-Squares Estimator," *Journal of Econometrics*.

Andrews, I., M. Gentzkow, and J. M. Shapiro (2017): "Measuring the Sensitivity of Parameter Estimates to Estimation Moments," *The Quarterly Journal of Economics*.

Asker, J., A. Collard-Wexler, and J. De Loecker (2014): "Dynamic Inputs and Resource (Mis)Allocation," *Journal of Political Economy*.

Aspelund, K. M. and A. Russo (2024): "Additionality and Asymmetric Information in Environmental Markets: Evidence from Conservation Auctions," *Working Paper*.

Bachelier, B. (2010): "Achieving a New Common Agricultural Policy," *Policy Report*.

Backus, M. (2020): "Why is Productivity Correlated with Competition?" *Econometrica*.

Becker, R. and V. Henderson (2000): "Effects of Air Quality Regulations on Polluting Industries," *Journal of Political Economy*.

Borusyak, K., P. Hull, and X. Jaravel (2022): "Quasi-Experimental Shift-Share Research Designs," *Review of Economic Studies*.

Bovenberg, L. A. and L. H. Goulder (1996): "Optimal Environmental Taxation in the Presence of Other Taxes: General-Equilibrium Analysis," *American Economic Review*.

Carlton, D. W. and G. C. Loury (1980): "The Limitations of Pigouvian Taxes as a Long-Run Remedy for Externalities," *Quarterly Journal of Economics*.

Carpentier, A. and P. Rainelli (1997): "Agriculture, Pesticides et Environnement: quelles politiques? Etude de cas de la France," *INRA Research Paper*.

CAS (2009): "An Economic Approach to Biodiversity and Ecosystem Services: Contribution to Public Decision-Making," *Centre d'Analyse Strategique du Premier Ministre*.

CGDD (2011): "Assessing Water Pollution Costs of Farming in France," *Studies and Documents, Department of the Commissioner General for Sustainable Development*.

Chabé-Ferret, S., A. Reynaud, and E. Tene (2021): "Water Quality, Policy Diffusion Effects and Farmers' Behavior," *Working Paper*.

Coeurdacier, N., C. Gaubert, and E. Henry (2026): "Farms, Forests and Cities: Structural Transformation of Land and the Carbon Balance," *Working Paper*.

Commission of the European Communities (1991): "The Agricultural Situation in the Community," *Policy Report*.

Costinot, A., D. Donaldson, and C. Smith (2016): "Evolving Comparative Advantage and the Impact of Climate Change in Agricultural Markets: Evidence from 1.7 Million Fields around the World," *Journal of Political Economy*.

Davis, L. W. (2017): "The Environmental Cost of Global Fuel Subsidies," *The Energy Journal*.

De Loecker, J., P. K. Goldberg, A. K. Khandelwal, and N. Pavcnik (2016): "Prices, Markups, and Trade Reform," *Econometrica*.

Dechezleprêtre, A., A. Fabre, T. Kruse, B. Planterose, A. S. Chico, and S. Stantcheva (2025): "Fighting Climate Change: International Attitudes toward Climate Policies," *American Economic Review*.

Dhyne, E., A. Petrin, V. Smeets, and F. Warzynski (2022): "Theory for Extending Single-Product Production Function Estimation to Multi-Product Settings," *NBER Working Paper*.

Dias, M., R. Rocha, and R. R. Soares (2023): "Down the River: Glyphosate Use in Agriculture and Birth Outcomes of Surrounding Populations," *Review of Economic Studies*.

Doraszelski, U. and J. Jaumandreu (2013): "R&D and Productivity: Estimating Endogenous Productivity," *The Review of Economic Studies*.

——— (2018): "Measuring the Bias of Technological Change," *Journal of Political Economy*.

EEA (2025a): "Emissions of the main air pollutants in Europe," .

——— (2025b): "Greenhouse gas emissions from agriculture in Europe," .

——— (2025c): "Pesticides in rivers, lakes and groundwater in Europe," .

Englander, G., J. Zhang, J. C. Villasenor-Derbez, Q. Jiang, M. Hu, O. Deschenes, and C. Costello (2023): "Input Subsidies and the Destruction of Natural Capital: Chinese Distant Water Fishing," *NBER Working Paper*.

European Court of Auditors (2020): "Biodiversity on farmland: CAP contribution has not halted the decline," *Special Report*.

——— (2024): "Common Agricultural Policy Plans: Greener, but not matching the EU's ambitions for the climate and the environment," *Special Report*.

FAO (2024): "Government expenditures in agriculture 2001–2022 – Global and regional trends," *FAOSTAT Analytical Briefs*.

Foster, A. D. and M. R. Rosenzweig (2022): "Are there too many Farms in the World? Labor Market Transaction Costs, Machine Capacities, and Optimal Farm Size," *Journal of Political Economy*.

Foster, L., J. Haltiwanger, and C. Syverson (2008): "Reallocation, Firm Turnover, and Efficiency: Selection on Productivity or Profitability?" *American Economic Review*.

Fowlie, M., M. Reguant, and S. P. Ryan (2016): "Market-Based Emissions Regulation and Industry Dynamics," *Journal of Political Economy*.

Ganapati, S., J. S. Shapiro, and R. Walker (2020): "Energy Cost Pass-Through in US Manufacturing: Estimates and Implications for Carbon Taxes," *American Economic Journal: Applied Economics*.

Goldsmith-Pinkham, P., I. Sorkin, and H. Swift (2020): "Bartik Instruments: What, When, Why and How," *American Economic Review*.

Gollin, D., D. Lagakos, and M. E. Waugh (2014): "The Agricultural Productivity Gap," *Quarterly Journal of Economics*.

Goulder, L. H. and I. W. Parry (2008): "Instrument Choice in Environmental Policy," *Review of Environmental Economics and Policy*.

Gouriéroux, C., A. Monfort, and E. Renault (1993): "Indirect Inference," *Journal of Applied Econometrics*.

Griliches, Z. and J. Mairesse (1998): "Production Functions: The Search for Identification," *Econometrics and Economic Theory in the Twentieth Century: The Ragnar Frisch Centennial Symposium*.

Grupp, T. E., P. Mishra, M. Reynaert, and A. A. van Benthem (2024): "An Evaluation of Protected Area Policies in the European Union," *Working Paper*.

Guillermet, J. (2015): "Couts de Production en Grandes Cultures," *Technical Report*.

Hopenhayn, H. (1992): "Entry, Exit, and firm Dynamics in Long Run Equilibrium," *Econometrica*.

Hsiao, A., J. Moscona, and K. A. Sastry (2026): "Food Policy in a Warming World," *Econometrica*.

Kislev, Y. and W. Peterson (1982): "Prices, Technology, and Farm Size," *Journal of Political Economy*.

Larsen, A., S. D. Gaines, and O. Deschênes (2017): "Agricultural pesticide use and adverse birth outcomes in the San Joaquin Valley of California," *Nature Communications*.

Larsen, A. E. and F. Noack (2017): "Identifying the Landscape Drivers of Agricultural Insecticide Use Leveraging Evidence from 100,000 Fields," *Proceedings of the National Academy of Sciences*.

Mayer, T., M. J. Melitz, and G. I. Ottaviano (2014): "Market Size, Competition, and the Product Mix of Exporters," *American Economic Review*.

Melitz, M. J. and S. Polanec (2015): "Dynamic Olley-Pakes productivity decomposition with entry and exit," *Rand Journal of Economics*.

Obolensky, M. (2025): "Protect or Prepare? Crop Insurance and Adaptation in a Changing Climate," *Working Paper*.

Olley, G. S. and A. Pakes (1996): "The Dynamics of Productivity in the Telecommunications Equipment Industry," *Econometrica*.

Orr, S. (2022): "Within-Firm Productivity Dispersion: Estimates and Implications," *Journal of Political Economy*.

Pakes, A. and D. Pollard (1989): "Simulation and the Asymptotics of Optimization Estimators," *Econometrica*.

Qi, J., X. Tang, and X. Xi (2021): "The Size Distribution of Firms and Industrial Water Pollution: A Quantitative Analysis of China," *American Economic Journal: Macroeconomics*.

Roberts, M. and W. Schlenker (2013): "Identifying Supply and Demand Elasticities of Agricultural Commodities: Implications for the US Ethanol Mandate," *American Economic Review*.

Seufert, V., N. Ramankutty, and J. A. Foley (2012): "Comparing the yields of organic and conventional agriculture," *Nature*.

Shapiro, J. S. (2021): "The Environmental Bias of Trade Policy," *The Quarterly Journal of Economics*.

Shapiro, J. S. and R. Walker (2018): "Why is Pollution from U.S. Manufacturing Declining? The Roles of Environmental Regulation, Productivity, and Trade," *American Economic Review*.

Shcherbak, I., N. Millar, and G. P. Robertson (2014): "Global metaanalysis of the nonlinear response of soil nitrous oxide (N2O) emissions to fertilizer nitrogen," *Proceedings of the National Academy of Sciences of the United States of America*.

Taylor, C. and G. Heal (2023): "Fertilizer and Algal Blooms: A Satellite Approach to Assessing Water Quality." *NBER Risks in Agricultural Supply*.

Tschofen, P., I. L. Azevedo, and N. Z. Muller (2019): "Fine particulate matter damages and value added in the US economy," *PNAS*.

Wooldridge, J. M. (2010): "Econometric Analysis of Cross Section and Panel Data," *MIT Press, Cambridge*.

# Supplemental Appendix:
## Unintended Environmental Policies:
## The Impact of European Agricultural Subsidies on Pollution

# A Additional Figures and Tables

## A.1 Figures

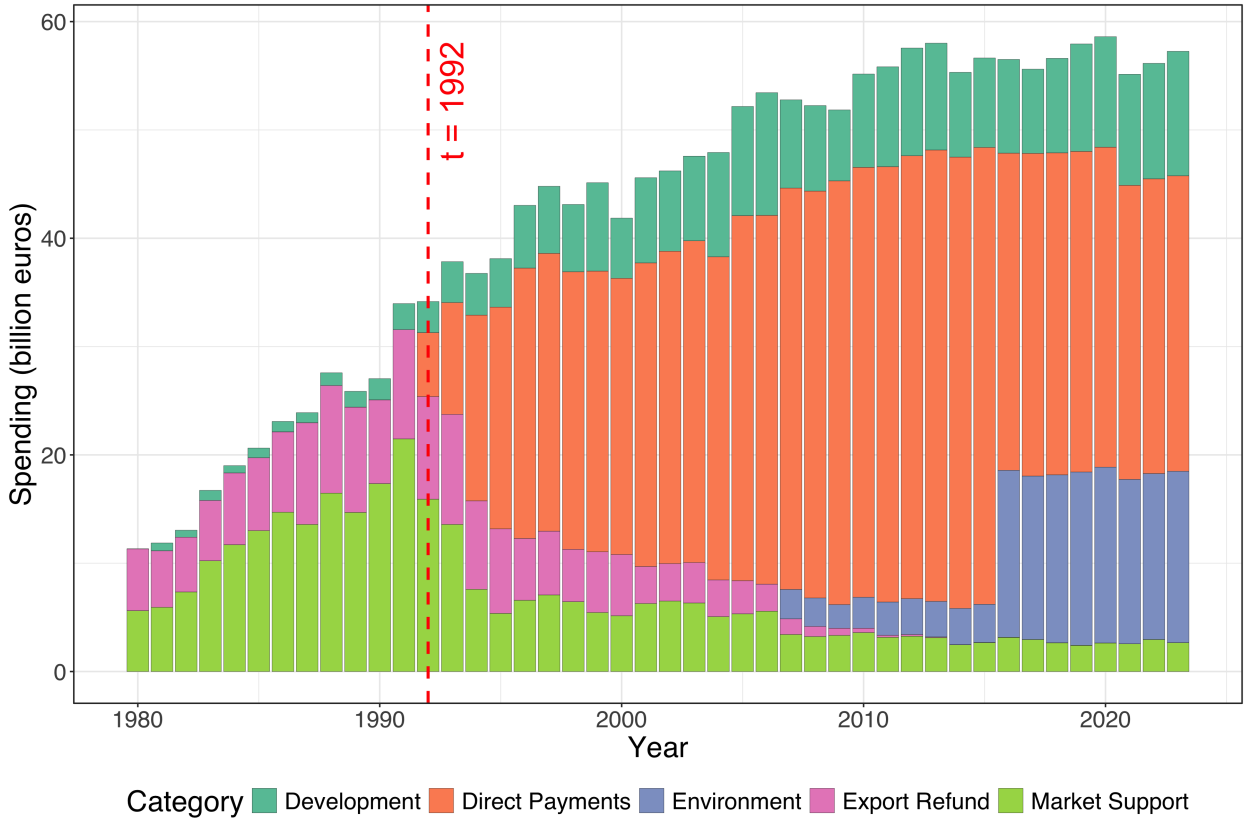



Figure A1: Breakdown of Common Agricultural Policy Spending

*Notes: Annual CAP spending data from the EU Commission Directorate General for Agricultural and Rural Development, broken down into its main components. Spending is expressed in 2025 euros. The vertical line indicates the start of the MacSharry reform, whose elements were sequentially implemented between 1992 and 1995.*

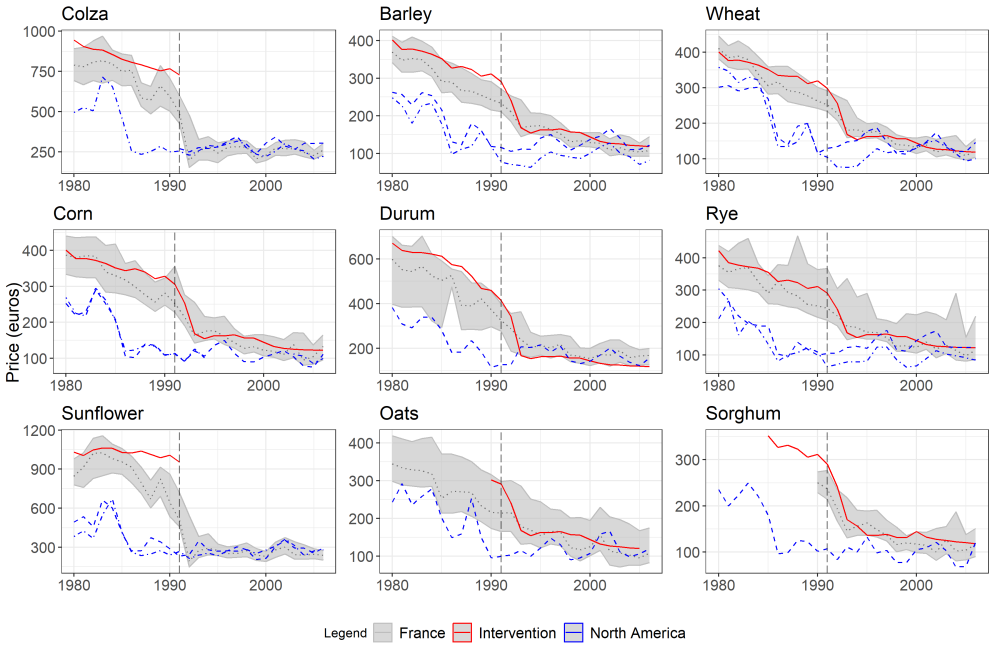



Figure A2: Convergence of French and North American Prices Following the MacSharry Reform

*Notes:* Evolution of French and North American farm-gate prices, as well as EU intervention prices for most cereal and oil crops. French farm-gate prices are shown with the grey ribbon, and taken from the FADN. The center dotted line corresponds to the weighted mean, and the edges to the bottom and top 25th and 75th centiles of the distribution of prices. Intervention prices are shown with the solid red line. The data is digitized, taken from the relevant EU directives published over time. U.S. and Canadian farm-gate prices are taken from FAOSTAT, and shown with the dashed and dot-dashed blue lines. All prices are converted into 2020 euros using the relevant exchange rates and correcting for inflation using the Insee's consumer price index.

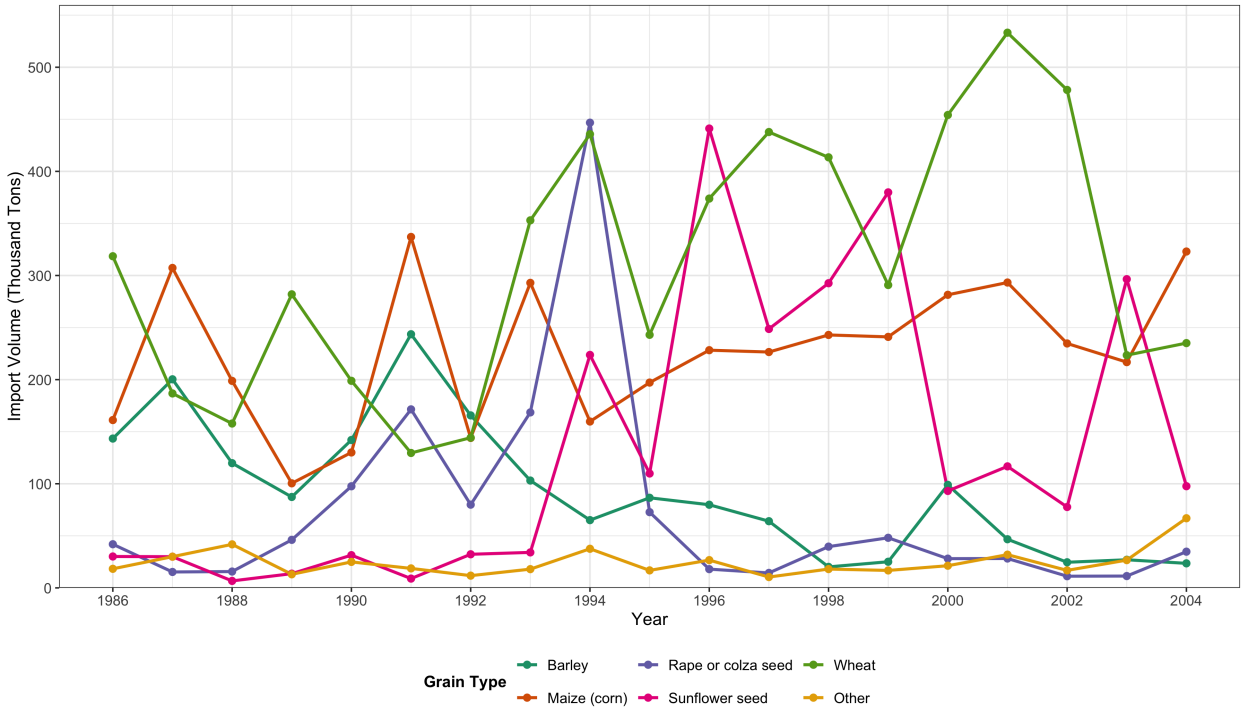



Figure A3: French Grain Imports

*Notes: This figure shows the evolution of French grain imports expressed in thousands of tons around the period of the MacSharry reform. The data comes from FAOSTAT.*

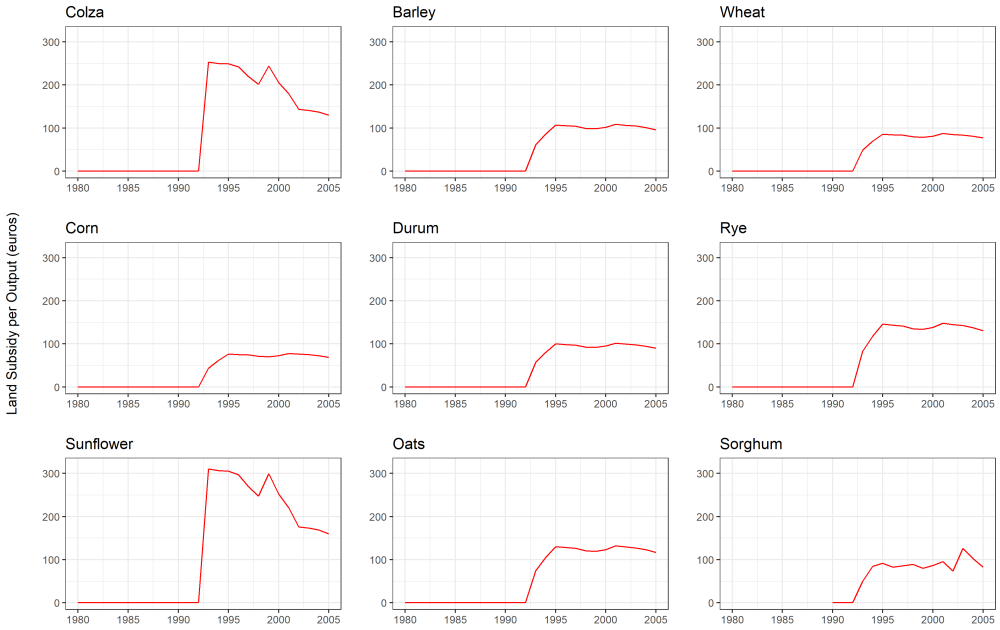



Figure A4: EU Land Subsidies per Crop under the CAP

*Notes: The figure shows the evolution of the average land subsidy per unit of output over time, as measured in the FADN. I measure the average subsidy per hectare for cereals and oil crops, dividing the farms' total subsidy for each crop category, by their land allocated to each category. I then divide this measure by the average crop-specific yield observed in France in that year, in order to recover a subsidy per unit of output. Subsidies per land are relatively stable across years, while yields increase, making it so that subsidies slightly decrease over time absent any policy adjustment.*

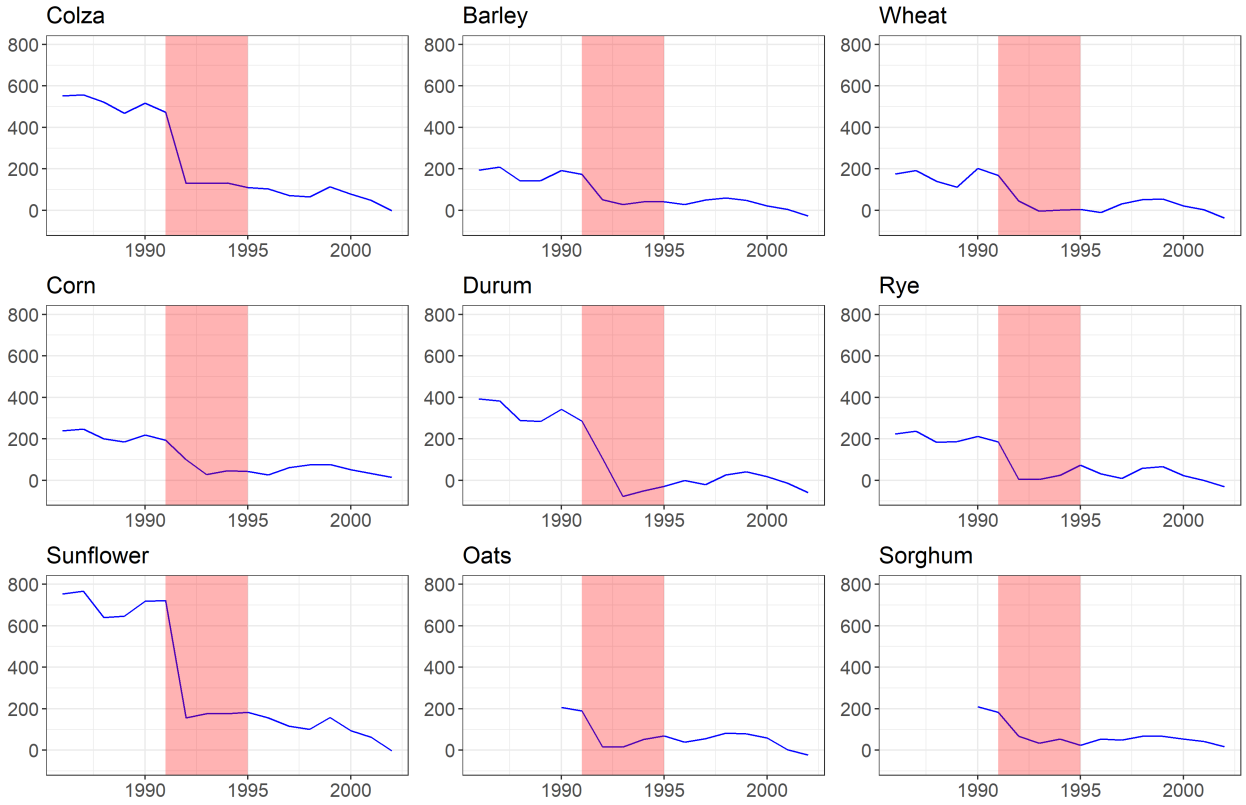



Figure A5: EU Subsidization per Unit of Output—Heterogeneity in Losses around the Reform

*Notes:* The figure plots crop-specific EU subsidies per unit of output over time, using data from the FADN, and digitized time series of EU intervention prices over time, as well as U.S. and Canadian farm-gate prices from FAOSTAT. The shaded area is for 1991-1995, going from the last year pre-reform to the last year of the reform. Data on oats and sorghum is only introduced in 1990 in the FADN.

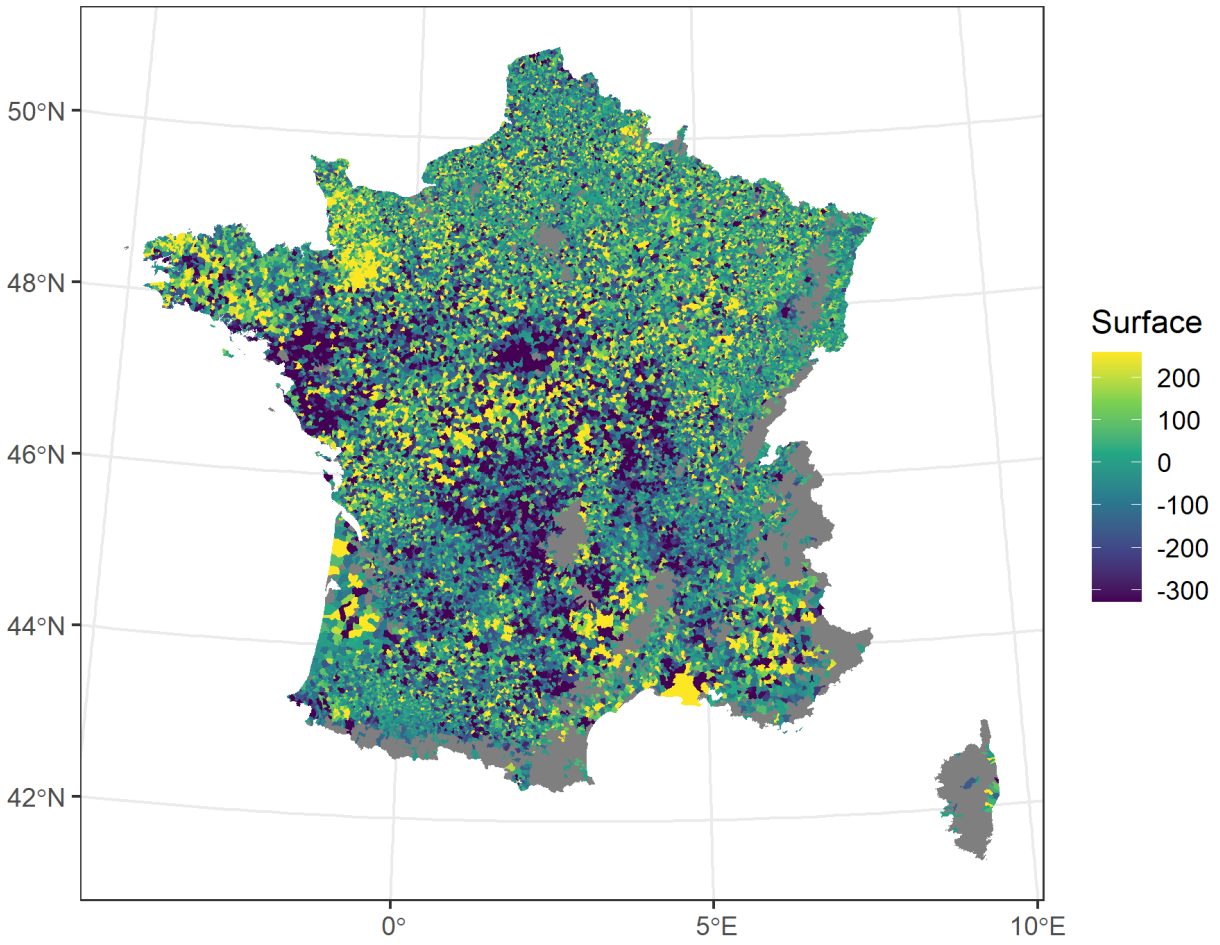



Figure A6: Evolution of Municipality-level Total Agricultural Area

*Notes: This map shows the evolution of within-municipality total agricultural area in hectares between 1988 and 2000. I use the data from the relevant waves of the French Agricultural Census. Decreases in agricultural area mainly happen in the center of France around the Massif Central, as well as in Vendée and possibly lower Brittany.*

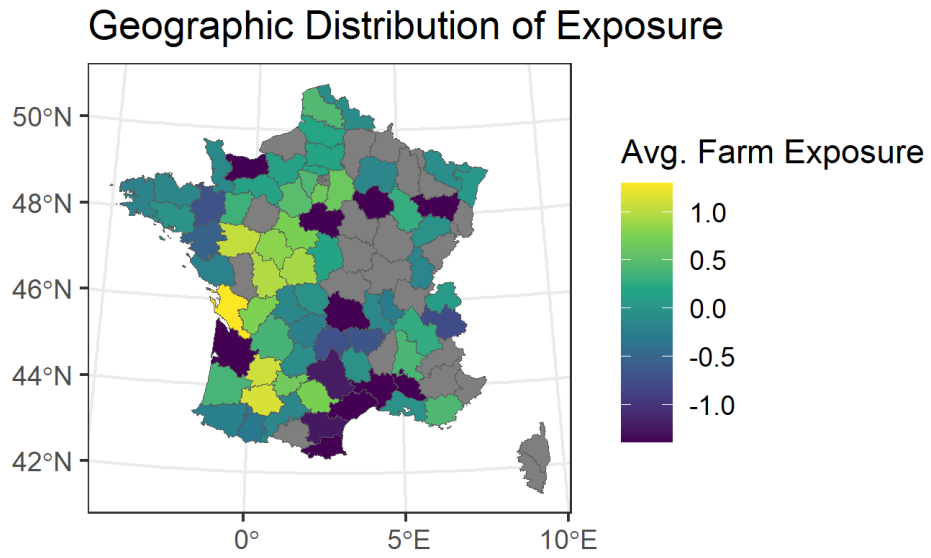



Figure A7: Department Weighted Averages of Farm-Level Exposure

*Notes: The figure shows the geographic distribution of the standardized measure of farm-level exposure. The exposure is computed using data from the FADN, and department-level averages are computed using the extrapolation weights provided there. I plot the geographic variation of the standardized exposure here.*

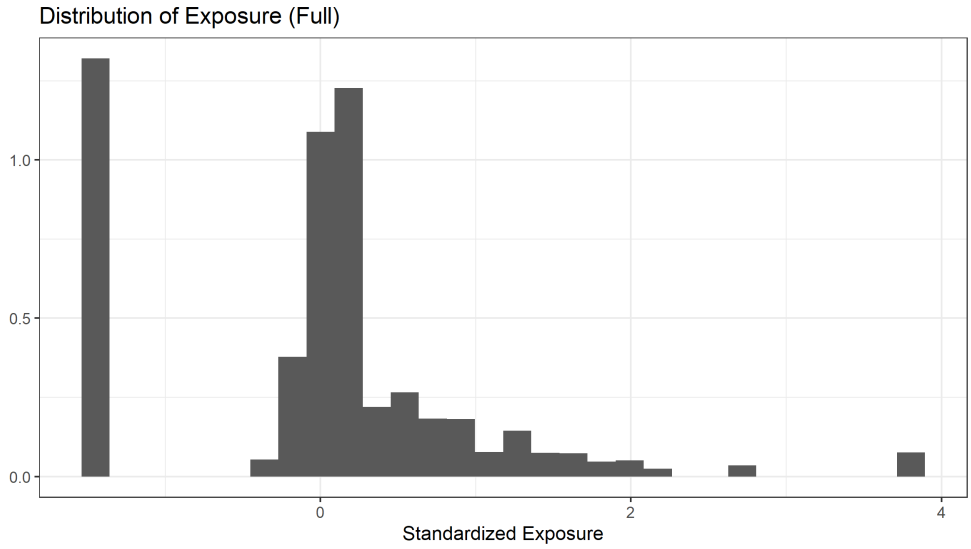



Figure A8: Distribution of Farm-Level Exposure

*Notes: The figure shows the distribution of the standardized measure of farm-level exposure. The exposure is computed using data from the FADN.*

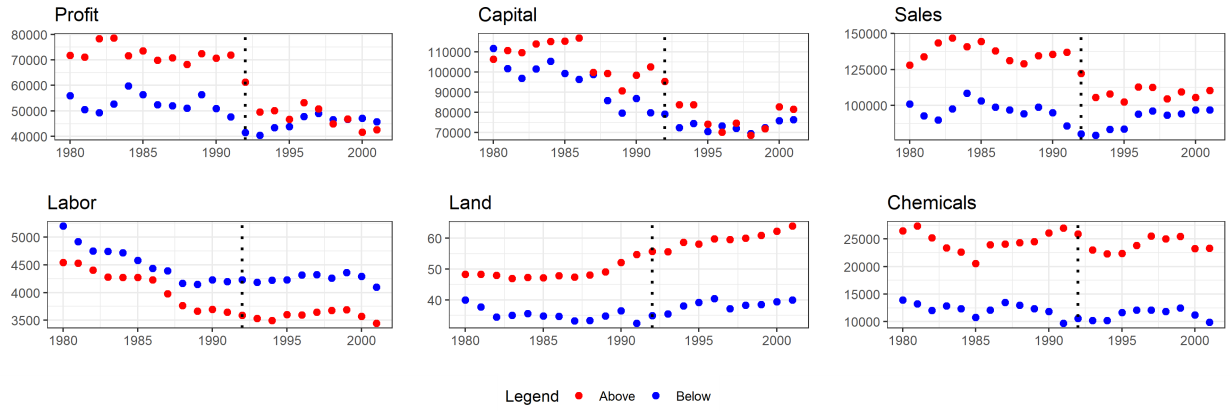



Figure A9: Balance in Levels and Trends

*Notes: These binscatters are computed using data from the FADN, and use sampling weights to compute bin-specific averages within each category. They compare farms above and below the median level of exposure. Profit corresponds to total farm profit, capital is the sum of total farm value in buildings and machinery, sales is total sales. All of these values are computed in 2020 euros. Labor is total labor in hours per year, land total utilized agricultural area in hectares, chemicals is the total deflated chemical expenditure.*

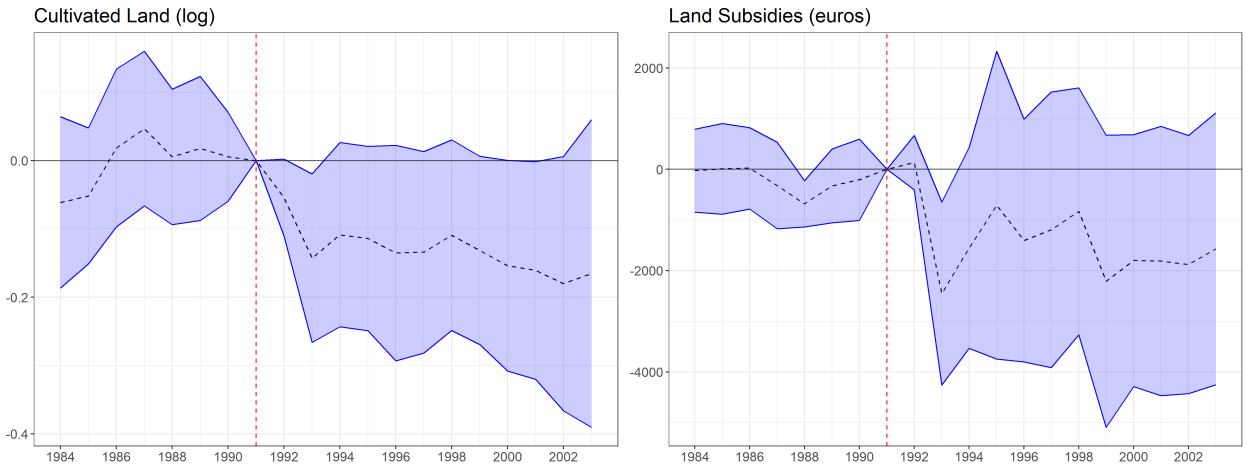
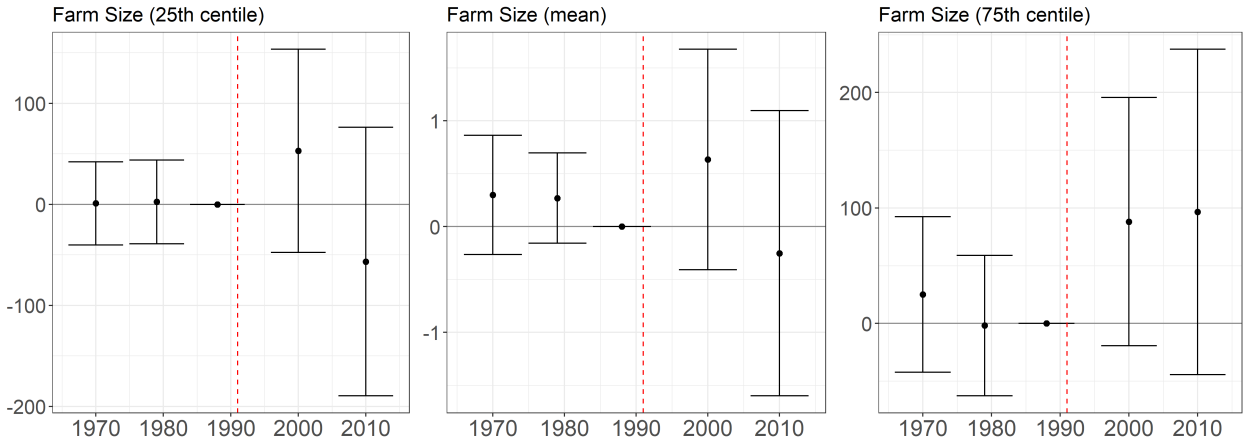



Figure A10: Exposure to the Reform—Farm-Level Event Study (Additional results)

*Notes:* The outcomes of the event studies are the following: the amount of land cultivated within the farm in log terms, and the amount of subsidies received from the land subsidy post-MacSharry reform in euros (this measure is 0 for all farms up to 1992, which is why it is measured in levels). All coefficients correspond to the year-specific coefficient associated to farm exposure, and give the effect of exposure in that given year. The vector of controls contains the following elements: farm 1991 capital stock, total farm profit, total labor used, total land use, their chemical use, the number of crops they grew and the evenness of their land allocation, their fertilizer-to-land and pesticides-to-land ratios, the share of their production which corresponds to oil crops (colza and sunflower), and the farms' 1983-1984 adoption trends in chemicals measured as the evolution in their chemical use. I add department-year and farm fixed effects, and cluster the standard errors one-way at the department level.




a) Within Municipality Farm Size

Figure A11: Additional Municipality Level Results

*Notes: This figure gives the results for the municipality-level event study. Outcomes are the following: the 25th, mean and 75th centile of the farm size distribution within a municipality, measured in ha cultivated land. The regression includes a series of controls set to their level in 1988 within the municipality, and interacted with time-varying coefficients, as well as municipality and department-by-year fixed effects. Standard errors are clustered at the department level.*

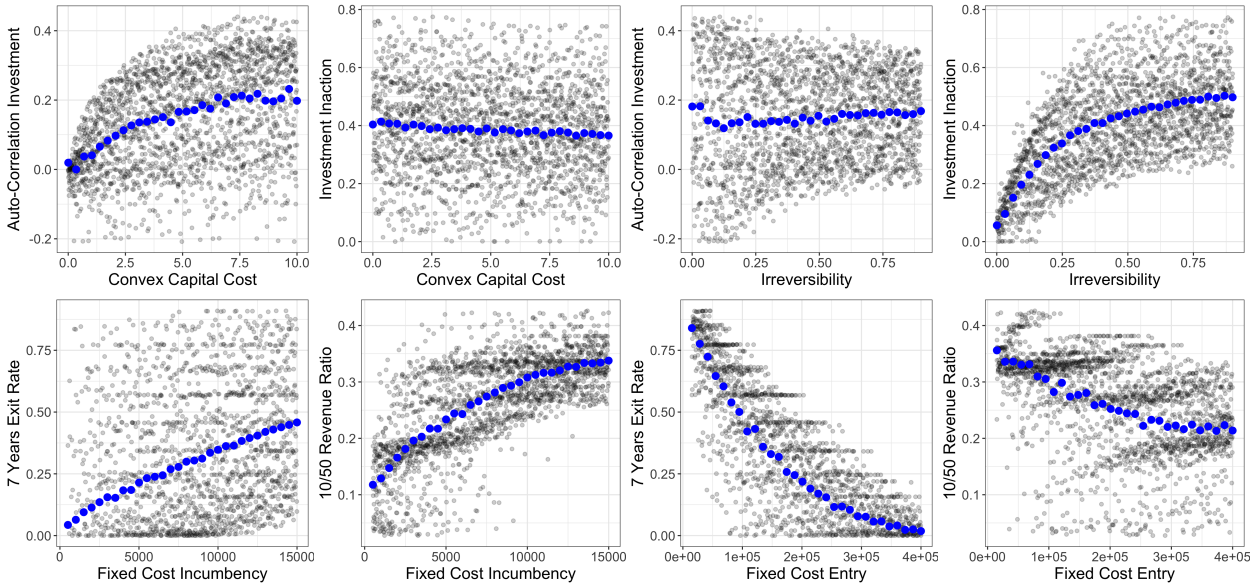



Figure A12: Relationship between Moments and Parameters (Simulated Method of Moments)

*Notes:* This figure shows a scatter plot of 2,048 runs of the full dynamic model where the four parameters (fixed costs of entry and incumbency, convex costs of capital adjustment and capital irreversibility) are drawn in a four-dimensional Sobol space, and then the associated equilibrium is computed and a simulated dataset is drawn from the equilibrium and run forward for T=200 periods. This drawing and simulation procedure is bootstrapped B=100 per parameter draw, and simulated moments are computed as the average over the bootstraps. The figure also shows the associated binscatter in dark blue, showing the relationship between parameters and the moments that identify them in the simulated method of moments. For legibility, scatter plots are shown for a sample trimmed for the top and bottom 1 percent of the moment values, while binscatters are computed on the entire sample. Parameters and simulated moments pairs which are not shown above do not share a substantial relation.

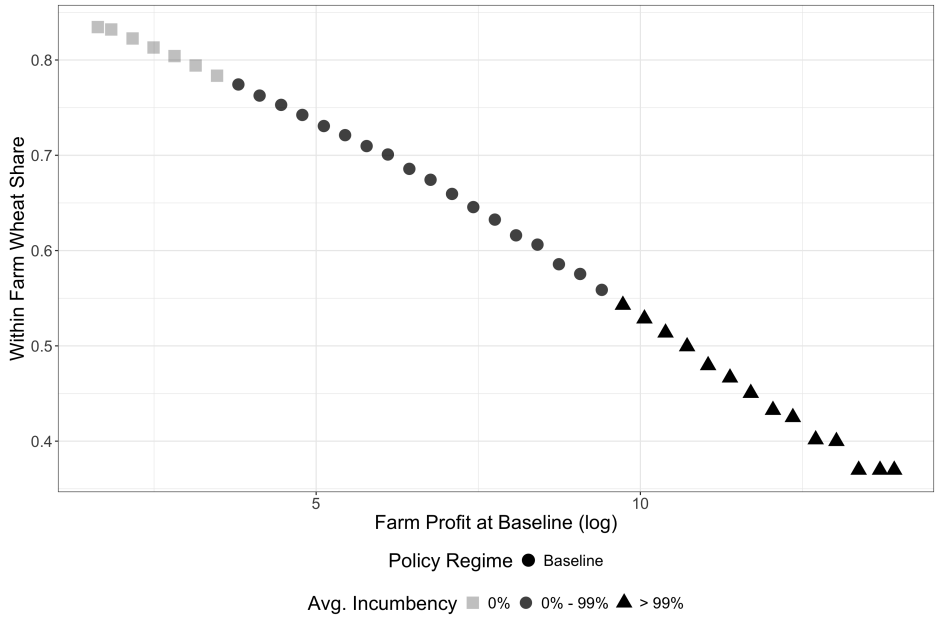



Figure A13: Systematic Variation in Crop Mix with Farm Profit

*Notes:* This figure plots on the x-axis a farm's total profit in the baseline no-policy equilibrium. This variable serves as a one-dimensional proxy for a farm's type (a combination of TFP, chemical-use efficiency and capital stock), and allows one to track farm behavior across equilibria. On the y-axis, the figure plots a farm's share of land allocated to wheat. Finally, changes in incumbency behavior are highlighted by changes in the shading and shapes of the curves: full triangles indicate a farm type proxy for which all farms are incumbents, circles indicate types for which at least some farms are incumbents, and squares types for which no farms remain incumbents.

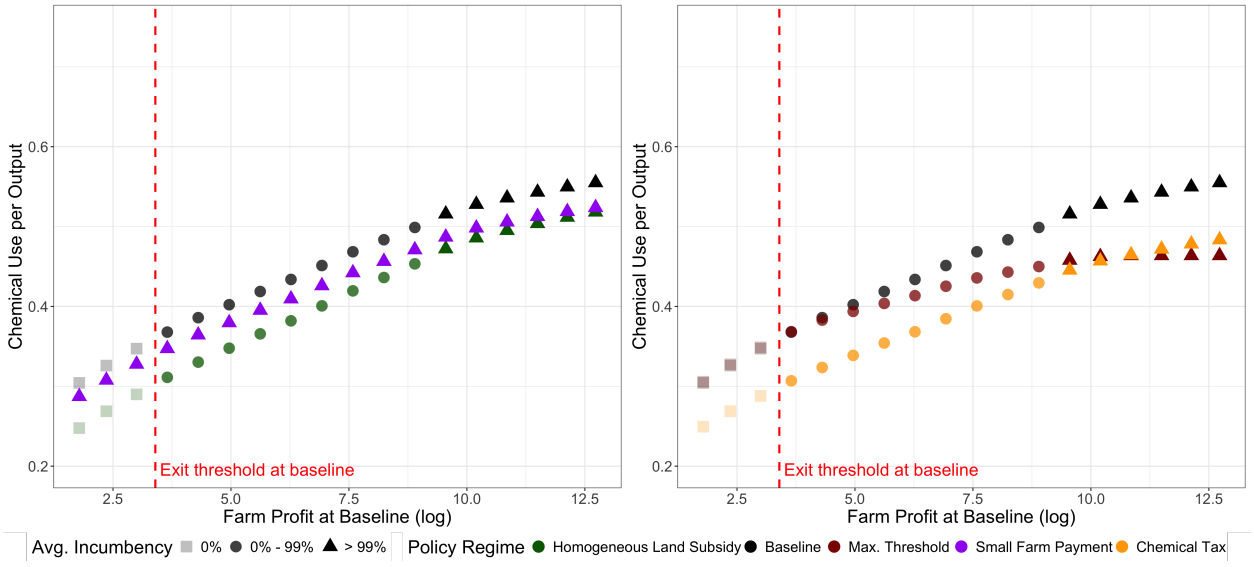



Figure A14: Intensive & Extensive Effects of Agricultural Policies (Wheat)

*Notes:* This figure plots on the x-axis a farm's total profit in the baseline no-policy equilibrium. This variable serves as a one-dimensional proxy for a farm's type (a combination of TFP, chemical-use efficiency and capital stock), and allows to track farm behavior across equilibria. On the y-axis, the figure plots a farm's chemical-use per unit of output for wheat production, under each relevant equilibrium. Finally, changes in incumbency behavior are highlighted by changes in the shading and shapes of the curves: full triangles indicate a farm type proxy for which all farms are incumbents, circles indicate types for which at least some farms are incumbents, and squares types for which no farms remain incumbents.

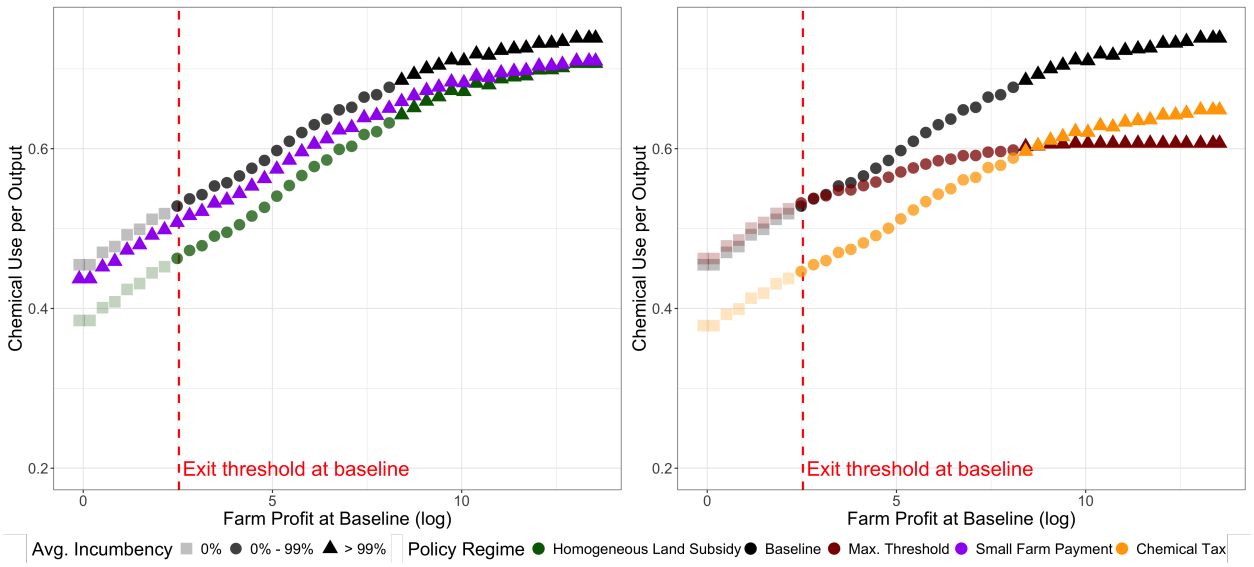



Figure A15: Intensive & Extensive Effects of Agricultural Policies (Other Aggregated Crop)

*Notes:* This figure plots on the x-axis a farm's total profit in the baseline no-policy equilibrium. This variable serves as a one-dimensional proxy for a farm's type (a combination of TFP, chemical-use efficiency and capital stock), and allows to track farm behavior across equilibria. On the y-axis, the figure plots a farm's chemical-use per unit of output for the second aggregated crop, under each relevant equilibrium. Finally, changes in incumbency behavior are highlighted by changes in the shading and shapes of the curves: full triangles indicate a farm type proxy for which all farms are incumbents, circles indicate types for which at least some farms are incumbents, and squares types for which no farms remain incumbents.

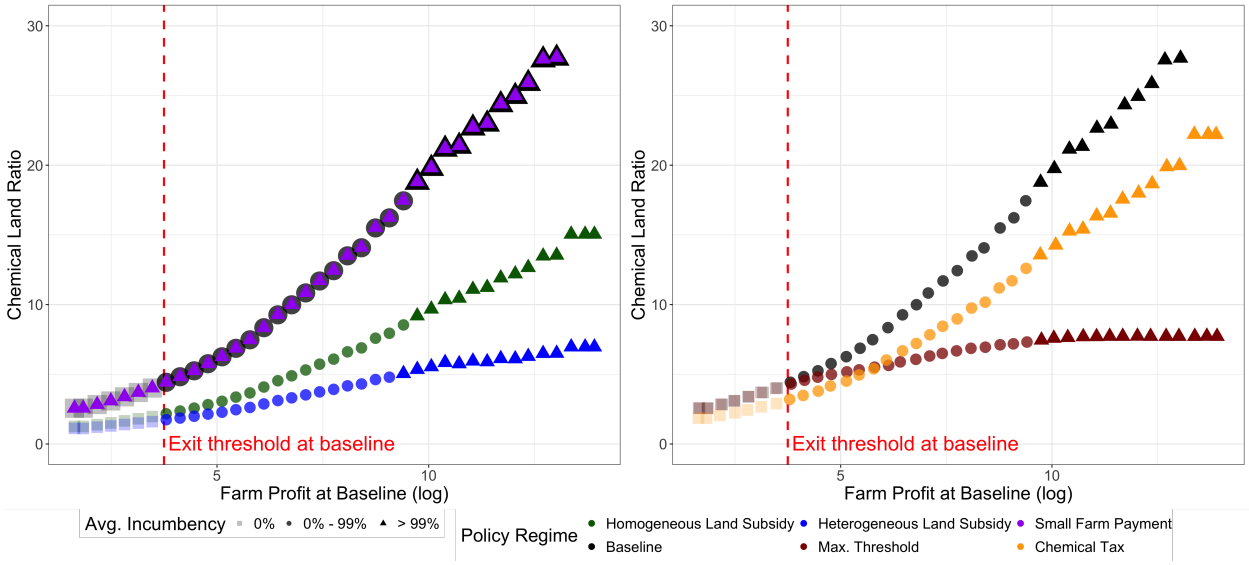



Figure A16: Intensive & Extensive Effects of Agricultural Policies (chemical use per hectare)

*Notes:* This figure plots on the x-axis a farm's total profit in the baseline no-policy equilibrium. This variable serves as a one-dimensional proxy for a farm's type (a combination of TFP, chemical-use efficiency and capital stock), and allows to track farm behavior across equilibria. On the y-axis, the figure plots a farm's farm-level average chemical-use per unit of land (hectares are used as units) under each relevant equilibrium. Finally, changes in incumbency behavior are highlighted by changes in the shading and shapes of the curves: full triangles indicate a farm type proxy for which all farms are incumbents, circles indicate types for which at least some farms are incumbents, and squares types for which no farms remain incumbents.

## A.2 Tables

Table A1: FADN Descriptive Statistics

| Statistic | N | Mean | St. Dev. |
|---|---|---|---|
| Output Volume—Sum (t) | 308,984 | 179 | 303 |
| Output Volume—Wheat (t) | 191,357 | 149 | 204 |
| Output Volume—Corn (t) | 95,925 | 116 | 205 |
| Output Volume—Sunflower (t) | 36,263 | 31.0 | 33.9 |
| Output Price Index (€/t) | 169,219 | 213.3 | 94.8 |
| Farm Surface (ha) | 308,982 | 71.0 | 67.6 |
| Capital (€) | 308,984 | 128,136 | 167,984 |
| Total Labor (hours) | 308,982 | 4,406 | 3,739 |
| Profit (€) | 273,777 | 78,133 | 79,688 |
| Phytosanitary (expenses in €) | 268,893 | 8,071 | 11,066 |
| Fertilizer (expenses in €) | 268,893 | 16,040 | 17,643 |

*Notes. This table shows descriptive statistics from the FADN dataset in panel format for years 1980-2015. When expressed, values in euros correspond to 2020 euros. These statistics are shown for the entire FADN sample for my period of analysis, which is further restricted for the production function estimation, as discussed in the production function estimation section.*

Table A2: Crop Mixes within French Farms

| Year | Only Wheat | Non-Wheat Only | Wheat + | Other + | Only Barley | Only Corn | Only Oats | Only Rye | Only Sunflower | Only Colza |
|---|---|---|---|---|---|---|---|---|---|---|
| 1979 | 0.041 | 0.19 | 0.64 | 0.14 | 0.048 | 0.11 | 0.011 | 0.0071 | 0.00020 | 0.00040 |
| 1988 | 0.094 | 0.18 | 0.64 | 0.092 | 0.054 | 0.085 | 0.014 | 0.011 | 0.0058 | 0.0023 |
| 2000 | 0.13 | 0.19 | 0.62 | 0.066 | 0.045 | 0.096 | 0.013 | 0.0051 | 0.0090 | 0.0022 |
| 2010 | 0.14 | 0.17 | 0.63 | 0.063 | 0.046 | 0.079 | 0.0078 | 0.0057 | 0.0078 | 0.0042 |

*Notes. The data is computed using the full count French Census of Agriculture for years 1979, 1988, 2000 and 2010.*

Table A3: Farm-Level Shift-Share Results

| Dependent Variables: | Output Price (log) | Sales (log) | Profit (log) | Chemical Expenditure (log) |
|---|---|---|---|---|
| Model: | (1) | (2) | (3) | (4) |
| *Variables* | | | | |
| $Exposure_j \times 1984$ | -0.2278 | -0.0485 | -0.1200 | -0.0504 |
| | (0.2068) | (0.1203) | (0.1554) | (0.1166) |
| $Exposure_j \times 1985$ | -0.0038 | -0.0133 | -0.0111 | -0.0158 |
| | (0.2447) | (0.0897) | (0.1703) | (0.1347) |
| $Exposure_j \times 1986$ | 0.5301 | -0.0081 | 0.0971 | -0.0374 |
| | (0.3720) | (0.1031) | (0.1238) | (0.1035) |
| $Exposure_j \times 1987$ | -0.0639 | 0.0255 | 0.0704 | -0.1787 |
| | (0.1798) | (0.0488) | (0.1497) | (0.1182) |
| $Exposure_j \times 1988$ | -0.0692 | 0.0330 | 0.2465 | 0.0476 |
| | (0.1596) | (0.0719) | (0.1669) | (0.0975) |
| $Exposure_j \times 1989$ | 0.1564 | 0.0228 | 0.1745 | -0.0332 |
| | (0.1854) | (0.0828) | (0.2214) | (0.0697) |
| $Exposure_j \times 1990$ | 0.0513 | -0.1275** | -0.1554 | 0.1067 |
| | (0.1226) | (0.0552) | (0.1702) | (0.0642) |
| $Exposure_j \times 1992$ | -0.2034* | -0.1370*** | -0.3424** | -0.0828 |
| | (0.1131) | (0.0478) | (0.1411) | (0.0820) |
| $Exposure_j \times 1993$ | -0.2790 | -0.1985* | -0.1827* | -0.1593 |
| | (0.1736) | (0.1023) | (0.1087) | (0.1269) |
| $Exposure_j \times 1994$ | -0.2486** | -0.1279 | -0.1053 | -0.1900 |
| | (0.1097) | (0.1156) | (0.2266) | (0.1389) |
| $Exposure_j \times 1995$ | -0.3789** | -0.2209* | -0.4566*** | -0.1974 |
| | (0.1462) | (0.1269) | (0.1545) | (0.1442) |
| $Exposure_j \times 1996$ | -0.6239*** | -0.2295 | -0.2339 | -0.2456 |
| | (0.1588) | (0.1520) | (0.1684) | (0.1507) |
| $Exposure_j \times 1997$ | -0.3100** | -0.2322* | -0.0670 | -0.1150 |
| | (0.1331) | (0.1385) | (0.1638) | (0.2252) |
| $Exposure_j \times 1998$ | -0.4480** | -0.1182 | -0.2958** | -0.2740** |
| | (0.1771) | (0.1559) | (0.1262) | (0.1268) |
| $Exposure_j \times 1999$ | -0.3481*** | -0.3201** | -0.1974 | -0.4082*** |
| | (0.0885) | (0.1226) | (0.1973) | (0.1192) |
| $Exposure_j \times 2000$ | -0.5517** | -0.2905** | -0.3045** | -0.2764 |
| | (0.2088) | (0.1335) | (0.1233) | (0.1679) |
| $Exposure_j \times 2001$ | -0.2311 | -0.3850*** | -0.5775*** | -0.3637* |
| | (0.1778) | (0.1440) | (0.1736) | (0.1936) |
| $Exposure_j \times 2002$ | -0.2828 | -0.3332** | -0.4189** | -0.2631 |
| | (0.2089) | (0.1420) | (0.1718) | (0.2085) |
| $Exposure_j \times 2003$ | -0.0658 | -0.4303*** | -0.5509*** | -0.3416*** |
| | (0.1506) | (0.1556) | (0.1531) | (0.1262) |
| *Fixed-effects* | | | | |
| Farm | Yes | Yes | Yes | Yes |
| Department-Year | Yes | Yes | Yes | Yes |
| *Fit statistics* | | | | |
| Observations | 1,910 | 2,978 | 2,945 | 2,978 |
| $R^2$ | 0.96634 | 0.95419 | 0.89874 | 0.94800 |

*Clustered (Department) standard-errors in parentheses*
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

*Notes. The event study outcomes are the following: a farm-level output price index (average price across row crops, using relative areas as weights), total sales, total farm profit and the farms' deflated chemical bill. All coefficients correspond to the year-specific coefficient associated to farm exposure, and give the effect of exposure in that given year relative to the effect in 1991. The vector of controls contains the following elements: farm 1991 capital stock, total farm profit, total labor used, total land use, their chemical use, the number of crops they grew and the evenness of their land allocation, their fertilizer-to-land and pesticides-to-land ratios, the share of their production which corresponds to oil crops (colza and sunflower), and the farms' 1983-1984 adoption trends in chemicals measured as the evolution in their chemical use. I add farm and department-year fixed effects, and cluster the standard errors one-way at the department level. The figure for these results is Figure 2.*

Table A4: Municipality-Level Shift-Share Results

| Dependent Variables: | Municipality Farm Count | Farm Crop Evenness | Municipality Crop Evenness |
|---|---|---|---|
| Model: | (1) | (2) | (3) |
| *Variables* | | | |
| $Exposure_k \times 1970$ | -0.0501 | -0.0051 | -0.0005 |
| | (0.5820) | (0.0033) | (0.0041) |
| $Exposure_k \times 1979$ | 0.2912 | -0.0040 | 0.0007 |
| | (0.3815) | (0.0030) | (0.0046) |
| $Exposure_k \times 2000$ | -0.7364** | 0.0112*** | 0.0049 |
| | (0.3016) | (0.0021) | (0.0032) |
| $Exposure_k \times 2010$ | -1.139*** | 0.0217*** | 0.0156*** |
| | (0.4017) | (0.0028) | (0.0036) |
| *Fixed-effects* | | | |
| Municipality | Yes | Yes | Yes |
| Department-Year | Yes | Yes | Yes |
| *Fit statistics* | | | |
| Observations | 147,157 | 147,157 | 147,157 |
| $R^2$ | 0.953 | 0.845 | 0.801 |

*Clustered (Department) standard-errors in parentheses*
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

*Notes. This table gives the results for the municipality-level event study. Outcomes are specified in levels and correspond to: the number of farms operating in the grain market within the municipality and the average farm-level index for the evenness of the distribution of land across crops. The regression includes a series of controls set to their level in 1988 within the municipality, and interacted with time-varying coefficients, as well as department-by-year and municipality fixed effects. Standard errors are clustered at the department level. Municipalities are weighted by their share of overall French agricultural land in 1988. The figure for these results is Figure 2.*

Table A5: County-Level Shift-Share Results: Algal Blooms

| Dependent Variable: | Algal Bloom (log) |
|---|---|
| Model: | (1) |
| *Variables* | |
| $Exposure_c \times 1986$ | -0.0458 |
| | (0.0322) |
| $Exposure_c \times 1987$ | -0.0700** |
| | (0.0265) |
| $Exposure_c \times 1988$ | -0.0293 |
| | (0.0301) |
| $Exposure_c \times 1989$ | 0.0239 |
| | (0.0323) |
| $Exposure_c \times 1990$ | -0.0224 |
| | (0.0407) |
| $Exposure_c \times 1992$ | -0.0717* |
| | (0.0408) |
| $Exposure_c \times 1993$ | -0.0454 |
| | (0.0277) |
| $Exposure_c \times 1994$ | -0.0467 |
| | (0.0299) |
| $Exposure_c \times 1995$ | -0.0440* |
| | (0.0253) |
| $Exposure_c \times 1996$ | -0.0687** |
| | (0.0317) |
| $Exposure_c \times 1997$ | -0.0905*** |
| | (0.0268) |
| $Exposure_c \times 1998$ | -0.0305 |
| | (0.0297) |
| $Exposure_c \times 1999$ | -0.1068*** |
| | (0.0286) |
| $Exposure_c \times 2000$ | -0.1662*** |
| | (0.0377) |
| $Exposure_c \times 2001$ | -0.0975*** |
| | (0.0278) |
| *Fixed-effects* | |
| County | Yes |
| Department-Year | Yes |
| *Fit statistics* | |
| Observations | 25,764 |
| $R^2$ | 0.88 |

*Clustered (Department) standard-errors in parentheses*
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

*Notes. This table gives the results for the county-level event study. The outcome is a Landsat-5 based index of algal bloom intensity on the within-county water bodies. Algal blooms are generally caused by the over-fertilization of agricultural land. The regression includes a series of controls set to their level in 1988 within the county (last year of the Census prior to the reform), and interacted with time-varying coefficients, as well as county and department-by-year fixed effects. Standard errors are clustered at the department level. The figure for these results is Figure 2.*

Table A6: Demand Estimation

| Dependent Variables: | Log Quantity (Wheat) | Log Quantity (Other) | Log Quantity (Wheat) | Log Quantity (Other) |
|---|---|---|---|---|
| Model: | (1) | (2) | (3) | (4) |
| *Variables* | | | | |
| Constant | 18.62*** | 18.71*** | 18.34*** | 17.73*** |
| | (0.4979) | (0.5629) | (0.5438) | (0.6642) |
| Log Price (wheat) | -0.3326*** | | -0.2843*** | |
| | (0.0861) | | (0.0961) | |
| Log Price (other) | | -0.3620*** | | -0.1964* |
| | | (0.0952) | | (0.1157) |
| Time Trend | 0.0091*** | 0.0106*** | 0.0103*** | 0.0144*** |
| | (0.0024) | (0.0028) | (0.0022) | (0.0027) |
| *Fit statistics* | | | | |
| Observations | 36 | 36 | 35 | 35 |
| $R^2$ | 0.819 | 0.734 | 0.831 | 0.769 |
| F-test (1st stage), Log Price (wheat) | 10.789 | | 8.2290 | |
| F-test (1st stage), Log Price (other) | | 13.525 | | 10.534 |

*Newey-West (L=1) standard-errors in parentheses*

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

*Notes. I obtain estimates of total quantity sold and average prices for France for the two crop categories using the FADN and its sampling weights. I then run regressions of quantities on prices, instrumenting the prices in turn using: 1) realized weather (realized GDD and realized HDD, as well as the yearly average in France of department-level deviations of GDD and HDD realizations from their thirty year GDD and HDD average), 2) a combination of realized weather and the land subsidy (the average deviations in GDD and HDD, and these averages deviations interacted with GDD and HDD realizations, and finally the yearly EU land subsidy applied in France). These are used as supply shifters orthogonal to demand. The second set of regressions does not include national realizations in weather, as average weather might be correlated across the EU and through this correlated with EU-wide demand shocks. I include a linear time trend in all regressions, and use the Newey-West correction for serial auto-correlation. When computing the intercept for the demand curves, I compute the constant in year 2015, i.e. the sum of the constant in this regression, and the time trend multiplied by the number of years from the start of the sample to 2015.*

Table A7: Additional Model Parameters

| Coefficient | Parameter | Value |
|---|---|---|
| **Other Estimated Parameters** | | |
| Competence Ladder | $\lambda$ | 0.82 |
| Mean TFP | $\mu$ | 2.60 |
| Auto-Correlation TFP | $\rho_h$ | 0.74 |
| Shape TFP Pareto Distribution | $\sigma_h$ | 0.26 |
| Location TFP Pareto Distribution | $\delta_h$ | 0.037 |
| Auto-Correlation Non-Hicksian shock | $\rho_{ch}$ | 0.86 |
| Shape Non-Hicksian Pareto Distribution | $\sigma_{ch}$ | 0.26 |
| Location Non-Hicksian Pareto Distribution | $\delta_{ch}$ | 0.011 |
| **Calibrated Parameters** | | |
| Time Preference | $\frac{1}{1-r}$ | 0.9615 |
| Capital Depreciation | $\delta_k$ | 0.08 |
| Price Capital (k €) | $p_k$ | 100 |
| Price Land | $p_s$ | 254 |
| Price Chemicals | $p_c$ | 100 |
| Price Labor | $p_l$ | 1078 |

*Notes: I use the recovered productivity shocks to estimate these other estimated parameters. Specifically, recovered TFP (which is demeaned) and chemical-use efficiency shocks from the production function are used to compute the following parameters: 1) the competence ladder parameter of specialization $\lambda$, which corresponds to the average ratio of the lowest crop TFP to highest crop TFP within a farm-year, across farms and years, 2) the Pareto distribution parameters (shape and location) are recovered across farms and years, 3) the auto-correlation in both shocks are computed using within-farm variation. For the additional calibrated parameters, the rate of capital depreciation is set to 8%. I do not observe chemical use quantities in the FADN, but rather the overall chemical expenditures. As such, I just deflate chemical prices by 100, the baseline value of the price index in 2015. The price of land corresponds to the rental price for 1ha of prime agricultural land in Eure-et-Loir, the main cereal producing department of France.[56] The price of labor is set to the wage of one worker for one month of labor. The average TFP is attributed to all farms in the dynamic model, and in that model the productivity shocks are modeled as AR(1) processes using the auto-correlation recovered here, with an evolution following a discretized Pareto distribution.*

[56] See "Arrêté concernant les valeurs locatives des terres, des bâtiments d'exploitation et des maisons d'habitation pour la campagne 2025/2026", Direction Départementale des Territoires d'Eure-et-Loir.

Table A8: Decomposition of Counterfactual Policy Changes in Market Average Farm-Level Chemical-use per Unit of Output (values expressed in levels, and then percent relative to baseline in parentheses)

| Policy Regime | Within-Farm, Within-Crop | Within-Farm, Across-Crop | Across-Farm | Selection (Entry) | Selection (Exit) | Total Change |
|---|---|---|---|---|---|---|
| Homogeneous Land Subsidy | -0.0505 (-8.4%) | 0.0010 (0.2%) | 0.0032 (0.5%) | 0.0000 (0.0%) | 0.0000 (0.0%) | -0.0462 (-7.7%) |
| Heterogeneous Land Subsidy | -0.0532 (-8.8%) | -0.0015 (-0.2%) | -0.0134 (-2.2%) | 0.0000 (0.0%) | 0.0000 (0.0%) | -0.0681 (-11.3%) |
| Small Farms Payment | -0.0246 (-4.1%) | 0.0004 (0.1%) | 0.0017 (0.3%) | -0.0151 (-2.5%) | 0.0000 (0.0%) | -0.0375 (-6.2%) |
| Chemical Tax | -0.0731 (-12.2%) | 0.0004 (0.1%) | -0.0081 (-1.3%) | 0.0000 (0.0%) | 0.0000 (0.0%) | -0.0808 (-13.4%) |
| Max. Threshold | -0.0361 (-6.0%) | 0.0013 (0.2%) | -0.0365 (-6.1%) | 0.0000 (0.0%) | 0.0000 (0.0%) | -0.0713 (-11.9%) |

*Notes: The table relies on a total decomposition of the change in the market average chemical use per unit of output at the farm level, when moving from a baseline no-policy equilibrium to each of the counterfactual policy regimes. The formula for the decomposition follows the lines of the decomposition shown in Appendix E. The modeled policies are as follows: a homogeneous 33% land subsidy, a heterogeneous land subsidy of 66% subsidizing only the aggregated crop, a 10k euros lump-sum payment to farms with profits under 3k euros, a 20% chemical tax, and a compulsory maximum chemicals-to-land ratio imposed on all farms, at the crop level, at the median of the chemical-to-land ratio observed in wheat production among incumbents in the baseline equilibrium. Channels are as follows: the within-farm within-crop channel corresponds to changes in input mix composition and scale of production, the within-farm across-crop channel corresponds to changes in the allocation of production across crops within the farm, the across-farms channel captures reallocation of production across farms that remain incumbents across equilibria, and finally the last two channels capture the role of entry and exit.*

# B Data Description

## B.1 French Agricultural Surveys & Censuses

**Structure des Exploitations Agricoles (ESEA):** I rely on the 2016 and 2023 ESEA to get an estimate of the exit rate of farms in the French grain market. The ESEA survey is a repeated cross-section run in between Census waves, containing about 60,000 farms in sample. The ESEA is more detailed than the Census, and is designed to track the overall French farm population. Importantly, the later waves of the ESEA contain a variable indicating when the current manager of the farm started operating the farm. Equating manager take-over with firm take-over, which is a common feature of the French grain market where a farmer will manage the same farm for their whole career until the farm is sold at retirement or passed to a family member, I can use this variable as proxy for when the farm started its activity. Using survey weights, I can then track the evolution of the farm population whose manager started their activity between 1982 and 2013, a sufficient number of years prior to the 2016 ESEA to make sure these younger farms could be sampled. I interpret this thirty year period as a period shorter than the average tenure of a French farmer, and long enough to prevent the random variation in the sampling of farms from driving the exit rate estimate. I use the change in these weighted counts as a proxy for farm exit over the seven years that separate these two ESEA waves. To the best of my knowledge, this is the most precise proxy I can obtain of the exit rate of farms in the French grain market, given that the Census of Agriculture does not allow one to track farms across waves, and does not include any information about farm cohort. Similarly, the FADN does not allow for the measurement of exit rates, as FADN survey exit is different from farm exit.

## B.2 Intervention Prices

For the purpose of this paper, and to the best of my knowledge, I created the first database of agricultural commodities' EU intervention prices covering the end of the 1960s to the early 2000s when intervention was completely removed. Intervention prices denominated in ECUs, euros or francs were gathered from the yearly and sometimes commodity-specific directives published by the European Union over that period of time. When prices were denominated in ECUs, I used the CAP-specific exchange rate which was then used to translate ECUs to each member state's currency. I similarly gathered data on commodity-specific land subsidies, which I later combine with farm-specific land subsidies as recorded in the FADN.

## B.3 Input Prices: Indices and Price Surveys

Input prices are drawn from the IPAMPA agricultural input price indices, and from the Enquête sur l'observation des Prix des Consommations Intermédiaires nécessaires aux Agricul-

teurs (EPCIA), the price survey used to develop the IPAMPA. I use the IPAMPA indices for pesticides and fertilizers interpolated by INSEE back to 1980.

The EPCIA has been run since 1996, and provides price data for pesticides and fertilizers. Its sample is based on a 1995 survey, which helped identify a sample of products representing 50% of the total sales within their product category. The sampling of establishments selling these products was done based on the establishments' respective market share within that product category. Finally, within a product and establishment, the series chosen corresponded to the product's most frequent sales conditions. As such, series are good-, company- and mode-of-sale-specific. When series disappear, they are replaced with their closest equivalent. The EPCIA is made of 4165 series, sold by 250 companies. The EPCIA is aggregated into national and regional price indices using 1990 plutocratic weights, and the following index of category-specific month-to-month price evolution:

$$i_t = i_{t-1}\left(\prod_i^n \frac{P_{i,t}}{P_{i,t-1}}\right)^{\frac{1}{n}} \tag{8}$$

I also recover agricultural hourly wages from the continuous labor survey "Enquête Emploi" up to 2015, which is a survey fielded for a sampled set of households, each household being drawn once and then followed for the next six trimesters. I consider the hourly wage data provided by workers whose occupation is classified as that of an agricultural worker or farmer.

Finally, I use land prices taken from the Land Market Value survey (Valeur Venale des Terres), a yearly and department-level survey, which is fielded every year by the statistical services of the French departmental administration for agriculture and forestry. These are based on data provided by the public company in charge of land management (SAFER), which authorizes agricultural land purchases and consolidations when transactions surpass a given threshold. This data is then complemented by data provided by local notaries, and several local administrations. I use this data for 1980-2015. The data was digitized from scanned data catalogues for the first years of the series.

### B.4 Weather Data: Realizations and Forecasts

My realized climate data comes from the European Centre for Medium-Range Weather Forecasts (ECMWF)'s ERA5 reanalysis product. ERA5 gives hourly estimates of climate data, out of which I use precipitation and temperature (temperature 2m above the surface of the Earth). ERA5 combines observational weather data with model-based data into a

0.25*0.25 gridded dataset.

I extract that data at the French department level, cropping the grid with department shapes, and averaging the data using simple area weights. Using the time separability assumption common in the literature, I aggregate the hourly data into growing-season observations [57]: growing degree days (GDD) and heat degree days (HDD) for temperature[58], and total precipitation for rainfall. In order to match the forecast dataset, I only use 4 daily measurements of temperature to compute the GDDs and HDDs, specifically at midnight, 6h, 12h and 18h.

The forecast data is taken from ECMWF's SEAS5 seasonal forecasting system. Forecasts are produced on the first of each month for the following 5,160 hours [59]. For temperature, they are produced at a 6h interval, and give an instantaneous prediction of temperature, while for rainfall, they give the accumulation of rainfall every 24h. As such, the rainfall forecasts for the second day of the month will be the following: a 24h ahead forecast, a 30 to 32 days ahead forecast produced on the first day of the previous month, and so on until the lead value exceeds 5,160 hours. Temperature forecasts work in a similar way, but are just produced with more granular time steps.

The main issue for my purpose is that given that forecasts are produced every first of the month, different days within a month will not be provided a forecast with the same lead. I would ideally like to build the forecast-equivalents of the growing-season aggregates for realized weather, for different lead times. For example, the rainfall forecast for the growing season, with a constant one-month lead throughout the season (or the equivalent of the farmer's knowledge about rainfall one month in advance throughout the growing season). I approximate this by bundling together forecasts produced one calendar month ago (the one month lead for the rest of the paper), produced two calendar months ago, up to five months ago. As such, the forecasts that I aggregate into growing season observations are not homogeneous in terms of lead value, but are the closest equivalent of it that I can get.

Similarly to the weather realization data, I extract the gridded data into department-level observations, using area weights.

### B.5 Landsat 5 Remote Sensing Data

The algal bloom index is based on Landsat 5 imagery. Landsat 5 was a low orbit satellite jointly managed by the U.S. Geological Survey and NASA, and ran between 1984 and 2013.

[57] I use an extensive definition of the agricultural growing season for France, running from October of the previous year, to July of the current one.

[58] Growing degree days are computed over the $[4°, 30°]$ degree interval, and heat degree days sum the realized temperature above 30°C.

[59] ECMWF provides an ensemble of 25 forecasts, which I average.

Its unusual longevity makes it a good source of data to build long time series. It had a repeat cycle of about 16 days, and was equipped among else with a Thematic Mapper (TM) and a Multi-Spectral Scanner.

My algal bloom index follows the methodology of Taylor and Heal (2023). For every year between 1986 and 2001, I filter Landsat data for images taken between June and August, which I crop for surface water and treat for clouds and cloud shadows. I then run the following function on each pixel:

$$Bloom = NIR - 1.03 * SWIR \tag{9}$$

NIR corresponds to the near infrared band of Landsat 5 (.77 to .90$\mu$m), and SWIR to the shortwave infrared band (1.55 to 1.75$\mu$m). In both cases, these come from atmospherically corrected surface reflectance data produced by the Landsat TM series. The index is then averaged across the selected months of the year, and averaged over surface water area at the county level.[60]

# C Robustness and Additional Results for the Reduced Form Analysis

I start this robustness section by showing the Rotemberg weights for the shift-share variable, following the formula provided by Goldsmith-Pinkham et al. (2020). The weights are crop-specific, and can be compared to both the crop-specific losses in subsidization from the reform, and the pre-reform variance in land shares across farms. These comparisons are useful to understand the source of the variation identifying the reform effects. I then compare them to the weights obtained without controls. This comparison highlights the role of controls in shifting the source of the variation used to identify the effect of farm reform exposure. I provide a more in-depth analysis of the conditional parallel-trends assumption driving the design. I show trends in the same set of variables as shown in Figure A9, but for farms with high or low shares for each crop used to build the shift-share.

The distribution of weights shown in Table A10 highlights the cross-farm heterogeneity in crop mix that drives the results, i.e. heterogeneity in shares of land allocated to sunflower, colza and durum wheat. Figure A9 provides balance tests at the farm level, Figure A19 at municipality level, and Appendix C.2 comparisons of farm level trends for farms with different crop mix compositions. These can be paired with the Rotemberg weights to gain confidence in my conditional exogeneity assumption, in the sense that farms with high and low pre-reform shares of the high weight crops had comparable trends prior to the reform. In this

[60] There are 2,054 counties in France, making them much smaller geographic entities than U.S. counties, and the smallest geographic entity above villages (communes).

section, I further address potential concerns about treatment heterogeneity by decomposing farm exposure into crop-specific variables in Appendix C.4. These establish that a high crop share in 1991 in any of high Rotemberg weight crop is also associated with similar effects on sales, profit, and chemicals as the ones identified by the aggregate exposure in the main specification. When computing the Rotemberg weights, I recover a few negative weights. The homogeneity of results when using the overall exposure variable, or the crop-per-crop exposures, imply that these negative weights should not bias the interpretation of the aggregate effect of farm exposure (Goldsmith-Pinkham et al. (2020)). I finally show the main results with a difference-in-difference framework, and using an alternative exposure variable which combines 1991 land and output shares, respectively to aggregate over pre-reform output subsidies and post-reform land use subsidies.

### C.1 Rotemberg Weights

I give the Rotemberg weights below—with and without the inclusion of the controls used in the farm-level regressions. This has the advantage of both showing what cross-crop variation is used for identification, and how this variation is influenced by the controls. Table A9 shows the Rotemberg weights obtained when including all the controls used in the main FADN event study specification. The first thing I note is that sunflower, colza and durum are the three crops with high positive weights, while other grains have small or negative weights. These three crops are also the high exposure crops, while other grains have lower and very homogeneous losses. Sunflower, colza and durum are also among the set of crops with highest land share variance. Only wheat has a comparable share variance. As such, comparing farms with high and low shares of sunflower, colza and durum wheat is a good proxy for the overall cross-farm heterogeneity in crop mix.

When using the exposure variable, I am making comparisons across farms that had a high share of their land allocated to sunflower/colza/durum, versus others who were more specialized in the remaining cereals. As such, worries about omitted variable biases should focus on the uncontrolled determinants of crop choice for these three specific crops. The balance tests for these three crops shown later are then of particular relevance to build trust in the design. The correlation matrix between weight values, shocks, and the variance of the shares in Table A9 shows that the exposure variable can be taken as a reasonable summary of the differential exposure to the reform across farms, coming from the heterogeneous effects of the reform across crops, and from heterogeneity across farms in crop mix composition.

Table A9: Rotemberg Weights: With Controls

| *Panel A* | Rotemberg Weights | | |
|---|---|---|---|
| Crop | Weight | Shock | Share Variance |
| Sunflower | 1.330 | 540 | 0.117 |
| Colza | 0.345 | 362 | 0.082 |
| Durum | 0.205 | 315 | 0.031 |
| Winter Barley | 0.038 | 133 | 0.015 |
| Sorghum | -0.008 | 160 | 0.024 |
| Rye | -0.031 | 113 | 0.006 |
| Oats | -0.063 | 121 | 0.010 |
| Spring Barley | -0.135 | 128 | 0.0001 |
| Wheat | -0.294 | 165 | 0.034 |
| Corn | -0.386 | 151 | 0.012 |
| *Panel B* | Correlation Matrix | | |
| Weights | 1 | 0.90 | 0.83 |
| Shocks | 0.90 | 1 | 0.95 |
| Variance Shares | 0.83 | 0.95 | 1 |

Notes. The weights are computed after residualizing exposure on the fixed effects and controls used in the farm-level regressions. These Rotemberg weights are obtained when using only land shares to measure farm exposure. The shock column indicates the value of the crop-specific shock, and the share variance corresponds to the variance observed in the FADN sample of the crop-specific share.

The second thing to note about the weights is that sorghum, rye, oats, spring barley, wheat and corn have negative weights. These negative weights could raise issues in the case of heterogeneous treatment effects. I investigate this possibility in Appendix C.4, by decomposing the shift-share variable into a series of crop-specific measures made of the farms' 1991 crop share interacted with the crop-specific shock. These results highlight that the effects associated with a high crop share for oats, spring barley, wheat and corn are of the same sign as the ones shown in the main event study. The only exception is the effect of a higher oats crop share on chemical use, which is non-significant and very close to zero.

Finally, I note a change in the ordering of weights when comparing weights obtained with and without controls: wheat plays a much larger role in instrument variation when no controls are included, and colza and durum play a much smaller role in the variation. The addition

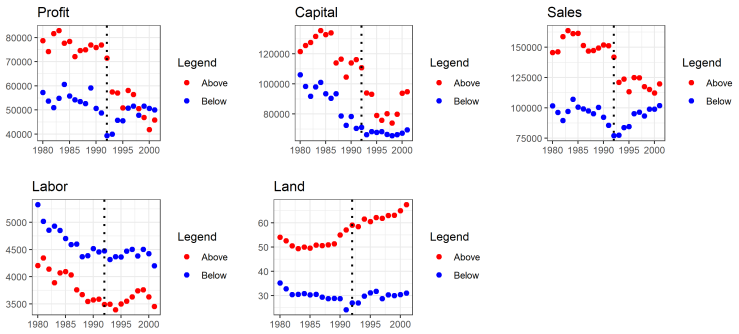
of controls shifts the variation in the instrument from being driven by the high-shock crop sunflower, and then wheat and corn which play a large role in the French grain market, to comparing farms that were growing high versus low shock crops.

Table A10: Rotemberg Weights: Without Controls

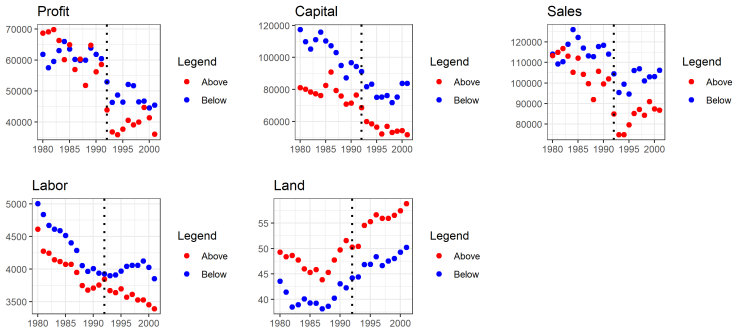
| Crop | Weight | Shock | Variance Land Share |
|---|---|---|---|
| Sunflower | 0.539 | 540 | 0.117 |
| Wheat | 0.18 | 165 | 0.034 |
| Durum | 0.088 | 315 | 0.031 |
| Corn | 0.076 | 151 | 0.012 |
| Colza | 0.065 | 362 | 0.082 |
| Spring Barley | 0.056 | 128 | 0.0001 |
| Winter Barley | 0.0047 | 133 | 0.015 |
| Sorghum | 0.0016 | 160 | 0.024 |
| Oats | 0.0015 | 121 | 0.010 |
| Rye | -0.0061 | 113 | 0.006 |

Notes. The weights are computed without residualizing exposure on the fixed effects and controls used in the farm-level regressions. These Rotemberg weights are obtained when using only land shares to measure farm exposure. The shock column indicates the value of the crop-specific shock, and the share variance corresponds to the variance observed in the FADN sample of the crop-specific share.

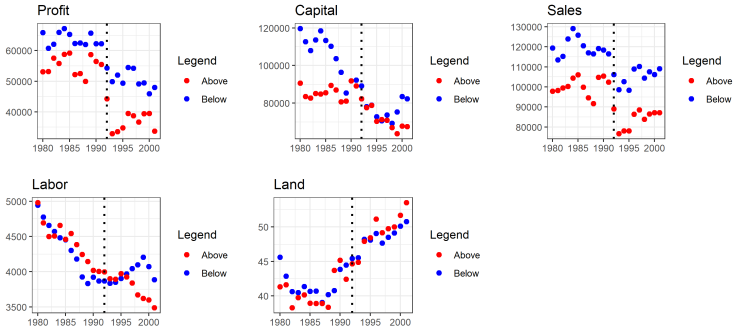
## C.2 Balance Tests: Across Crop Shares




((a)) Balance Tests - Wheat



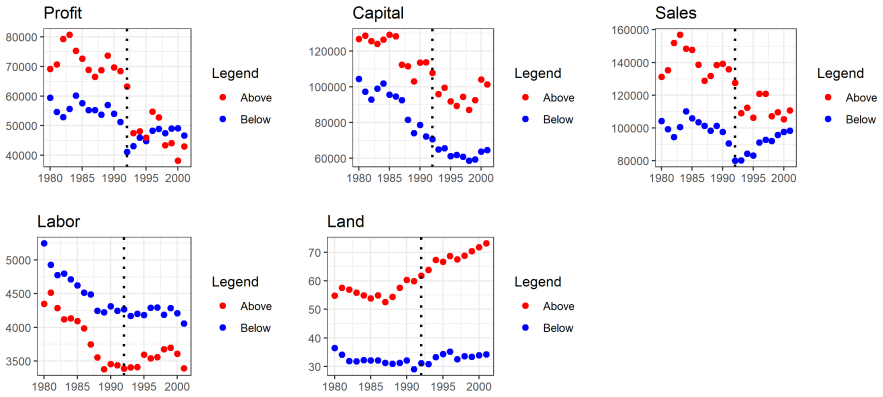

((b)) Balance Tests - Sunflower



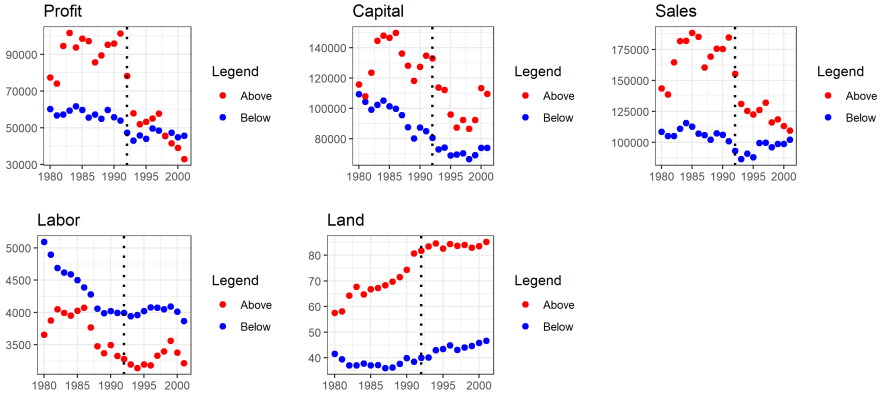

((c)) Balance Tests - Corn

Figure A17: Trends per Category

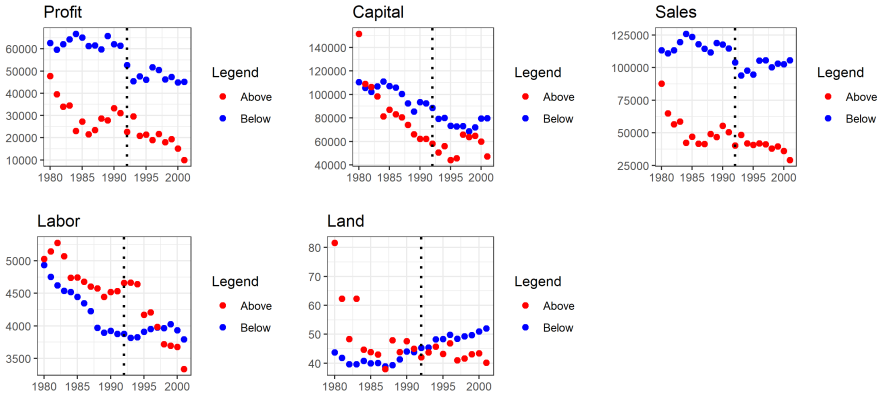
*Notes: The figures show the trends of relevant farm characteristics for the farms above and below the median in terms of their share of land allocated to the relevant crop in 1991.*




((a)) Balance Tests - Barley




((b)) Balance Tests - Colza




((c)) Balance Tests - Rye

Figure A18: Trends per Category (continued)

*Notes: The figures show the trends of relevant farm characteristics for the farms above and below the median in terms of their share of land allocated to the relevant crop in 1991.*

Next, I show the balance tests across farms with an above and below median share of wheat, corn, winter barley, rye, sunflower and colza. The main threat to identification lies with the presence of unobservables that are linked with heterogeneity in exposure, and which affect differential farm-level growth. The following elements stand out: across all crops, capital is the one variable that differs the most in trends across groups prior to the reform—making it a necessary control in the design. Profit, sales and labor show relatively comparable trends prior. Trends in land use also show some notable differences in the cases of rye and colza. While the differences for rye are quite significant, I also know from the previous Rotemberg weights that rye plays an almost absent role in the variation of the shift-share variable, and hence should not drive results too much. Differences in colza are potentially more important, again justifying the control for the share of production coming from oil crops pre-reform.

### C.3 Additional Farm-Level Results

It is useful to recast the event study design in a pure difference-in-difference design in order to recover a unique coefficient summarizing the effect of the reform. I can do so using a modified estimating regression of the following form:

$$Y_{jt} = \alpha_j + \eta_{d(j)t} + \beta Exp_j Post_t + X_j Post_t \Gamma + \varepsilon_{jt}. \tag{10}$$

I also run this design with an alternative construction of exposure, where price intervention shocks are averaged across crops using output-based weights, and land subsidy ones using land weights. In both cases, results match the ones obtained using the event study design.

Table A11: Difference-in-Difference Results

| Dependent Variables: | Output Price (log) | Sales (log) | Profit (log) | Chemical Expenditure (log) |
|---|---|---|---|---|
| Model: | (1) | (2) | (3) | (4) |
| *Variables* | | | | |
| $Exposure_j \times Post_t$ | -0.2380* | -0.2376** | -0.3117*** | -0.2473 |
| | (0.1334) | (0.1041) | (0.0996) | (0.1482) |
| Mean Outcome in Level | 224 | 109k | 50k | 17k |
| *Fixed-effects* | | | | |
| Farm | Yes | Yes | Yes | Yes |
| Department-Year | Yes | Yes | Yes | Yes |
| *Fit statistics* | | | | |
| Observations | 1,929 | 2,978 | 2,945 | 2,978 |
| $R^2$ | 0.94 | 0.94 | 0.88 | 0.93 |

*Clustered (Department) standard-errors in parentheses*

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes. Outcomes are the following: a farm-level output price index (average price across row crops, using relative areas as weights), total sales, profit, and the farms' deflated chemical bill. Coefficients correspond to the post-reform coefficient associated to farm exposure, and give the effect of exposure after the reform. The vector of controls contains the following elements: farm 1991 capital stock, total labor used, total land use, profit, their chemical use, the number of crops they grew and the evenness of their land allocation, their fertilizer-to-land and pesticides-to-land ratios, the share of their production which corresponds to oil crops (colza and sunflower), and the farms' 1983-1984 adoption trends in chemicals measured as the evolution in their chemical use. I add farm and department-year fixed effects, and cluster the standard errors at the department level.

Table A12: Difference-in-Difference Results (Alternative Exposure)

| Dependent Variables: | Output Price (log) | Sales (log) | Profit (log) | Chemical Expenditure (log) |
|---|---|---|---|---|
| Model: | (1) | (2) | (3) | (4) |
| *Variables* | | | | |
| $Exposure_j^{alt} \times$ Post | -0.1928* | -0.2322** | -0.3228*** | -0.2357 |
| | (0.1042) | (0.1042) | (0.1032) | (0.1422) |
| Mean Outcome in Level | 224 | 109k | 50k | 17k |
| *Fixed-effects* | | | | |
| Farm | Yes | Yes | Yes | Yes |
| Department-Year | Yes | Yes | Yes | Yes |
| *Fit statistics* | | | | |
| Observations | 1,929 | 2,978 | 2,945 | 2,978 |
| $R^2$ | 0.94 | 0.94 | 0.88 | 0.94 |

*Clustered (Department) standard-errors in parentheses*

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes. Outcomes are the following: a farm-level output price index (average price across row crops, using relative areas as weights), total sales, profit, and the farms' deflated chemical bill. Coefficients give the effect of exposure after the reform. The vector of controls contains the following elements: farm 1991 capital stock, total labor used, total land use, profit, their chemical use, the number of crops they grew and the evenness of their land allocation, their fertilizer-to-land and pesticides-to-land ratios, the share of their production which corresponds to oil crops (colza and sunflower), and the farms' 1983-1984 adoption trends in chemicals measured as the evolution in their chemical use. I add farm and department-year fixed effects, and cluster the standard errors at the department. Exposure is computed using land shares for land subsidies, and output shares for price intervention.

## C.4 Independent Exposures

Table A13: Heterogeneity in Crop-Specific Effects

| Dependent Variables: | Sales (log) | Profit (log) | Chemical Expenditure (log) |
|---|---|---|---|
| Model: | (1) | (2) | (3) |
| *Variables* | | | |
| $Wheat_j \times Post_t$ | -0.0024 | -0.0030* | -0.0032 |
| | (0.0016) | (0.0015) | (0.0019) |
| $WinterBarley_j \times Post_t$ | 0.0002 | -0.0003 | -0.0010 |
| | (0.0025) | (0.0025) | (0.0030) |
| $Corn_j \times Post_t$ | -0.0024** | -0.0020 | -0.0024 |
| | (0.0012) | (0.0015) | (0.0019) |
| $Rye_j \times Post_t$ | -0.0053*** | 0.0005 | -0.0023 |
| | (0.0018) | (0.0021) | (0.0020) |
| $Sunflower_j \times Post_t$ | -0.0014 | -0.0050*** | -0.0010 |
| | (0.0020) | (0.0013) | (0.0027) |
| $Colza_j \times Post_t$ | -0.0030 | -0.0075*** | -0.0022 |
| | (0.0019) | (0.0019) | (0.0027) |
| $Durum_j \times Post_t$ | -0.0093*** | -0.0077*** | -0.0069*** |
| | (0.0025) | (0.0018) | (0.0026) |
| $Sorghum_j \times Post_t$ | 0.0161 | 0.0746*** | 0.0060 |
| | (0.0147) | (0.0128) | (0.0219) |
| $SpringBarley_j \times Post_t$ | -0.0006 | -0.0015 | -0.0010 |
| | (0.0008) | (0.0014) | (0.0024) |
| $Oats_j \times Post_t$ | -0.0009 | -0.0046*** | 0.0015 |
| | (0.0021) | (0.0016) | (0.0018) |
| *Fixed-effects* | | | |
| Farm | Yes | Yes | Yes |
| Department-Year | Yes | Yes | Yes |
| *Fit statistics* | | | |
| Observations | 2,978 | 2,945 | 2,978 |
| $R^2$ | 0.94 | 0.88 | 0.94 |

*Clustered (Department) standard-errors in parentheses*

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes. I show the heterogeneity of the difference-in-difference results across crop-specific shocks. Exposure is decomposed in multiple crop-specific measures. I include the crop shares in levels, but do not show the coefficients in the table. The vector of controls contains the following elements: farm 1991 capital stock, total labor used, total land use, profit, their chemical use, the number of crops they grew and the evenness of their land allocation, their fertilizer-to-land and pesticides-to-land ratios, the share of their production which corresponds to oil crops (colza and sunflower), and the farms' 1983-1984 adoption trends in chemicals measured as the evolution in their chemical use. I add farm and department-year fixed effects, and cluster the standard errors at the department level.

I decompose the shift-share variable into crop-specific variables. Each crop-specific exposure measure corresponds to the crop-specific change in subsidization between 1991 and 1995,

interacted with the land share allocated to that crop by the farm in 1991. The regression is otherwise similar to the difference-in-difference regressions presented previously. The coefficients for the crop exposures interacted with the post-1991 dummy isolate the effect of crop-specific exposure to the reform after the reform happened. All crop-specific exposures but sorghum and winter barley have a negative effect on sales, and these two are not statistically significant. Sorghum is the one crop with a positive effect for profit. This means that the shift-share exposure hides some degree of heterogeneity in terms of the effects of exposure. I note however, that sorghum production in France corresponds to about 1.5% of the overall production of the grain market, and hence does not correspond to a very significant share of the market. The crop-by-crop results otherwise describe a high level of homogeneity in the consequences of exposure to the reform.

### C.5 Municipality-Level Results

In this section, I first present balance tests comparing municipalities with relatively more and relatively less exposure to the reform, and then the distribution of the measure of exposure used for the municipality-level design.

Starting with balance tests, I show the trends (averages for each wave of the census) for six municipality characteristics, splitting them nationally between municipalities with a median exposure above or below the French median in 1988. Trends are overall similar, apart from the evenness variable. In the estimation design, I control for this evenness, both at the municipality and farm level.

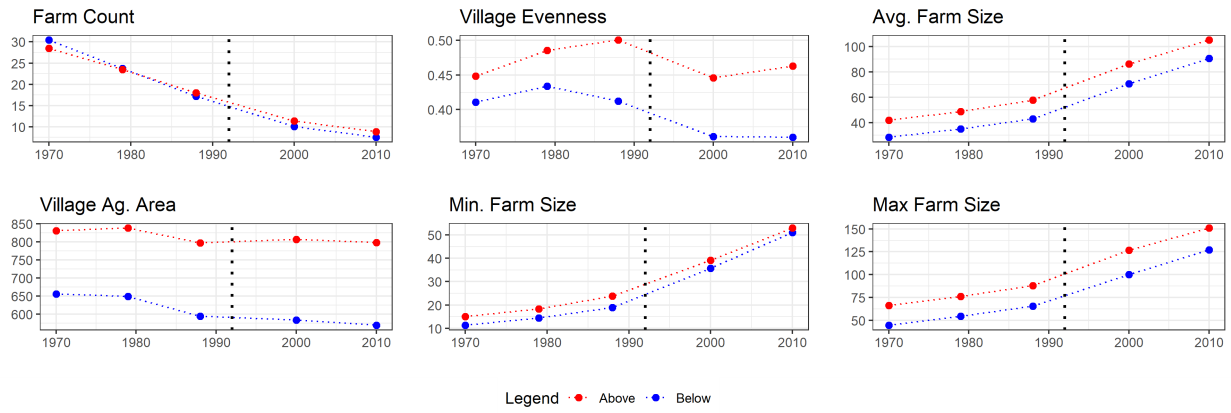



Figure A19: Municipality-Level Trends

*Notes:* This figure gives the evolution of municipality level outcomes over time, for the group of municipalities for which median exposure is above or below the median exposure in France as a whole.

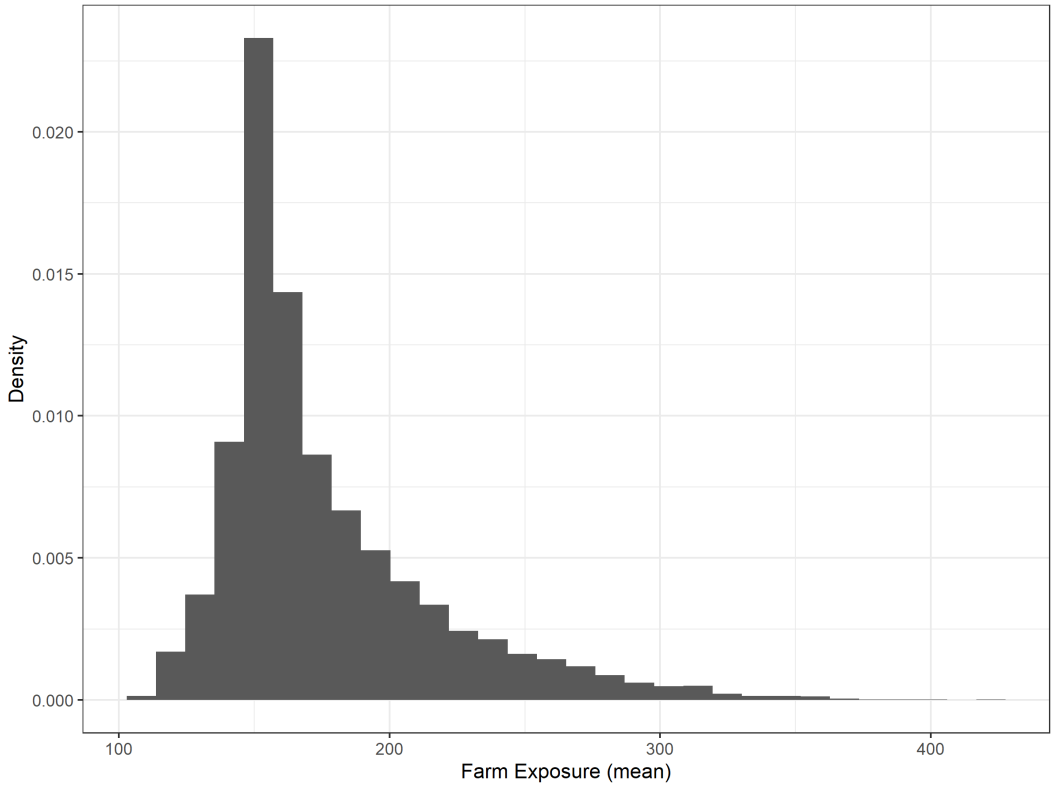



Figure A20: Distribution of Mean Exposure

*Notes:* This figure gives the distribution of the municipality-level exposure to the MacSharry reform.

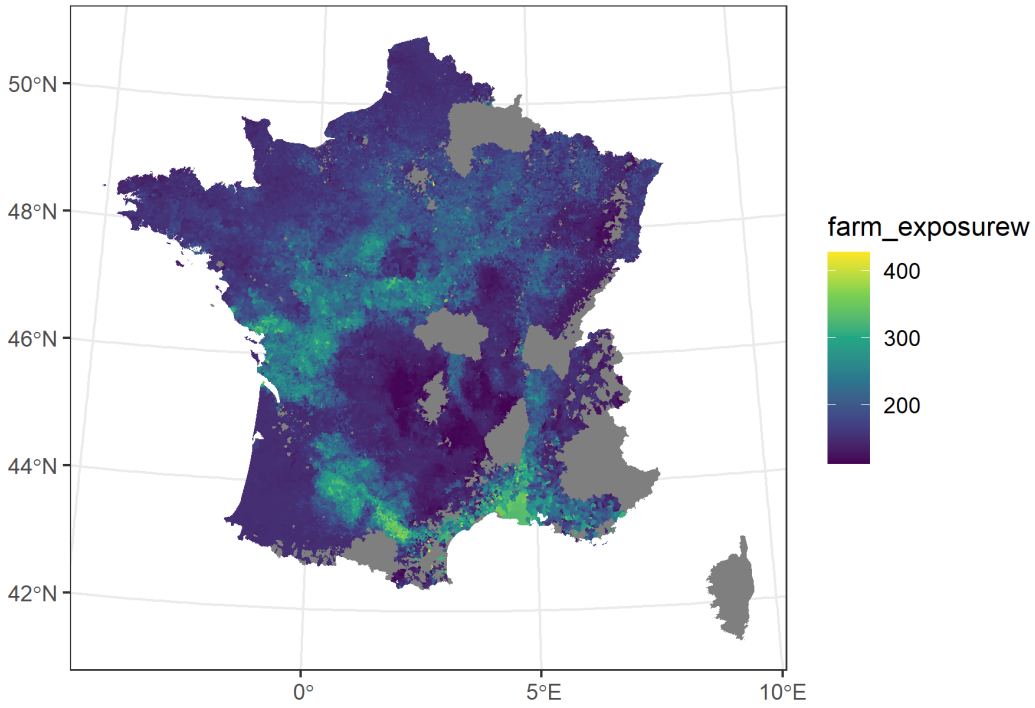



Figure A21: Geographic distribution of Mean Exposure

*Notes:* This figure gives the distribution of the municipality-level exposure to the MacSharry reform.

Next, I show the distribution of mean municipality exposure. As mentioned previously, all the measures of municipality-level exposure are computed after winsorizing the farm-level measure of exposure in the Census for the bottom and top 1ppt values. Exposure is mostly distributed between 100 and 200€ per output unit, with a large right tail which likely corresponds to municipalities which have some of their land allocated either to colza, durum and sunflower, which are the crops with a shock value higher than 300 € per unit. These are mostly located in two areas of France, as shown in Figure A21, along the Mediterranean coast, as well as the main grain region of France around the Beauce region below Paris.

# D Model Details and Extensions

## D.1 Formal Model Presentation

I start with the Bellman equations describing the farms' decisions—both incumbent and entrant farms—which correspond to the general outline of the model from Section 4.2.

The incumbent's decision can be characterized by:

$$V(\Upsilon_{jt};\Omega_{jt}) = \max_{\{X_{jct}\}_{\mathbb{C}_{jt}},\xi^x_{jt},K_{jt+1},\mathbb{C}_{jt+1}} \underbrace{\Pi_{jt}}_{\text{Static Profit}} - \underbrace{C(K_{jt},K_{jt+1})}_{\text{Capital Adjustment}}$$
$$+\xi^x_{jt}\underbrace{(1-\tau_k)P^K_{jt}K_{jt}}_{\text{Scrap Value if Exit}} + (1-\xi^x_{jt})\Bigg(\underbrace{\beta\mathbb{E}\Big[V(\Upsilon_{jt+1};\Omega_{jt+1})|\Upsilon_{jt},\Omega_{jt}\Big] - f_{inc}}_{\text{Expected Value of Continuation}}\Bigg)$$

And I can further write static profit as:

$$\Pi_{jt} = \sum_{c\in\mathbb{C}_{jt}} \underbrace{\Big[P_{jct}Q_{jct} - P^L_{jt}L_{jct} - P^S_{jt}S_{jct} - P^{chem}_{jt}Chemicals_{jct}\Big]}_{\text{Crop Specific Profit: } \pi_{jct}}$$

The entrant's problem takes the form:

$$V_e(\Omega_{jt}) = \max_{\xi^e_{jt},\xi^x_{jt},K_{jt+1},\mathbb{C}_{jt+1}} \mathbb{E}\Bigg[\xi^e_{jt}\Bigg\{-f_e - \underbrace{C(0,K_{jt+1})}_{\text{Capital Adjustment}}$$
$$+(1-\xi^x_{jt})\underbrace{\beta\mathbb{E}\Big[V(\Upsilon_{jt+1};\Omega_{jt+1})|\Upsilon_{jt},\Omega_{jt}\Big]}_{\text{Expected Value of Continuation}}\Bigg\}|\Omega_{jt}\Bigg]$$

The timing assumptions which relate to these Bellman equations are represented in Figure A22 and Figure A23 below.

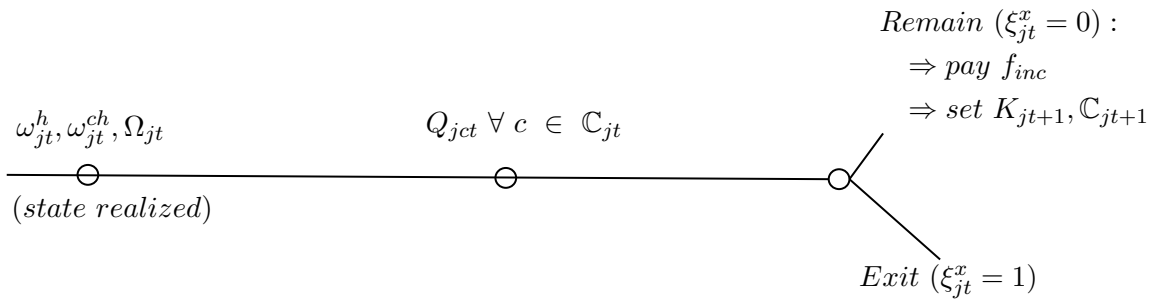



Figure A22: Timing for Incumbent Farms

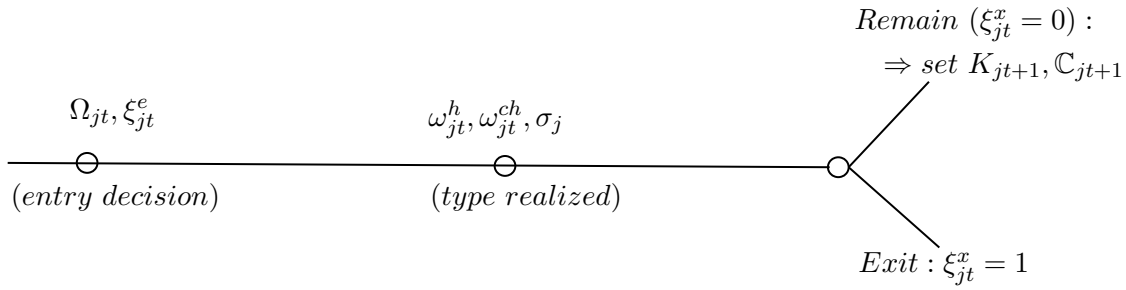



Figure A23: Timing for Entrants

After the simplifying assumptions regarding the homogeneity of $\sigma_j$ and output prices, I can further write the transition operators which describe the evolution of entrant and incumbents farms as: $\phi_t(\omega', \omega^{ch'}, K', \mathbb{C}'|\omega, \omega^{ch}, K, \mathbb{C})$ and $\phi_{e,t}(\omega', \omega^{ch'}, K', \mathbb{C}'|\omega, \omega^{ch})$. They require the additional introduction of the transition operator $J^h(.|.)$ and $J^{ch}(.|.)$ for both shocks, and of the policy functions $\kappa^k$ for incumbent capital investment, $\kappa^{\mathbb{C}}$ for crop choice, and $\kappa^k_e$ for entrant capital investment. These transition operators now write as:

$$\begin{aligned}\phi_t(\omega', \omega^{ch'}, K', \mathbb{C}'|\omega, \omega^{ch}, K, \mathbb{C}) =& J^h(\omega'|\omega) J^{ch}(\omega^{ch'}|\omega^{ch}) \mathbb{1}\{K' = \kappa^k(\omega, \omega^{ch}, K|\Omega_t)\} * \\ & \mathbb{1}\{\mathbb{C}' = \kappa^{\mathbb{C}}(\omega, \omega^{ch}, K|\Omega_t)\} \mathbb{1}\{\xi^x(\omega, \omega^{ch}, K|\Omega_t) = 0\}\end{aligned}$$

I note that crop choice is a dynamic decision simply because of the timing assumption—which allows me to address the selection bias induced by focusing on single crop farms for estimation—but crop choice has no impact on a farm's transition. The current state of a farm's crop mix in a given period has no bearing on the dynamic decisions made by the farm regarding the next period, including its crop mix decision. $\phi_t(\omega', \omega^{ch'}, K', \mathbb{C}'|\omega, \omega^{ch}, K, \mathbb{C})$ could as such be re-written as $\phi_t(\omega', \omega^{ch'}, K', \mathbb{C}'|\omega, \omega^{ch}, K)$, though I will keep the inclusion of $\mathbb{C}$ in the conditioning for clarity.

The analogue transition operator for entrants writes $\phi_{e,t}(\omega', \omega^{ch'}, K', \mathbb{C}'|\omega, \omega^{ch})$:

$$\begin{aligned}\phi_{e,t}(\omega', \omega^{ch'}, K', \mathbb{C}'|\omega, \omega^{ch}) =& J(\omega'|\omega)J^{ch}(\omega^{ch'}|\omega^{ch})\mathbb{1}\{K' = \kappa_e^k(\omega, \omega^{ch}, 0|\Omega_t)\}* \\ & \mathbb{1}\{\mathbb{C}' = \kappa^{\mathbb{C}}(\omega, \omega^{ch}, 0|\Omega_t)\}\mathbb{1}\{\xi^x(\omega, \omega^{ch}, 0|\Omega_t) = 0\}\end{aligned}$$

Once I have defined these, I can then define the law of motion for the measure $\mu_t$. $G_1(.)$ and $G_2(.)$ are the distributions from which entrants draw resp. their original TFP and chemical productivity. With $N_t^e$ the mass of entrants, the allocation of farms over the market space is then fully described by the measure $\mu_t$, whose transition operation $T(.)$ can be written as:

$$\begin{aligned}\mu_{t+1}(\omega', \omega^{ch'}, K', \mathbb{C}') = & \int \phi_t(\omega', \omega^{ch'}, K', \mathbb{C}'|\omega, \omega^{ch}, K, \mathbb{C})d\mu_t(\omega, \omega^{ch}, K, \mathbb{C}) + \\ & N_t^e \int\int \phi_{e,t}(\omega', \omega^{ch'}, K', \mathbb{C}'|\omega, \omega^{ch})dG_1(\omega)dG_2(\omega^{ch})\end{aligned}$$

**Stationary Equilibrium:** With all these elements, I can finally define the stationary equilibria on which I will focus as follows. I look at stationary competitive equilibria composed of the tuple of $\Omega^\star = \{\mu^\star, N^{e,\star}, \{P_c^\star\}_{\mathbb{C}}, \{P^x\}_x, Policy\}$, such that—for a given policy state $(Policy)$ and given input prices $(\{P^x\}_x)$:

- **Zero Ex-Ante Profit**: $V^e(\Omega^\star) \leq 0$, and $V^e(\Omega^\star) = 0$ if $N^{e,\star} > 0$.
- **Stationary Market**: $\mu^\star = T(\Omega^\star)$.
- **Output Market Clearing**: $P^\star$ s.t. $Q_c^S(\Omega^\star) = Q_c^D(P_c^\star) \ \ \forall c \in \mathbb{C}$.

### D.2 Optimal Static Farm Behavior

It is useful to derive the expressions for the average cost of production, marginal cost of production, and optimal farm profit, as these highlight how the assumption of a homogeneous and frictionless market impacts equilibrium heterogeneity across farms. By homogeneity and lack of frictions, I mean how farms are single agents facing unique input and output prices, and how the model does not imply any misallocation of production in the usual economic sense. For simplicity, I will remove the subscripts $\{j, c, t\}$ in the following derivations, and take $\varepsilon = 0$.

In a first step, I will recover the input price for the composite input $X$, which is defined as the total expenditure required to produce one unit of $X$. Because the composite input nest $X$ has constant returns to scale, I can recover this cost by computing the cost of producing

the first unit of $X$, which offers a simple derivation from the following minimization problem:

$$\min_{S,Chemicals} Cost_S = P_s S + P_{ch} Chemicals,\ s.t.\ 1 = \left[\delta_s S^\rho + (1-\delta_s)\{e^{\omega^{ch}} Chemicals\}^\rho\right]^{\frac{1}{\rho}}$$
$$\Leftrightarrow \min_{S,Chemicals} Cost_S = P_s S + P_{ch} Chemicals,\ s.t.\ 1 = \delta_s S^\rho + (1-\delta_s)\{e^{\omega^{ch}} Chemicals\}^\rho$$

The Lagrangian can hence be written as:

$$\mathcal{L} = P_s S + P_{ch} Chemicals - \lambda\left(\left[\delta_s S^\rho + (1-\delta_s)\{e^{\omega^{ch}} Chemicals\}^\rho\right] - 1\right).$$

Combining the first order conditions of the problem, I can write the price of the composite input $P_X$ as:

$$P_X = P_S S + P_{ch} Chemicals = \lambda\rho.$$

Introducing the elasticity $\sigma = \frac{1}{1-\rho}$, I then solve for the expression of $\lambda$:

$$\lambda = \rho^{-1}\left[\delta_s^\sigma P_s^{1-\sigma} + (1-\delta_s)^\sigma (P_{ch} e^{-\omega^{ch}})^{1-\sigma}\right]^{\frac{1}{1-\sigma}}.$$

This implies finally:

$$P_X = \left[\delta_s^\sigma P_s^{1-\sigma} + (1-\delta_s)^\sigma (P_{ch} e^{-\omega^{ch}})^{1-\sigma}\right]^{\frac{1}{1-\sigma}}.$$

Now, I can consider the profit maximization problem faced by the farm in a given production period. Within the period, the farm takes its capital stock $K$ as given. This implies that the production function has decreasing returns in the flexible inputs (labor and the composite input). The cost minimization problem writes:

$$\min_{L,X} C = P_l L + P_X X \quad s.t.\ \ Q = e^{\omega^h} K^{\alpha_k} L^{\alpha_l} X^{\alpha_s}$$

It is useful to write $\alpha = \alpha_l + \alpha_s$, as well as: $\alpha_l' = \frac{\alpha_l}{\alpha}$, and $\alpha_{s'} = \frac{\alpha_s}{\alpha}$. The normalized production function with constant returns to scale $Y = Q^{\frac{1}{\alpha}}$ has a constant marginal cost of production:

$$c_y = \left(e^{-\omega^h} K^{-\alpha_k}\right)^{\frac{1}{\alpha}} \left(\alpha_l'^{-\alpha_l'} \alpha_s'^{-\alpha_s'}\right) P_l^{\alpha_l'} P_X^{\alpha_s'}.$$

Given the transformation, the same set of inputs which produces $Y$ with the normalized function will also produce $Q$ with the main function. With $C_Q(.)$ the cost function of

function $Q$, and $C_Y(.)$, the cost function of function $Y$, one always has that:

$$C_Q(Q) = C_Y(Y) = c_y Y = c_y Q^{\frac{1}{\alpha}}$$

This implies that the average cost is $AC(Q) = \frac{C_Q(Q)}{Q} = c_y Q^{\frac{1-\alpha}{\alpha}}$ which is upward sloping in the quantity produced, and that the marginal cost is $MC(Q) = \frac{c_y}{\alpha} Q^{\frac{1-\alpha}{\alpha}}$, which is also upward sloping. The average cost is then always a fraction of the marginal cost such that $AC(Q) = \alpha MC(Q)$.

The model has no distortions, the market is competitive and all farms face a unique price for a given crop. This implies that at equilibrium, all farms will produce such that their marginal cost is equal to the market price $P$. This implies that the average cost of production will be equal across all farms, and will correspond to a fraction of the market price:

$$AC(Q^\star) = \alpha P = \Big(\alpha_l + \alpha_s\Big) P.$$

Next, I derive the expression for the farm-crop-level optimal profit. It is useful here to consider the crop-level static profit maximization problem of the farm, where maximization happens over flexible inputs, and where I directly write the problem as a function of the composite land input $X_{jct}$:

$$\underset{L,X}{max}\ \pi = PQ - P^l L - P^X X$$

Parametric assumptions guarantee the existence of an interior solution, and first order conditions follow the usual Cobb-Douglas form:

$$L = \alpha_l \frac{PQ}{P^l}$$
$$X = \alpha_s \frac{PQ}{P^X}$$

This yields the optimal output level:

$$Q^\star = \left[ e^{\omega_h} K^{\alpha_k} P^{\alpha} \left(\frac{\alpha_l}{P^l}\right)^{\alpha_l} \left(\frac{\alpha_s}{P^X}\right)^{\alpha_S} \right]^{\frac{1}{1-\alpha}}$$

Substituting out the expression for $Q^\star_{jct}$ into the expressions for $L$ and $X$, and using the formula for farm profit, this finally yields:

$$\pi^\star = (1-\alpha) P^{\frac{1}{1-\alpha}} \left[ e^{\omega^h} K^{\alpha_k} \left(\frac{\alpha_l}{P^l}\right)^{\alpha_l} \left(\frac{\alpha_s}{P^X}\right)^{\alpha_s} \right]^{\frac{1}{1-\alpha}}.$$

This expression can be used to determine the effect of an increase in $\omega^{ch}$ on farm-crop-level profits. First, note that an increase in $\omega_{ch}$ reduces the price of the effective land input $X$:

$$\frac{\partial P_X}{\partial \omega^{ch}} = -(1-\delta_s)^{\sigma}\left(e^{-\omega_{ch}}P_{ch}\right)^{1-\sigma}P_X^{\sigma} < 0.$$

Then, relying on the chain rule and this expression, the partial derivative of farm-crop-level profits with respect to the same shock writes:

$$\frac{\partial \pi^{\star}}{\partial \omega_{ch}} = \frac{\alpha_s}{1-\alpha}\frac{\pi^{\star}}{P_X}\left(1-\delta_s\right)^{\sigma}\left(e^{-\omega_{ch}}P_{ch}\right)^{1-\sigma}P_X^{\sigma} > 0.$$

This further implies that farm-level profits are also increasing in $\omega_{ch}$, indeed farm-level profit are a direct sum of farm-crop profits. As a farm faces a higher chemical use efficiency, its profits necessarily increase, in turn this will make this farm less likely to exit the market.

Lastly, I derive the expression for optimal land use, which is then used to compute the log-ratio of land use between two crops $c$ and $c'$ within a farm. Relying again on the first order condition for land this time, rather than the composite input, the following expression is obtained:

$$S^{\star} = \frac{\alpha_s}{1-\alpha P_S}\pi^{\star}\delta_s^{\sigma}\left(\frac{P_X}{P_s}\right)^{\sigma-1}$$

Substituting for the expression of optimal profit, this then writes as:

$$S^{\star} = \alpha_s^{\frac{1-\alpha_l}{1-\alpha}}P^{\frac{1}{1-\alpha}}e^{\frac{\omega_h}{1-\alpha}}\left(\frac{\alpha_l}{P^l}\right)^{\frac{\alpha_l}{1-\alpha}}K^{\frac{\alpha_k}{1-\alpha}}\delta_s^{\sigma}P_s^{-\sigma}P_X^{\sigma-1\frac{\alpha_s}{1-\alpha}}.$$

The log-ratio can then be obtained as, reintroducing the $c$ and $c'$ markers for clarity:

$$\begin{aligned}\log\left(\frac{S^{\star}_{jct}}{S^{\star}_{jc't}}\right) = &+\left[\frac{1}{1-\alpha_c}\log P_{jct} - \frac{1}{1-\alpha'_c}\log P_{jc't}\right] + \left[\frac{\alpha_{l,c}}{1-\alpha_c}\log(\frac{\alpha_{l,c}}{P^l_{jt}}) - \frac{\alpha_{l,c'}}{1-\alpha_{c'}}\log(\frac{\alpha_{l,c'}}{P^l_{jt}})\right]\\
&+\left[\left(\sigma-1-\frac{\alpha_{s,c}}{1-\alpha_c}\right)\log(P^X_{jct}) - \left(\sigma-1-\frac{\alpha_{s,c'}}{1-\alpha_{c'}}\right)\log(P^X_{jc't})\right] + \left[\frac{1}{1-\alpha_c}\omega^h_{jct} - \frac{1}{1-\alpha_{c'}}\omega^h_{jc't}\right]\\
&+\left[\frac{\alpha_{k,c}}{1-\alpha_c} - \frac{\alpha_{k,c'}}{1-\alpha_{c'}}\right]\log(K_{jt}) + \underbrace{\left[\frac{1-\alpha_{l,c}}{1-\alpha_c}\log\alpha_{s,c} - \frac{1-\alpha_{l,c'}}{1-\alpha_{c'}}\log\alpha_{s,c'}\right] + \left(\log\delta_{s,c} - \log\delta_{s,c'}\right)}_{\equiv A_{cc'}}\end{aligned}$$

I focus on the determinants of crop mix heterogeneity across farms in the main text, and use $A_{cc'}$ for the terms that are constant across farms and only vary with crop-specific input elasticities.

One can further observe that the crop-level chemical-to-output ratio writes as follows:

$$\frac{Chemicals^{\star}_{jct}}{Q^{\star}_{jct}} = \frac{P_{jct}}{P^{ch}_{jt}} \frac{\alpha_{x,c}\big(1-\delta_{s,c}\big)}{\delta_{s,c}\left[\frac{P^{s}_{jt}(1-\delta_{s,c})e^{\omega^{ch}_{jt}}}{P^{ch}_{jt}\delta_{s,c}}\right]^{\frac{\rho}{\rho-1}} + \big(1-\delta_{s,c}\big)}.$$

Here, one can understand the land subsidy as directly impacting $P^{s}_{jt}$ in the denominator, with a larger subsidy lowering the effective price of land faced by farmers. A larger land subsidy will impact the chemical-to-output ratio in a way that depends on the parameter of substitution $\rho$, which relates land and chemicals in production. If land and chemicals are substitutes, or $\rho > 0$, a larger land subsidy will reduce this ratio. On the contrary, a larger subsidy will increase this ratio if the inputs are complements. For $\rho \approx 0$, the input nest would reduce to a Cobb-Douglas function where the ratio would be independent from prices.

### D.3 Extended Production Function

The production function for quantity $Q_{jct}$ of crop $c$ for farm $j$ at time $t$ remains Cobb-Douglas in its main inputs:

$$Q_{jct} \;=\; e^{\omega^{h}_{jct}}\; K^{\alpha^{c}_{k}}_{jt}\; L^{\alpha^{c}_{l}}_{jct}\; X^{\alpha^{c}_{s}}_{jct}\; e^{\varepsilon_{jct}}, \tag{11}$$

with $L_{jct}$ as labor, $\alpha^{c}_{k} + \alpha^{c}_{l} + \alpha^{c}_{x} = 1$, and $\varepsilon_{jct}$ is a mean-zero, i.i.d. shock realized after input decisions are made. The key mechanisms of substitution and biased productivity are embedded in the composite input $X_{jct}$, which combines land and chemicals. This composite input is now modeled as two nested constant elasticity of substitution functions, in order to allow for flexible substitution patterns and chemical-biased productivity shocks:

$$X_{jct} = \big\{\delta^{c}_{s} S^{\rho}_{jct} + (1-\delta^{c}_{s})\big(e^{\omega^{ch}_{jt}}\big\{\delta^{c}_{p} Pest^{\rho_2}_{jct} + (1-\delta^{c}_{p}) Fert^{\rho_2}_{jct}\big\}^{\frac{1}{\rho_2}}\big)^{\rho}\big\}^{\frac{1}{\rho}}, \tag{12}$$

This function has two layers. The inner nest combines pesticides ($Pest_{jct}$) and fertilizers ($Fert_{jct}$) into a chemical aggregate, with an elasticity of substitution governed by $\rho_2$. The outer nest then combines physical land ($S_{jct}$) with this chemical aggregate, with an elasticity of substitution governed by $\rho$. The chemical-biased productivity shock ($\omega^{ch}_{jt}$) still acts as an augmenting shifter on the entire chemical aggregate.

The second step of the estimation is similar to the one used in the main estimation procedure described in the paper. I obtain the following results:

Table A14: Alternative Production Function (Additional Nest)

| Coefficient | Parameter | Estimate | Std. Error |
|---|---|---|---|
| Substitution Land-Chemicals | $\rho$ | 0.58 | (0.12) |
| Substitution Fertilizers-Pesticides | $\rho_2$ | -1.92 | (0.43) |
| Land Share (Wheat) | $\delta_{s,wheat}$ | 0.16 | (0.21) |
| Land Share (Others) | $\delta_{s,other}$ | 0.27 | (0.17) |
| Pesticide Share (Wheat) | $\delta_{p,wheat}$ | 0.01 | (0.10) |
| Pesticide Share (Others) | $\delta_{p,other}$ | 0.07 | (0.06) |
| Labor Share (Wheat) | $\alpha_{l,wheat}$ | 0.42 | (0.08) |
| Labor Share (Others) | $\alpha_{l,other}$ | 0.29 | (0.07) |
| Land-Nest Share (Wheat) | $\alpha_{x,wheat}$ | 0.42 | (0.10) |
| Land-Nest Share (Others) | $\alpha_{x,other}$ | 0.45 | (0.08) |
| N | | 13,281 | |
| Unique Farms | | 3,901 | |
| N - Wheat | | 4,089 | |
| N - Other | | 9,192 | |

*Notes: The parameters are obtained from a two-step estimation, each step performed by GMM. I keep farms observed for at least three periods in a row*[61]*, and that produce either only crops in the wheat group, or in the other crops group. I remove farms that are not observed with positive input values for the set of considered inputs (land, labor, capital, fertilizers, pesticides). standard errors are obtained using a block bootstrap procedure, where I draw all the observations related to a farm at a time, using* $B = 1000$.

**Recovering Input Allocations:** As discussed in Section 4.3, I estimate the production parameters on single crop-group firms, following De Loecker et al. (2016). Because I observe farm-crop level land allocations, knowing the parameters of the production specification, I can then recover both input allocations and Hicksian and non-Hicksian productivity shocks for multi-product firms. The production function has the following shape:

$$Q_{jct} = e^{\tilde{\omega}^h_{jct}} K_{jt}^{\alpha^c_K} L_{jct}^{\alpha^c_L} \left\{ \delta^c_s S^{\rho}_{jct} + (1-\delta^c_s) \left( e^{\omega^{ch}_{jt}} \left[ (1-\delta^c_p) Fert^{\rho_2}_{jct} + \delta^c_p Pest^{\rho_2}_{jct} \right]^{\frac{1}{\rho_2}} \right)^{\rho} \right\}^{\frac{\alpha^c_S}{\rho}}$$

With: $\tilde{\omega}^h_{jct} = \omega^h_{jct} + \varepsilon_{jct}$

Drawing from the first-order conditions, I can write the following relation between pesticides and fertilizers:

$$Fert^{1-\rho_2}_{jct} = \frac{P^P_{jt}}{P^F_{jt}} \frac{(1-\delta^c_p)}{\delta^c_p} Pest^{1-\rho_2}_{jct} \tag{13}$$

[61] This is done to smooth potential measurement issues, and follows from De Loecker et al. (2016) and Doraszelski and Jaumandreu (2018).

I can use the relation to recover a relation between land and pesticide allocations:

$$\frac{P^s_{jt}S^{1-\rho}_{jct}}{\delta^c_S} = \frac{P^P_{jt}}{\delta^c_p(1-\delta^c_s)}\left(e^{-\omega^{ch}_{jt}}\right)^{\rho} Pest^{1-\rho}_{jct}\left[\left(1-\delta^c_p\right)\left(\frac{P^P_{jt}(1-\delta^c_p)}{P^F_{jt}\delta^c_p}\right)^{\frac{\rho_2}{1-\rho_2}} + \delta^c_p\right]^{1-\frac{\rho}{\rho_2}}$$

For farms producing the two crop groups, I can then take the ratio of this expression for their two crop categories $c$ and $c'$:

$$\frac{Pest^{1-\rho}_{jct}}{Pest^{1-\rho}_{jc't}} = \frac{S^{1-\rho}_{jct}}{S^{1-\rho}_{jc't}}\frac{\delta^c_p(1-\delta^c_s)\delta^{c'}_s}{\delta^{c'}_p(1-\delta^{c'}_s)\delta^c_s}\frac{\left[\left(1-\delta^c_p\right)\left(\frac{P^P_{jt}(1-\delta^c_p)}{P^F_{jt}\delta^c_p}\right)^{\frac{\rho_2}{1-\rho_2}} + \delta^c_p\right]^{\frac{\rho}{\rho_2}-1}}{\left[\left(1-\delta^{c'}_p\right)\left(\frac{P^P_{jt}(1-\delta^{c'}_p)}{P^F_{jt}\delta^{c'}_p}\right)^{\frac{\rho_2}{1-\rho_2}} + \delta^{c'}_p\right]^{\frac{\rho}{\rho_2}-1}}$$

Once I have estimated the parameters of the production function, the right hand side of this expression is fully known. As I know total farm pesticide use, I can then recover the volumes of pesticide used for each of the crops that they grow. I can then use Equation (13) to recover the fertilizer allocations as well. With these, I can construct the crop-specific land nest values, and use the outer Cobb-Douglas shape of the production function to recover crop-specific labor allocations, from these crop-specific land nests.

### D.4 Full CES Production Function

In this section, I consider an alternative production function where a CES nest relates capital, labor and the composite input $X_{jct}$. This production function is strictly more general and nests the Cobb-Douglas specification considered in the main text. I outline here under which conditions the results obtained in the Cobb-Douglas case hold with a more general CES specification. I show that the Cobb-Douglas approach will yield results that are of the same sign and similar magnitude as the CES—specifically Cobb-Douglas results will be a lower bound for the CES results—while offering a more transparent framework and a simpler estimation procedure. The function writes as:

$$Q_{jct} = e^{\omega^h_{jct}}\left(\tilde{\delta}_k K^{\rho_0}_{jt} + \tilde{\delta}_l L^{\rho_0}_{jct} + (1-\tilde{\delta}_k-\tilde{\delta}_l)X^{\rho_0}_{jct}\right)^{\frac{1}{\rho_0}} e^{\varepsilon_{jct}}$$

I first derive the expression for optimal chemical-to-output ratio. For simplicity, I remove the $j$, $c$ and $t$ subscripts. In a first step, I recover the optimal ratio of chemicals and the composite input, and I do so by considering the composite input cost minimization problem:

$$\mathcal{L} = P_s S + P_{ch}Chemicals - \lambda\left(\left[\delta_s S^{\rho} + (1-\delta_s)\left\{e^{\omega^{ch}}Chemicals\right\}^{\rho}\right]^{\frac{1}{\rho}} - X\right).$$

Now, with $\lambda = P^X$, I get from the first order condition relative to chemical use:

$$\frac{Chemicals^\star}{X^\star} = \left(\frac{P^X(1-\delta_s)}{P^{ch}}\right)^{\frac{1}{1-\rho}} exp\left(\frac{\rho}{1-\rho}\omega^{ch}\right)$$

In the same way, but using the profit maximization problem, one can obtain:

$$\frac{X^\star}{Q^\star} = \left(\frac{P(1-\tilde{\delta}_k-\tilde{\delta}_l)}{P^X}\right)^{\frac{1}{1-\rho_0}} exp\left(\frac{\rho_0}{1-\rho_0}\omega^{h}\right)$$

Then, the optimal chemical-to-output ratio can be written as:

$$\begin{aligned}\frac{Chemicals^\star}{Q^\star} =& \left[P\big(1-\tilde{\delta}_k-\tilde{\delta}_l\big)\right]^{\sigma_0}\left[\frac{1-\delta_s}{P^{ch}}\right]^{\sigma} * \\ & exp\left(\big(\sigma_0-1\big)\omega^h+\big(\sigma-1\big)\omega^{ch}\right)\left[\delta_s^\sigma(P^s)^{1-\sigma}+(1-\delta_s)^\sigma(P^x e^{-\omega^{ch}})^{1-\sigma}\right]^{\frac{\sigma-\sigma_0}{1-\sigma}}.\end{aligned}$$

**Chemical Use Efficiency and Chemical Use Intensity:** There are two main differences with the Cobb-Douglas case. First, $\omega^h$ plays a role in shifting the chemical-to-output ratio, which depends on the value of $\rho_0$. If land, capital and the composite input are substitutes, then an increase in $\omega^h$ will also raise the ratio. Second, heterogeneity in chemical-use efficiency now plays two different roles in shifting that ratio: by changing incentives to use land relative to chemicals, but also by shifting the incentives for the use of the composite input relative to labor and capital. In this set-up, an increase in $\omega^{ch}$ can increase the ratio if $\rho > 0$, as is the case with the Cobb-Douglas function, but also in some cases where $\rho$ is not too small relative to $\rho_0$. In that sense, the conditions for a positive correlation between $\omega^{ch}$ and the chemical-to-output ratio considered in the main text are strictly more restrictive than in the CES case. This can be seen by introducing the share of chemicals in the composite input, which writes as:

$$s_{ch} = \frac{(1-\delta_s)^\sigma(P^x e^{-\omega^{ch}})^{1-\sigma}}{\delta_s^\sigma(P^s)^{1-\sigma}+(1-\delta_s)^\sigma(P^x e^{-\omega^{ch}})^{1-\sigma}}$$

And by noting that $\frac{\partial \log P^X}{\partial \omega^{ch}} = -s_{ch}$. Then:

$$\begin{aligned}\frac{\partial}{\partial \omega^{ch}}\log\big(\frac{Chemicals^\star}{Q^\star}\big) =& (\sigma-1)+(\sigma-\sigma_0)\frac{\partial \log P^X}{\partial \omega^{ch}} \\ =& (\sigma-1)-(\sigma-\sigma_0)s_{ch}\end{aligned}$$

Then:

$$\frac{\partial}{\partial \omega^{ch}} \log\left(\frac{Chemicals^{\star}}{Q^{\star}}\right) > 0 \Rightarrow \sigma(1-s_{ch}) + \sigma_0 s_{ch} > 1$$

Because $s_{ch} \in [0,1]$, there are cases where land and chemicals are complements, but where the cost share of chemicals $s_{ch}$ is large enough, and where labor, capital and the composite input are substitutes enough that an increase in $\omega^{ch}$ will raise the farm's optimal chemical-to-output ratio at the crop-level.

**Profitability and Chemical Use Intensity:** Now, in the CES case $\omega^{h}$ will influence both profits and the chemical-to-output ratio. Because of this, it might matter for the profitability-to-chemical-intensity correlation where increases in profitability within the farm are coming from. Note first that if $\sigma(1-s_{ch}) + \sigma_0 s_{ch} > 1$ and $\rho_0 > 0$, or for $\sigma(1-s_{ch}) + \sigma_0 s_{ch} < 1$ and $\rho_0 < 0$, then farm profitability and chemical-use intensity will always have the correlation given by the relation between $\omega^{ch}$ and chemical-use intensity. In these two cases, the sign of the correlation between farm profits and chemical-use intensity obtained in the Cobb-Douglas case maps one-to-one to the sign of the correlation obtained in a CES case. The Cobb-Douglas specification loses in generality in the case where an increase in profitability coming from $\omega^{h}$ or an increase in profitability coming from $\omega^{ch}$ do not have the same implications for the correlation between chemical-use-intensity and farm profits. For this, I need that $\rho_0 < 0$, $\sigma(1-s_{ch}) + \sigma_0 s_{ch} > 1$. In this special case, I can study whether an increase in chemical-use efficiency can raise the chemical-intensity more than a similar increase in TFP will decrease the ratio. This writes as:

$$\begin{aligned} & \left|\frac{\partial}{\partial \omega^{h}} \log\left(\frac{Chemicals^{\star}}{Q^{\star}}\right)\right| > \left|\frac{\partial}{\partial \omega^{ch}} \log\left(\frac{Chemicals^{\star}}{Q^{\star}}\right)\right| \\ \Leftrightarrow \; & |(\sigma_0 - 1)| > |(\sigma - 1) - (\sigma - \sigma_0) s_{ch}| \end{aligned}$$

This in turn leads to the following condition:

$$\frac{1-s_{ch}}{1-\rho} + \frac{1+s_{ch}}{1-\rho_0} < 2$$

That is, if I have low substitutability between land and chemicals, but a high degree of complementarity between labor, capital and the composite input.

**Testing the Condition:** The Cobb-Douglas remains my favored specification: first because it relates profits and chemical-use intensity in a more transparent way, and second because it yields a linear second estimation step which is easier to precisely estimate. However, in order to make sure that the parametric constraints brought by the Cobb-Douglas are not biasing the sign of the correlation between farm profits and chemical-use intensity, I also

estimate this CES specification. For the CES specification, the first estimation step detailed in Equation (6) remains correct, and so I keep the estimated parameters for the $X_{jct}$ from the main text. In order to estimate the rest of the parameters, I rely on the first order condition for static profit maximization with respect to labor. This yields, reintroducing the $j$, $c$ and $t$ subscripts for precision:

$$\frac{\partial \pi_{jct}}{\partial L_{jct}} = 0 \Leftrightarrow e^{\omega^h_{jct}} \tilde{\delta}_{l,c} L_{jct}^{\rho_0 - 1} \left( \tilde{\delta}_{k,c} K_{jt}^{\rho_0} + \tilde{\delta}_{l,c} L_{jct}^{\rho_0} + (1 - \tilde{\delta}_{k,c} - \tilde{\delta}_{l,c}) X_{jct}^{\rho_0} \right)^{\frac{1-\rho_0}{\rho_0}} P_{jct} = P^l_{jt}$$

This in turn yields the following equation for the TFP process $\omega^h_{jct}$:

$$\omega^h_{jct} = \log\left(\frac{P^l_{jt}}{P_{jct}}\right) + \frac{\rho_0 - 1}{\rho_0} \log\left(\tilde{\delta}_{k,c} K_{jt}^{\rho_0} + \tilde{\delta}_{l,c} L_{jct}^{\rho_0} + (1 - \tilde{\delta}_{k,c} - \tilde{\delta}_{l,c}) X_{jct}^{\rho_0}\right) + \left(1 - \rho_0\right)_{jct} - \log \tilde{\delta}_{l,c}$$

I then follow the same approach as the one outlined for the second step of the estimation in Section 4.3.2, assuming that the TFP follows a first-order Markov process, and accounting for the selection bias introduced by focusing on single product farms and using the selection correction. I obtain the following parameters, where $\{\rho, \delta_{s,wheat}, \delta_{s,other}\}$ are naturally similar to the estimates of Section 4.3.2, as the first part of the estimation is identical.

Table A15: Alternative Production Function (Full CES)

| Coefficient | Parameter | Estimate | Std. Error |
|---|---|---|---|
| Substitution Land-Chemicals | $\rho$ | 0.44 | (0.19) |
| Substitution Labor-Capital-Land Nest | $\rho_0$ | 0.18 | (0.14) |
| Land Share (Wheat) | $\delta_{s,wheat}$ | 0.45 | (0.22) |
| Land Share (Others) | $\delta_{s,other}$ | 0.40 | (0.20) |
| Labor Share (Wheat) | $\tilde{\delta}_{l,wheat}$ | 0.46 | (0.076) |
| Labor Share (Others) | $\tilde{\delta}_{l,other}$ | 0.43 | (0.057) |
| Land-Nest Share (Wheat) | $\tilde{\delta}_{x,wheat}$ | 0.25 | (0.013) |
| Land-Nest Share (Others) | $\tilde{\delta}_{x,other}$ | 0.23 | (0.071) |
| N | | 13,281 | |
| Unique Farms | | 3,901 | |
| N - Wheat | | 4,089 | |
| N - Other | | 9,192 | |

*Notes: The parameters are obtained from a two-step estimation, each step performed by GMM. The estimation is run on the FADN French sample restricted to 1980-2015. Observations prior to 1980 are removed, as they do not contain output price data. Observations post-2015 are removed to match the length of my agricultural hourly wage panel. I keep farms observed for at least three periods in a row[62], and that produce either only crops in the wheat group, or in the other crops group. I*

[62] This is done to smooth potential measurement issues, and follows from De Loecker et al. (2016) and Doraszelski

*remove farms that are not observed with positive input values for the set of considered inputs (land, labor, capital, fertilizers, pesticides). standard errors are obtained using a block bootstrap procedure, where I draw all the observations related to a farm at a time, using $B = 1000$. The two steps of the production function estimation are included in the bootstrap, in order for the standard errors of the second step to account for the fact that parameters from the first step are also estimated, rather than known.*

Importantly, the estimation yields $\rho_0 = 0.18$ (0.14). At the point estimate, this translates into some substitutability between labor, capital and the land nest, and means that the condition detailed above holds, i.e. an increase in profitability, whether it comes from an increase in TFP or chemical-use efficiency, will correlate positively with an increase in chemical-use intensity. However, this coefficient is not statistically significant, and this estimation cannot reject the Cobb-Douglas structure relating labor, capital and the land-chemicals composite input. The point estimate of $\rho_0$ implies a small degree of substitutability between these inputs, which means that the Cobb-Douglas used in the main specification will be a good approximation of this more general production framework.

In that sense, the Cobb-Douglas provides a lower bound for the correlation between farm profit and chemical-use intensity, and for the effects of farm exit and selection on profitability on overall chemical-use intensity. In this context, the Cobb Douglas remains my favored specification, as it provides a very transparent framework where the sign of this correlation is driven by a single parameter with a clean interpretation.

### D.5 Joint Production Estimation Framework

Here I describe an alternative joint production framework, for which I also provide estimates. This approach combines the work of Dhyne et al. (2022) and of Doraszelski and Jaumandreu (2018). From the first, I use the idea that inputs can be shared across product lines, and that the parametrization of a firm-level transformation function will define the degree of penalization that accompanies this sharing of inputs. I then draw from Doraszelski and Jaumandreu (2018) to model and estimate non-Hicksian shocks, within that multi-product production framework. Because the recovery of non-Hicksian shocks relies heavily on the first-order conditions taken from the firm-level optimization problem, I move away slightly from the exact specification used by Dhyne et al. (2022).

In effect, consider a firm $j$ which produces the set of crops $\mathbb{C}_{jt}$ within a production period $t$. The firm is endowed with product-specific input-neutral shocks $\tilde{\omega}_{jct}$. I write the transformation function as:

$$f_{jct} = exp(\tilde{\omega}_{jct}) g_{jct}^{\gamma_{cc}} \prod_{c' \in \mathbb{C}_{jt}, c' \neq c} g_{jc't}^{\gamma_{cc'}}$$

and Jaumandreu (2018).

Here $f_{jct}$ is the quantity of crop $c$ produced. That quantity depends on a specific TFP shock, and on product-specific functions $g_{jct}$ which are evaluated at the level of inputs used for each crop grown in the season. The parameters $\gamma_{cc}$ parametrize the rivalry or complementarity in production from jointly producing $c$ and $c'$, and potentially sharing some inputs across these two product lines. I keep the functions $g_{jct}$ fairly general for the moment, and parametrize them further later. One additional set of assumptions I make is that:

$$g_{jct} = g_c(K_{jt}, L_{jt}, Fert_{jt}, Pest_{jt}, S_{jct}, \omega^p_{jt}, \omega^f_{jt})$$

This means that each function $g_{jct}$ depends on the firm-level chosen amounts of capital $K_{jt}$, labor $L_{jt}$, fertilizer $Fert_{jt}$, pesticides $Pest_{jt}$, on a crop-specific chosen amount of land $S_{jct}$, and on firm-level pesticide and fertilizer productivity shocks that I denote $\omega^p_{jt}$ and $\omega^f_{jt}$. This implies that all inputs but land are selected at the firm-level, and shared across product-lines according to the $\gamma$ matrix. I also consider that the $g_c$ function only varies across crops, but not periods or farms, and that these two non-Hicksian shocks fully account for the heterogeneity in these functions across farms. This assumption of land being a crop-specific input, while others are shared with the same penalty $\gamma$ allows me to derive an estimating equation to recover the non-Hicksian shocks.

With this specification, the firm-level profit is as follows:

$$\Pi_{jt} = \left( \sum_{c \in \mathbb{C}_{jt}} P_{jct}(f_{jct}, D_{jct}) f_{jct} - P^S_{jt} S_{jct} \right) - P^L_{jt} L_{jt} - P^F_{jt} Fert_{jt} - P^p_{jt} Pest_{jt}$$

Taking $\{L_{jt}, Fert_{jt}, Pest_{jt}, S_{jct}\}$ as flexible inputs, their setting will only impact within period profit. I write $P_{jct}(f_{jct}, D_{jct})$ for the inverse demand function, known to the farmers, which depends on the volume produced and sold $f_{jct}$, and a demand shock $D_{jct}$. Writing $X_{jt}$ for any of the public inputs $\{L_{jt}, Fert_{jt}, Pest_{jt}\}$, I get the following expression:

$$\frac{\partial \Pi_{jt}}{\partial X_{jt}} = 0 \Leftrightarrow P^X_{jt} = \sum_{c \in \mathbb{C}_{jt}} \frac{\partial g_{jct}}{\partial X_{jt}} g^{-1}_{jct} \sum_{c' \in \mathbb{C}_{jt}} \gamma_{cc'} f_{jc't} P_{jc't} \left[ 1 - \frac{1}{\eta_{jc't}} \right]$$

I denote by $\eta_{jct}$ the absolute value of the elasticity of demand. I then look at land, the only private input in this set-up:

$$\frac{\partial \Pi_{jt}}{\partial S_{jct}} = 0 \Leftrightarrow P^S_{jt} = \frac{\partial g_{jct}}{\partial S_{jct}} g^{-1}_{jct} \sum_{c' \in \mathbb{C}_{jt}} \gamma_{c'c} f_{jc't} P_{jc't} \left[ 1 - \frac{1}{\eta_{jc't}} \right]$$

Combining these two equations, and under symmetry of the $\gamma$ matrix, I get:

$$\frac{P^X_{jt}}{P^S_{jt}} = \sum_{c \in \mathbb{C}_{jt}} \frac{\frac{\partial g_{jct}}{\partial X_{jt}}}{\frac{\partial g_{jct}}{\partial S_{jct}}}$$

This relation is quite intuitive, the relative allocation of inputs has to be such that the ratio of their marginal impacts on production across production lines equals the ratio of their prices.

To go further, I assume a specific shape for the $g_{jct}$ function[63]. I take:

$$g_{jct} = K_{jt}^{\alpha_k} L_{jt}^{\alpha_L} \left\{ \delta_s^c S_{jct}^{\rho} + \delta_p^c \left[ e^{\omega^p_{jt}} Pest_{jt} \right]^{\rho} + \delta_f^c \left[ e^{\omega^f_{jt}} Fert_{jt} \right]^{\rho} \right\}^{\frac{\alpha_s}{\rho}}$$

Without relying on a translog specification, this is the simplest specification which allows me to model non-Hicksian productivity shocks. I note that the heterogeneity in the $g_{jct}$ across crops only relies on the $\{\delta_s^c, \delta_f^c, \delta_p^c\}_c$ parameters. Because I am mostly worried about variations in crop-composition impacting the chemicals-to-land ratios at the farm-level, this is the main form of heterogeneity I am interested in. A more expansive heterogeneity could specify crop-specific elasticities $\rho_c$, which I assume away here.

With this parametrization, I obtain the following estimating equation, where lower-case letters represent logs, with $x \in \{pest, fert\}$ :

$$p^x_{jt} - p^s_{jt} = \rho \omega^x_{jt} + (\rho - 1) x_{jt} + \log \Big( \sum_{c \in \mathbb{C}_{jt}} \frac{\delta_x^c}{\delta_s^c} S_{jct}^{1-\rho} \Big)$$

This equation mirrors the first stage estimating equation from Doraszelski and Jaumandreu (2018), adapted to the parametrized multi-product setting. Here, the input ratio can either be affected by variations in the input price ratio, by changes in the establishment's production mix, or by changes in the establishment's proficiency in using input $x$ in the production process. Similarly to them, when the inputs of interest are substitutes, and holding everything else constant, an increase in $\omega^x_{jt}$ will increase the ratio of $x_{jt}$ to land.

I propose to recover the $\{\delta_s^c, \delta_f^c, \delta_p^c\}_c$, and $\rho$ in a first estimation step, and to recover the remaining parameters in a second step. To simplify the estimation process, I assume that $\gamma_{cc} = 1$, and $\gamma_{cc'} = \gamma$, and verify post-estimation that $\gamma_{cc'} \in ]-\frac{1}{dim(\mathbb{C})}, 0[$ to guarantee that the second order conditions from the optimization problem are met.

[63] Note that this framework can also accommodate a full CES specification. Given the difficulty of estimating the parameters within a CES nest, I keep the CES structure to a minimum - the land nest being itself estimated within an equation which is linear in part of the parameters.

**First Step:** I specify a structure for the two input-biased productivity shocks. For each of them, I assume an AR(1) structure such that, with $\zeta^x_{jt+1}$ an exogenous i.i.d. innovation shock:

$$\omega^x_{jt+1} = \mathbb{E}[\omega^x_{jt+1}|\omega^x_{jt}] + \zeta^x_{jt+1} = g^x(\omega^x_{jt}) + \zeta^x_{jt+1}$$

I take $g^x(.)$ to be a third-order polynomial. I use the following moments, with $A^1_{jt}$ the matrix of instruments:

$$\mathbb{E}[(\zeta^p_{jt} + \zeta^f_{jt})A^1_{jt}] = 0$$

My instruments match the production function literature, and correspond to lagged firm-level observables presumably uncorrelated to the productivity innovations, as well as the current values of farm-level land prices and hourly agricultural wages.

**Second Step:** For the second step, I take the log of the production function, and re-order the elements to get:

$$\tilde{\omega}_{jct} = \log(f_{jct}) - \left(\sum_{c' \in \mathbb{C}_{jt}} \gamma_{cc'}\right)\left[\alpha_K k_{jt} + \alpha_L l_{jt}\right] - \sum_{c' \in \mathbb{C}_{jt}} \gamma_{cc'} X_{jc't}$$

Where $X_{jc't}$ is the log of the land-fertilizer-pesticide nest, which I can compute using the estimates from the first step. Now, I use crop-specific policy cushions, as well as land prices and agricultural wages as instruments denoted $A^2_{jt}$ in a GMM estimation, with moments:

$$\mathbb{E}\left[\tilde{\omega}_{jct} A^2_{jt}\right] = 0$$

I obtain the following results:

Table A16: Joint Production - Parameters

| Coefficient | Parameter | Estimate | Std. Error |
|---|---|---|---|
| Capital Share | $\alpha_k$ | 0.2415 | 0.0006 |
| Labor Share | $\alpha_l$ | 0.4812 | 0.0015 |
| Land Share | $\alpha_s$ | 0.0931 | $2.2e^{-5}$ |
| Penalty | $\gamma$ | -0.0904 | 0.00013 |
| Substitution Land-Chemicals | $\rho$ | 0.3886 | 0.00073 |
| Fertilizer Share | $\delta_f$ - Cereals | 0.2428 | 0.0027 |
| Fertilizer Share | $\delta_f$ - Oil/Protein | 0.2916 | 0.0034 |
| Fertilizer Share | $\delta_f$ - Industrial | 0.0282 | 0.0002 |
| Pesticide Share | $\delta_p$ - Cereals | 0.1789 | 0.0025 |
| Pesticide Share | $\delta_p$ - Oil/Protein | 0.2925 | 0.004 |
| Pesticide Share | $\delta_p$ - Industrial | 0.1024 | 0.002 |

*Notes:* The parameters are obtained from a two-step estimation, each step performed by GMM. The estimation is run on the FADN French sample. Standard errors from the second step are corrected for the two-step procedure, following Doraszelski and Jaumandreu (2018), and the coefficients associated to the input-biased productivity processes are concentrated out, and not estimated.

I note a negative value for $\gamma$ indicating that the more you share the public inputs across production lines, the less a given crop will benefit from that input (either because of an increase in scope, or an increase in the scale of the other product's production levels). I also note that $\rho \in ]0,1[$, indicating that land, pesticides and fertilizers are substitutes in production, which was expected.

### D.6 Selection Correction for the Production Function Estimation

I need to account for selection while building moments for the second step of the estimation procedure. Using $\Xi_{jct} = 1$ a dummy for the fact that $j$ only produces crop $c$ in period $t$, I want to build moments based on the following corrected process:

$$\omega^h_{jct+1} = \mathbb{E}[\omega^h_{jct+1} | \omega^h_{jct}, \Xi_{jct+1} = 1] + \xi^h_{jct+1} \tag{14}$$

With $I_{jt}$ the farm's information set at the end of period $t$, I want to base the moments on the following derivation—with $\overline{\omega}^h_{jcc't+1}$ the threshold used at the end of period $t$ to decide on the inclusion of crop $c'$ on top of $c$. This $\overline{\omega}^h_{jcc't+1}$ corresponds to the minimum value the farm's TFP must take ($\omega^h_{jt}$), for $c'$ to be added in the crop mix as a second crop, when $c$ is the first crop included in the mix. Because the crop mix is set at the end of the period for the next, $\omega^h_{jt}$ rather than $\omega^h_{jt+1}$ is the TFP value compared to the threshold $\overline{\omega}^h_{jcc't+1}$. This threshold will be a function of both input and output prices, agricultural subsidies, the farm's capital stock and chemical-use efficiency.

Here $c$ is the ex-ante most profitable crop to grow in period $t$, and $c'$ the second (out of two crops). The farms' competence ladder is both farm-specific and time-constant, which means that there will be a time-constant relation between the farm's TFP between $c$ and $c'$. I also note that because demand and input prices play a role in crop choice—the core competence of a farm might not always match the identity of the unique crop it grows. I can write, using the log of the production function and the process specified in Equation (14):

$$\begin{aligned}
\xi^h_{jct+1} + \varepsilon_{jct+1} &= q_{jct+1} - \alpha^c_k k_{jt+1} - \alpha^c_l l_{jct+1} - \alpha^c_s X_{jct+1} - \mathbb{E}[\omega^h_{jct+1} | I_{jt}, \Xi_{jct+1} = 1] \\
&= q_{jct+1} - \alpha^c_k k_{jt+1} - \alpha^c_l l_{jct+1} - \alpha^c_s X_{jct+1} - \mathbb{E}[\omega^h_{jct+1} | \omega^h_{jct}, \overline{\omega}^h_{jcc't+1}] \\
&= q_{jct+1} - \alpha^c_k k_{jt+1} - \alpha^c_l l_{jct+1} - \alpha^c_s X_{jct+1} - g_h(\omega^h_{jct}, \overline{\omega}^h_{jcc't+1}).
\end{aligned}$$

Where moving from line 1 to 2 relies on the first-order Markov process assumption, and the definition of the threshold $\overline{\omega}^h_{jcc't+1}$.[64] I can then specify $g_h(.)$ as a polynomial of its two terms. While Equation (7) provides an expression for $\omega^h_{jct}$, I also need one for the threshold $\overline{\omega}^h_{jcc't+1}$. For this I derive an expression for the conditional probability of producing only crop $c$ as a function of farm productivity and its state, focusing on the crop-addition threshold.[65] The competence ladder structure means that the threshold for the production of the second crop $c'$ can be expressed using the farm's observed TFP shock for crop $c$ combined with its competence ladder $\sigma_j$—specifically by its rank on the farm's ladder ($\sigma_j^{-1}(c)$), as well as additional variables that influence the value of the threshold—input and output prices, capital stock and chemical productivity:

$$\begin{aligned}
\mathbb{P}\Big(\Xi_{jct+1} = 1\Big) &= \mathbb{P}\Big(\omega^h_{jct} \leq \overline{\omega}^h_{jcc't+1}(\omega^{ch}_{jct}, K_{jt+1}, \sigma_j^{-1}(c), \Omega_{jt}) | \overline{\omega}^h_{jcc't+1}(.), \omega^h_{jct}\Big) \\
&= h(\overline{\omega}^h_{jcc't+1}(.), \omega^h_{jct}) \\
&= h(\omega^h_{jct}, \omega^{ch}_{jct}, K_{jt+1}, \sigma_j^{-1}(c), \Omega_{jt}) \\
&= h(\{X_{jct}\}, \omega^{ch}_{jct}, K_{jt+1}, \sigma_j^{-1}(c), \Omega_{jt}) \\
&= SP_{jt}
\end{aligned}$$

Equation (7) is used to move from the third to the fourth line. While I do not observe $\sigma_j^{-1}(c)$, I approximate it by the interaction between the farm's location and a dummy for the unique crop it produces—assuming competence ladders are determined by local soil and climatic factors. $SP_{jt}$ is recovered non-parametrically, and as in Olley and Pakes (1996) and De Loecker et al. (2016), under some regularity conditions on the density of $\omega^h_{jct}$, I can invert $SP_{jt}$ to obtain a proxy for the threshold $\overline{\omega}^h_{jcc't+1}$. I can then use it in the polynomial

[64] Because of the assumed competence ladder structure, firms face a unique TFP process that propagates across their crops. The process $g_h$ is similar across crops as a direct consequence of this.

[65] Using a balanced panel of farms helps to address the selection bias induced by farm exit, when no grain production is observed in the FADN.

approximation of $g_h(.)$, which depends both on the threshold and on the farm TFP.

### D.7 Robustness for the Estimation of the Production Function

Table A17: Production Function (Estimation Robustness)

| Coefficient | Parameter | Estimate | Std. Error | Estimate (R) | Std. Error (R) |
|---|---|---|---|---|---|
| Substitution Land-Chemicals | $\rho$ | 0.44 | (0.19) | 0.55 | (0.11) |
| Land Share (Wheat) | $\delta_{s,wheat}$ | 0.45 | (0.22) | 0.29 | (0.24) |
| Land Share (Others) | $\delta_{s,other}$ | 0.40 | (0.20) | 0.51 | (0.20) |
| Labor Share (Wheat) | $\alpha_{l,wheat}$ | 0.38 | (0.022) | 0.39 | (0.14) |
| Labor Share (Others) | $\alpha_{l,other}$ | 0.44 | (0.019) | 0.41 | (0.16) |
| Land-Nest Share (Wheat) | $\alpha_{x,wheat}$ | 0.40 | (0.024) | 0.38 | (0.18) |
| Land-Nest Share (Others) | $\alpha_{x,other}$ | 0.39 | (0.022) | 0.44 | (0.13) |
| N | 13,281 | | | | |
| Unique Farms | 3,901 | | | | |
| N - Wheat | 4,089 | | | | |
| N - Other | 9,192 | | | | |

*Notes: Production function parameters are obtained from a two-step GMM estimation. Both steps are done on the FADN French sample restricted to 1980-2015. Prior to 1980 the FADN does not contain output price data, while my agricultural hourly wage data stops in 2015. I keep farms observed for at least three periods in a row*[66]*, and that produce either only wheat or the other aggregated crop. Relying on interior solutions from profit optimization, I remove farms that are not observed with positive input values for the set of considered inputs (land, labor, capital, fertilizers, pesticides). Standard errors for the production function are obtained with a block bootstrap procedure at the farm level, using* $B = 1000$*. The two steps of the production function estimation are included in the bootstrap so that second step standard errors account for first step uncertainty. The two sets of estimates compare results when using/not using my two subsidy measures (the land subsidy and the average farm exposure to EU subsidy) as instruments. Estimates obtained without the subsidy instruments are in the column marked by (R) for robust.*

### D.8 Monopolistic Competition

An alternative to perfect competition is to adopt a monopolistic competition framework—which preserves the single agent setting—but allows for price dispersion and some horizontal differentiation across goods. While agricultural products are commodities usually thought of as homogeneous, exogenous local conditions can lead to similar crops having different moisture or nutrient content, making them more or less suitable for different uses. In that sense, a model with differentiated products might be preferable. Monopolistic competition can also allow for mark-ups that vary across producers, and along with changes in market structure.

I adopt for this section the quadratic demand of Mayer et al. (2014). The demand structure

goes as follows: in a closed economy, a representative consumer has the following utility at period $t$:

$$U_t = q_{0,t} + \alpha \int_{i\in\Omega} q_{i,t}^c di - \frac{1}{2}\gamma \int_{i\in\Omega} (q_{i,t}^c)^2 di - \frac{1}{2}\eta \Big( \int_{i\in\Omega} q_{i,t}^c di \Big)^2 \tag{15}$$

$q_{0,t}$ corresponds to the quantity of outside good consumed. Agricultural varieties $i \in \Omega$ are differentiated. $\alpha$ and $\eta$ control the substitution between differentiated varieties and the outside good, while $\gamma$ accounts for the degree of differentiation across varieties. Here, I note that two varieties can be two types of corn grown by two different farms, or wheat and corn grown by the same farm. I discuss next a clustering of the agricultural market, where each market corresponds to a crop type, and firms produce different varieties within each crop market.

I take the outside good to be the numeraire. This preference structure implies the following demand for variety $i$ - with $\overline{p}_t$ the average price of agricultural varieties on the market, and $M_t$ the mass of consumed varieties:

$$q_{i,t} = \frac{\alpha}{\eta M_t + \gamma} - \frac{1}{\gamma} p_{i,t} + \frac{\eta M_t}{\eta M_t + \gamma} \frac{1}{\gamma} \overline{p}_t$$

I see that $M_t$ and $\overline{p}_t$—market-level statistics on which atomistic farms have no individual impact—determine the characteristics of the demand curve faced by farms. From this relation, I can also derive a choke price $p_t^{max}$ above which farms will not face any positive demand.

**Estimating Demand:** I use the following expression to recover the parameters of demand. From now on I denote by $m$ the crop grown, to match the competence ladder structure that I introduce next. Because crops are differentiated in the model, each of a farm's varieties is uniquely produced by that farm.

$$q_{mt} = \frac{1}{\gamma}\alpha - \frac{1}{\gamma} p_{mt} - \frac{\eta}{\gamma} Q_t$$

I take for $Q_t$ total agricultural production sold in France, approximated using the weighted sum of sold output in the FADN, for the set of crops that I consider. $q_{jmt}$ and $p_{jmt}$ are respectively the quantity sold, and associated price, for each variety produced by a farm in a given period. I estimate the following regression using a two-stage least squares strategy, with exogenous shocks to firm supply as instruments. Specifically, I use local realized and forecasted weather shocks as supply shocks.

$$q_{jmt} = \beta_0 + \beta_1 p_{jmt} + \beta_2 Q_t + \varepsilon_{jmt}$$

I recover the following parameters. I only use data post-2003 for the estimation, in order to obtain demand parameters that are not impacted by intervention pricing.

Table A18: Demand - Parameters

| Coefficient | Parameter | Estimate | Std. Error |
|---|---|---|---|
| Substitution with Numeraire | $\eta$ | 1.32e-07 | 6.92e-08 |
| Substitution within Varieties | $\gamma$ | 3.19e-02 | 1.01e-02 |
| Substitution with Numeraire | $\alpha$ | 203.5 | 71.42 |

*Notes: I recover the parameters from a 2SLS regression, using FADN data from 2003-2020, and using realized and forecasted weather data constructed at the department-year level as supply shocks. The realized weather data is constructed from the ECMWF ERA-5 hourly weather data series, and the forecasts come from ECMWF SEAS-5. Both are aggregated into growing season observations.*

The standard errors for the transformed parameters are computed using the delta method. This yields a decreasing choke price over time, which drops significantly around the time of the reform.

### D.9 Opening to Trade

My main model considers the effect of agricultural policies within a closed French economy. In this sub-section, I consider an extension of the model where both France and a representative producer for the rest of the world together produce grains consumed by a world consumer. This extension highlights two channels through which trade can interact with the effect of agricultural subsidies. First, European agricultural policy can increase (in the case of subsidies) or decrease (in the case of environmental regulation) the French market share of world grain production. This is a frequently debated feature of European agricultural subsidies. Second, if France is integrated into world markets, one can expect that French producers face a relatively more elastic residual demand curve than they would in a closed economy. This is the case if world demand is relatively inelastic, and crops from different origins are close substitutes, which is likely for agricultural commodities. The consequence of the first point is that regulations can lead to pollution leakages, and by driving production outside of France, only decrease local pollution in France at the cost of increased local pollution abroad. The second point implies a shift in the incidence of policies: as the elasticity of the residual demand faced by France increases, producers receive more of the gains of subsidies, but also bear more of the costs of environmental regulation. This second feature changes the political economy of subsidies and regulations, making subsidies relatively more favorable in the eyes of producers, and as such possibly easier to implement.

I consider a world consumer with an iso-elastic demand for each crop $c$, removing any time subscript $t$ for simplicity:

$$Q^D_{W,c} = \exp(\gamma_{W,c}) P_c^{\beta_{W,c}},$$

where $P_c$ is the world price for crop $c$, $\beta_{W,c}$ the overall elasticity of world demand for crop $c$, and $\gamma_{W,c}$ a term scaling overall world demand. $Q^D_{W,c}$ is then written as a CES composite of French and rest-of-the-world production of crop $c$:

$$Q^D_{W,c} = \left( \omega_c^{\frac{1}{\sigma}} (Q^D_{F,c})^{\frac{\sigma-1}{\sigma}} + (1-\omega_c)^{\frac{1}{\sigma}} (Q^D_{ROW,c})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}},$$

with $\omega_c$ representing the relative weight of French grain in world demand for grain $c$, $\sigma$ the elasticity of substitution between the two origins of grain $c$ assumed to be homogeneous across crops, and $Q^D_{F,c}$ and $Q^D_{ROW,c}$ resp. the demand for grain $c$ from France, and from the rest of the world.

Finally, I consider that the rest of the world has an iso-elastic supply curve for each grain $c$:

$$Q^S_{ROW,c} = \exp(\alpha_{ROW,c}) P_{ROW,c}^{\delta_{ROW,c}},$$

with $\delta_{ROW,c}$ the elasticity of world supply, and $\alpha_{ROW,c}$ scaling the volume of production. In this simple model, there are no trade costs, and substitution between the two sources of production is done solely according to their respective prices and given the elasticity of substitution $\sigma$.

Solving the model yields a residual demand curve for French grain, for each crop $c$:

$$Q^D_{F,c} = \omega_c \left( \frac{P_{F,c}}{P_c} \right)^{-\sigma} Q^D_{W,c},$$

with $P_c = \left( \omega_c P_{F,c}^{1-\sigma} + (1-\omega_c) P_{ROW,c}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}$ the CES price index for crop $c$. I integrate this extension into the model by relying on several calibrations:

- $\sigma = 5.4$ is calibrated using the elasticity estimated by Costinot et al. (2016),
- $\beta_{W,c} = -0.054$ and $\delta_{ROW,c} = 0.087$, identical across crops, calibrated from Roberts and Schlenker (2013) who estimate iso-elastic demand and supply curves for the four main grain commodities,
- $\omega_c$ is set such that at baseline the simulated and observed French market share of world grain production as observed in FAOSTAT data in 2015 match, respectively 5.8% for wheat and 2.5% for the other grains,

- Finally the $\alpha_{ROW,c}$ and $\gamma_{W,c}$ are set such that at the output prices obtained in my closed-economy baseline model without any agricultural policy, France produces the right share of the world output, and the same amount of output as observed in this baseline equilibrium. These are set at (19.17, 20.34) for the $\alpha_{ROW,c}$, and (19.98, 21.11) for the $\gamma_{W,c}$ parameters.

The model is run without any policy to establish a new baseline point of comparison, and then with each policy presented in Table 3, highlighting changes in rest of the world production relative to baseline in order to illustrate potential leakages, and the evolution of French producer surplus, to capture the effect of openness to trade on the incidence of agricultural policy.

# E Decomposition

I propose a slight modification of the dynamic decomposition of Melitz and Polanec (2015) to highlight the role of each channel in changing the market's chemical-use intensity: input mix changes, crop mix changes, reallocations across infra-marginal incumbents and market selection. Here market selection means a change in the support of the equilibrium measure $\mu$ describing the equilibrium allocation of farms over the potential farm state. The decomposition proposed by Melitz and Polanec (2015) has the advantage of presenting easily interpretable terms for the contribution of market selection. For example, this term will be negative if the extension of the support is such that new farm types that can now survive in the market have a lower chemical-use intensity than infra-marginal incumbents. I augment the decomposition to further highlight the role of changes in crop mix composition within the farm.

I decompose the evolution of the market-level average chemical-use per unit of output between $t$ and $t+1$, with $\mathcal{A}$ denoting the set of farm types for which the equilibrium mass is positive across both equilibria (infra-marginal incumbents), $\mathcal{B}$ the set of farms which only select into the market in period $t$, and $\mathcal{C}$ for farms which only select into the market in period $t+1$. I write:

$$\begin{aligned}
\Delta\Phi = & \frac{1}{N_{\mathcal{A}}} \sum_{j \in \mathcal{A}} \left( \omega_{jt}^{c_1} \Delta\phi_j^{c_1} + (1 - \omega_{jt}^{c_1}) \Delta\phi_j^{c_2} \right) \\
& + \frac{1}{N_{\mathcal{A}}} \sum_{j \in \mathcal{A}} \left( \phi_{jt+1}^{c_1} - \phi_{jt+1}^{c_2} \right) \Delta\omega_j^{c_1} \\
& + \Delta\mathbb{C}\text{ov}_{\mathcal{A}}\left(\tilde{s}_j, \phi_j\right) \\
& + s_{\mathcal{C}t+1}\left(\Phi_{\mathcal{C}t+1} - \Phi_{\mathcal{A}t+1}\right) \\
& + s_{\mathcal{B}t}\left(\Phi_{\mathcal{A}t} - \Phi_{\mathcal{B}t}\right).
\end{aligned}$$

Here, $\Delta\Phi$ is the change in the market average chemical-use per unit of output, $s_{Gt} = \sum_{j \in G} s_{jt}$ is the market share of the group of farms $G$ in period $t$, and market shares are computed using total output levels. $\Phi_{Gt} = \sum_{j \in G} \left(\frac{s_{jt}}{s_{Gt}}\right) \phi_{jt}$ is group $G$'s average chemical-use per unit of output, with $\phi_{jt}$ standing for farm $j$'s average chemical-use per unit of output. $\omega_{jt}^{c_1}$ is the share of output within farm $j$ coming from wheat, and $\phi_{jt}^{c_1}$ is farm $j$'s chemical-use per unit of output specifically for wheat. Finally, $\mathbb{C}\text{ov}_{\mathcal{A}}\left(\tilde{s}_{jt}, \phi_{jt}\right) = \sum_{j \in \mathcal{A}} (\tilde{s}_{jt} - \bar{s}_{\mathcal{A}t})(\phi_{jt} - \bar{\phi}_{\mathcal{A}t})$ represents the cross-farm sample covariance between farm's chemical-use intensity and its normalized intra-group market share ($\tilde{s}_{jt} = s_{jt}/s_{\mathcal{A}t}$), specifically for infra-marginal farms, isolating the effect of reallocations across continuing farms.

I use this accounting equivalence to decompose the changes produced by policies analyzed in Section 5.1, as well as for the analysis focusing on the MacSharry reform shown in Appendix G. I also show the crop-level equivalent decompositions for Section 5.1.

# F Sensitivity Analysis for the Production Function Estimation

I rely on the work of Andrews et al. (2017) to discuss the role of the identifying assumptions regarding instrument exogeneity in driving the estimation of the production function parameters. The purpose of the method developed by Andrews et al. (2017) is to recover the sensitivity of model parameters to the moments used in structural estimation, and in this extending the omitted variable bias formula to structural cases. They propose to recover a matrix they call the "sensitivity" matrix, which maps the relation between the parameters of the model, and the moments used for their estimation. When combined with alternative assumptions about instrument validity—recast as the impact of alternative assumptions on moment values—this sensitivity matrix allows one to predict the impact of alternative assumptions on the estimated parameters. In other words, this analysis aims at shedding light on the role of identifying assumptions in driving estimation.

I denote by $\hat{\theta}$ the parameter vector minimizing the criterion function:

$$\hat{g}(\theta)' \hat{W} \hat{g}(\theta)$$

In this case, $\hat{\theta}$ can be the estimate for the first stage and second stage of the production function estimation. For any $a$ in the set A of alternative assumptions, they define a local perturbation of the model in the direction of $a$ such that the estimate $\hat{\theta}$ has first-order asymptotic bias—where $\Lambda$ is the sensitivity matrix:

$$\mathbb{E}[\tilde{\theta}(a)] = \Lambda \mathbb{E}[\hat{g}(a)]$$

An alternative assumption $a$ should be interpreted as an alternative assumption regarding

the relation between the chosen instruments and the structural errors for resp. the first and second step of estimation. The sensitivity matrix can be written as:

$$\Lambda = -\left(G'WG\right)^{-1} G'W$$

Where $W$ is the probability limit of $\hat{W}$, the weight matrix used in the GMM criterion, and $G$ is the Jacobian of the probability limit of $\hat{g}(\theta)$ evaluated at the true parameter vector $\theta_0$. $\Lambda$ serves as a local approximation to the mapping from moments to estimated parameters.

I provide an estimate of $\Lambda K$ for each of the estimation step for the production function, where $K$ is a weighting matrix based on the standard deviation of the relevant moments. As Andrews et al. (2017) discuss, the units of $\Lambda$ are contingent on the units of the different moments (and implicitly on the range of variation of the instruments). The weighting ensures that $\Lambda K$ elements can be read as the effect of a one standard deviation violation of the given moment condition on the asymptotic bias in $\hat{\theta}$.

Table A19: Sensitivity for First Step

| | Lag Chemicals-Land Ratio | Lag Capital | Lag Chemical-Land Price Ratio | Land Subsidy | Land Price | Farm Exposure |
|---|---|---|---|---|---|---|
| $\rho$ | 0.2950 | -0.0511 | -0.0479 | 0.0705 | 0.1030 | 0.0212 |
| $\delta_s$ (Wheat) | 2.8234 | -0.8465 | -1.4053 | 1.3430 | 1.9662 | 0.3819 |
| $\delta_s$ (Others) | 1.3004 | -0.0686 | 0.2710 | 0.0388 | 0.0657 | 0.0050 |

Table A20: Sensitivity for Second Step

| | Constant | Capital | Lag Capital | Lag Labor | Lag Wage | Lag Land Price |
|---|---|---|---|---|---|---|
| $\alpha_l$ (group 1) | -0.0515 | -0.0002 | 0.0002 | 0.0001 | 0.0000 | 0.0000 |
| $\alpha_s$ (group 1) | 0.0435 | 0.0001 | -0.0002 | -0.0000 | -0.0000 | -0.0000 |
| $\alpha_l$ (group 2) | 0.0223 | 0.0001 | -0.0001 | 0.0000 | -0.0000 | -0.0000 |
| $\alpha_s$ (group 2) | -0.0202 | -0.0001 | 0.0001 | 0.0000 | 0.0000 | 0.0000 |
| | Lag Chemicals-Land Ratio | Lag Land Composite | Lag Chemicals-Land Price Ratio | Lag Output Price Index | Land Subsidy | Farm Exposure |
| $\alpha_l$ (group 1) | 0.0001 | 0.0000 | -0.0005 | -0.0001 | 0.0000 | -0.0002 |
| $\alpha_s$ (group 1) | -0.0002 | 0.0001 | 0.0003 | 0.0001 | -0.0000 | 0.0002 |
| $\alpha_l$ (group 2) | -0.0000 | -0.0002 | 0.0002 | 0.0000 | 0.0000 | 0.0001 |
| $\alpha_s$ (group 2) | 0.0001 | 0.0001 | -0.0001 | -0.0000 | 0.0000 | -0.0001 |

# G A Simplified MacSharry: Budget Equivalent Government Purchasing vs. Land Subsidies

The MacSharry reform consisted both of the end of intensive output price intervention via government purchasing on the European agricultural market, the removal of import tariffs and export subsidies, and of the introduction of land subsidies. As my model does not capture the role of trade, it is ill-suited to fully model the consequences of the reform. In light of this, in this exercise, I focus on one of the dimensions central to the reform, which is the move from government purchasing, which I model as purchasing at the market price, to a land subsidy. Specifically, I first model the current baseline EU land subsidy of 51%, and then search for the amount of grain purchasing that generates a similar government expenditure. I find the budget equivalent level of purchasing to be of 2.7 million tons of grain for each crop. While data on the quantities directly purchased in Europe through government intervention are hard to obtain, several reports track the evolution of the intervention stocks for cereals pre MacSharry, with stocks estimated around 9 million tons in 1989, 12 million in 1990, to over 30 in 1992.[67] Given the substantial share of European grain production coming from France, this implies that the levels of intervention I model are in line with pre MacSharry purchasing. Table A21 shows the consequences of the policy change.

Table A21: Government Purchasing vs. Land Subsidization (Budget Equivalent)

| *Panel A* | Market Outcomes | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | P (Wheat) | P (Other) | Q (Wheat) | Q (Other) | Avg. Profit | Effective Avg. Cost (wheat, other) | Chemical/Q (Wheat) | Chemical/Q (Other) |
| Government Purchasing | 153 €/t | 203 €/t | 34.7 Mt | 31.6 Mt | 37.8 k€ | (119, 169) | 0.51 | 0.70 |
| Land Subsidy | 143 €/t | 196 €/t | 32.6 Mt | 29.2 Mt | 33.5 k€ | (128, 178) | 0.42 | 0.62 |
| *Panel B* | Welfare Implications | | | | | | | |
| | Consumer Surplus | | Total Profit | | Chemical Use (units of deflated chemical input) | | | |
| Δ Level | + 535.7 M€ | | -96.4 M€ | | -8.2 M | | | |
| *Panel C* | Decomposing the Change in Market Average Chemical-per-Output (% relative to pre-reform) | | | | | | | |
| | Crop-Level (Within) + | | Crop Mix (Within) + | | Reallocation (Infra-Marginal Farms) | | Entry + Exit | |
| | -15.5% | | +0.26% | | 0.28% | | +0.0% | |

*Notes:* I recover the equilibrium under a homogeneous (51%, 51%) land subsidy. I then perform a bisection to find the level of government purchasing for which government expenditure is equivalent. This is achieved for a symmetric government purchase of 2.7 Mt of grain on each crop market. Both equilibria have a government expenditure of 973.2M€. This table compares these two equilibria. Average costs reflect the effective resource costs of production, exclusive of subsidies, in order to reflect the efficiency of production.

**Reform Evaluation:** The change in subsidy design brought by the reform has a large impact on both output prices: the price of wheat falls by about 6.5%, while the price of the second crop decreases by 3.4%. This decrease in output prices is combined with a fall in produced quantities: the quantity of wheat produced decreases by 6.0%, while the quantity of the aggregated crop decreases by 7.6%. Because the aggregated crop is produced relatively more

[67] See respectively "The Agricultural Situation in the Community", a 1990 report by the Commission of the European Communities (Commission of the European Communities (1991)), https://aei.pitt.edu/31386/1/CM5890934ENC_002.pdf, and "Achieving a New Common Agricultural Policy" (Bachelier (2010)).

by high TFP and high chemical-use efficiency producers in the model, the additional output that was purchased by the government was accommodated by a smaller price increase, which here translates into a smaller price decrease for that aggregated crop (relative to wheat) as the policy is removed.

The reform raises consumer surplus, while it decreases producer surplus. Under the land subsidy, Table A21 shows that demand clears the market at much lower output levels than at the levels observed under government purchasing. Under government purchasing, consumers are not willing to consume all the quantity of grain produced, and in the model, this additional output is discarded. As modeled then, the policy does not benefit consumers at all. The wedge in output prices between the two equilibria captures the magnitude of the consumer surplus loss induced by purchasing relative to a land subsidy. As the market moves towards a land subsidy, consumer surplus expands substantially by 536 million euros. Producers, however, get to produce more at higher prices under government purchasing, in turn generating large profits. While the land subsidy reduces their costs of production, part of these cost reductions are passed into output prices. Overall, this implies that producers gain more from government purchasing than from a land subsidy, and the reform generates a 96 million euros loss in producer surplus.

Producer surplus shrinks substantially, and flexible farm profit tracks this decrease and falls by 11%. While the land subsidy reduces costs of production, the elasticities of demand imply that most of these cost reductions are then passed through to consumers as lower output prices. I then track costs of production, exclusive of the land subsidy, in order to obtain a measure of resource efficiency. The increase in the average costs of production illustrates the extent to which an input subsidy distorts optimal input choices, while government purchasing is less distortive to the farm-level production process.

Finally, the reform leads to a large decrease in chemical-use per unit of output for both crops. This decrease is decomposed in Panel C of Table A21, using an extension of the dynamic decomposition proposed by Melitz and Polanec (2015) shown in Appendix E. This decomposition is a total decomposition of the change in market average chemical-use per unit of output between: 1) within-farm within-crop changes in chemical-use per unit of output, 2) within-farm across crops reallocations of production within the farm, 3) changes in the covariance between market share and chemical-use per unit of output within infra-marginal incumbents that are active in both equilibria (or reallocations across infra-marginal incumbents), 4) and finally the impact of market selection changing the set of farms that can remain active on the market. The decomposition highlights that all the decrease in chemical-use intensity of the reform comes from the introduction of a land subsidy which modifies input choice for each crop and all producers, and from a decrease in production scale that

also lowers the chemical-use per unit of output ratio.[68] The decomposition further highlights that the reform has very little effect on market selection, measured here as an extension or a reduction in the set of farm types that can survive in the market. In other words, government purchasing leads to entry and increases the mass of producers on the market, but it does so without changing market selection, or the type of farm that can survive on the market. This is because the increase in average profits observed under government purchasing is not sufficient to have lower type farms overcome the fixed cost of incumbency.

In conclusion, the removal of government purchasing and its replacement by a land subsidy produces a decrease in output quantities, in prices, and a decrease in farm profit. The reform also leads to a decrease in chemical-use intensity through changes within the farm and within the crop in production practices, while it has little effect on chemical-use intensity through the farm selection channel. These findings match the results highlighted by the shift-share analysis in Section 3.2.

[68] See the expression for the optimal chemical-use per unit of output shown in Section 4.2.